Structural-Physical Theory: from Mathematical Structures to Physics, Chemistry, Life, Language, and Formal Knowledge

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Preliminary Note

The present text is part of a large book developed according to the roadmap fixed in the file `physics_theory_book_roadmap_v1.md`. The book advances in order of increasing system complexity: first the language of mathematical structures is introduced, then structural and physical objects are built, after that quantum, statistical, and field-theoretic regimes are analyzed, then the many-body, chemistry, biology-like, cognition/language, and formal-symbolic levels are developed, and finally the program is audited through its Standard Model specialization, theorem chain, failure modes, and formalization boundary. Special microphysical models, including the kernel-residue formalization of the Standard Model, are embedded into this general ladder rather than replacing it.

The canonical manuscript is organized according to the scheme "one chapter — one file," so that the structure of the book, navigation through the text, and future formalization in Lean remain transparent.

Chapter 1

Preface, Notation, and the Logical Protocol of the Book

1.1 The Place of the Book in the General Program

This book is built around one central idea: between the raw symbolic representation of an object and its final semantic meaning there exists an intermediate structural layer that is neither mere syntax nor an already formed mathematical or physical entity. It is precisely this intermediate layer that carries information about the origin of the object, about admissible transformations, about its internal composition, and about those invariants that disappear under excessively coarse factorization.

In traditional mathematical practice, a symbolic expression is often treated as a temporary shell for meaning. In traditional physical practice, an observable quantity is often treated as the main object of the theory, while deeper levels are viewed merely as technical machinery for its calculation. This book adopts the opposite methodological stance. Here structural organization precedes both the semantic quotient and the observable numerical value. Accordingly, the theory is built not according to the scheme

$$\text{expression} \to \text{meaning} \to \text{observable number},$$

but according to the longer and more informative scheme

$$\text{symbolic structure} \to \text{structural equivalence} \to \text{compositional organization} \to \text{kernel/resolvent descript}$$

This decision has two consequences. First, within the framework of this manuscript, the book does not begin with the Standard Model, quantum mechanics, or thermodynamics in their usual form, because all those languages are treated here as late, coarse-grained, and observer-dependent regimes of a more general construction. Second, the book is not organized as a collection of autonomous essays. It is built as a sequential chain

of definitions, lemmas, propositions, and theorems in which each new chapter depends explicitly on previously introduced objects.

1.2 The Main Goal of the Book

The main goal of the book is to construct a unified rigorous theory in which:

1. symbolic structures are treated as the original mathematical objects;
2. physical histories arise as admissible causal rewritings of those structures;
3. the observer is introduced as a special regime of projection and decategorification rather than as an external philosophical entity;
4. energy, momentum, inertia, transport, fields, many-body phases, chemical bonds, life-like organization, language-like symbolic systems, and formal knowledge receive a unified conceptual language;
5. the book advances by increasing system complexity and culminates in the re-emergence of mathematics as a special class of stable symbolic fixed points;
6. all major blocks of the text are prepared from the outset for formalization in Lean.

It is important to emphasize that the term "unified theory" does not mean a hasty mixing of different subject areas into one rhetorical text. It means only that all those areas must be placed within one common hierarchy of dependencies. Those are fundamentally different things. The former leads to vagueness; the latter leads to rigor.

1.3 The Result of the Book in Standard Language

Although the book is built from within the structural-physical program, the reader must understand from the beginning what final image of the theory is intended. In standard language that image can be formulated as follows.

> There exists a class of mathematical systems in which symbolic objects, admissible transformations, compositional structures, kernel/resolvent representations, and spectral invariants are organized in such a way that classical physics, quantum physics, statistical mechanics, transport theory, many-body physics, chemistry, life-like adaptive organization, language-like symbolic coordination, and formal knowledge appear as successive observer-level regimes of one deeper structural dynamics.

This formulation is intentionally programmatic for now. Its task is not to declare as already proved what still has to be developed, but to set the scale of the book immediately and to remove the false expectation that the project concerns only a special model of microphysics.

Remark 1.1 (Reader orientation)**.** The early chapters deliberately use an abstract structural language. Their role is architectural rather than rhetorical: later kernel, residue, observer, and effective-field constructions depend on this separation between symbolic organization, equivalence, and observer-accessible coarse structure.

1.4 Basic Levels of Description

Throughout the book three levels of description are systematically distinguished.

Definition 1.2 (Structural level)**.** The structural level is the level at which symbolic objects, admissible transformations, compositions, normalizations, equivalence relations, and functorial transitions between such data are given.

Definition 1.3 (Spectral level)**.** The spectral level is the level at which compositional structure is encoded through kernels, resolvents, marked coefficients, pole/residue data, dominant branches, and other spectral invariants.

Definition 1.4 (Observer level)**.** The observer level is the level at which structural and spectral data pass through observer-dependent projection, coarse-graining, and decategorification and become familiar physical, chemical, biological, linguistic, or logical quantities.

1.5 Statuses of Statements

Definition 1.5 (Definition/Axiom)**.** Definition/Axiom is the basic building block of the book. A definition introduces an object, relation, map, or invariant. An axiom fixes a foundational principle of the theory wherever that principle is not derived from earlier data.

Definition 1.6 (Derived Proposition/Theorem)**.** Derived Proposition/Theorem is a statement that must be derived inside the already fixed framework, without invoking additional programmatic hypotheses.

Definition 1.7 (Target Theorem)**.** Target Theorem is a substantive goal of the theoretical program. It must either be proved later in the book or honestly left as an open goal if a full proof has not yet been constructed at the present stage.

Definition 1.8 (Failure Mode)**.** Failure Mode is an explicitly identified point at which the program may break down. Such blocks are needed not to weaken the theory but to increase its rigor and falsifiability.

1.6 Notation

- X, Y, and W denote sets or classes of symbolic or structural states.
- $[\![x]\!]$ denotes the semantic image of the symbolic object x.
- R denotes a relation of admissible rewriting or admissible transformation.
- $\otimes$ denotes composition of objects at the structural level.
- c denotes normalization, when such a map is given.
- H denotes the structural pattern extractor.
- EFU denotes elementary functorial units.
- K denotes a kernel, and $(I - K)^{-1}$ the corresponding resolvent.
- $\mathcal{O} = (\Pi_{\mathcal{O}}, \Theta_{\mathcal{O}}, \chi_{\mathcal{O}})$ denotes an observer.
- Γ denotes a history, that is, a compatible chain of rewritings.

1.7 A Minimal Example of the Three Levels of Description

Example 1.9 (One object on three levels)**.** Suppose we are given three strings

$$x_1 = ((ab)c), \qquad x_2 = (a(bc)), \qquad x_3 = abc.$$

Assume that the semantic interpretation forgets the bracketing structure and depends only on the linear word. Then

$$[\![x_1]\!] = [\![x_2]\!] = [\![x_3]\!].$$

At the semantic level all three objects coincide. At the structural level, however, they are different: the first two objects carry different binary compositional geometries, while the third is already a normalized image. If one now introduces a kernel that records the cost of normalization and the order in which subwords are glued, then at the spectral level these three expressions also cease to be identical, even if their final semantics agrees.

1.8 Lean Orientation of the Text

1. Definitions must precede any informal use of a term.
2. Every statement of existence, uniqueness, and functoriality must be separated from motivational comments.
3. If the same word is used at the structural level and at the observer level, that fact must be marked explicitly.
4. No implicit "obvious" jumps across several levels of abstraction at once are allowed.

1.9 The Reading Order of the Book

Although the book is organized linearly, it has six large logical bands.

1. Chapters 0–3 form the closed structural core.
2. Chapters 4–8 build the observer, geometric, and dynamical basis.
3. Chapters 9–15 develop the quantum, informational, statistical, continuum, relativistic, and field-theoretic layers.
4. Chapters 16–20 treat many-body physics, chemistry, biology-like organization, cognition/language, and formal knowledge as higher regimes of the same theory.
5. Chapters 21–26 specialize the framework to the kernel-residue formalization of the Standard Model.
6. Chapter 27 closes the book with the final theorem audit, failure modes, and formalization boundary.

1.10 Conclusion of the First Chapter

The preface has fulfilled three tasks. First, it fixed the basic methodology of the book. Second, it separated the structural, spectral, and observer levels. Third, it established the language of statement statuses, without which further progress through the book would remain logically opaque.

The next chapter begins the theory proper. There the minimal structural framework will be built: symbolic structures, the semantic quotient, sequential objects, and the tower of equivalences without which no later physical interpretation has a rigorous foundation.

Chapter 2

Symbolic Structures and Sequential Objects

2.1 Why the Theory Begins with Symbolic Structures

If a single language is supposed to describe physics, chemistry, life-like organization, language, and formal knowledge, then the book cannot begin from a specialized object such as a particle, a field, a graph, an automaton, or a logical formula. Any such choice would bind the whole program in advance to one narrow class of examples. The starting point must therefore be maximally general while still being mathematically sharp enough not to collapse into empty philosophical declaration.

The approach adopted in this book is to treat as fundamental neither a "value" nor a "material carrier", but a symbolic structure together with its semantics and its admissible transformations. This choice has several immediate advantages.

First, it applies equally well to strings, formulas, programs, graph encodings, history descriptions, assembly words, proof objects, and observer-level symbolic records. Second, it allows us from the outset to speak about several layers of equivalence and thereby control exactly which information is preserved and which is lost. Third, this starting point is the most natural one for Lean: first the type of objects is fixed, then semantics, then the family of admissible morphisms, and only after that the subtler constructions.

2.2 Symbolic Structures

Definition 2.1 (Symbolic structure)**.** A symbolic structure is a triple

$$S = (X, [\![\cdot]\!], R),$$

where:

- X is a class or set of admissible symbolic objects;
- $[\![\cdot]\!] : X \to Y$ is a semantic interpretation map into a space of values Y;
- $R \subseteq X \times X$ is a relation of admissible transformations.

This definition is intentionally broad. It does not assume that X consists only of strings, that semantics must be a numerical evaluation, or that R is a reduction relation in the narrow rewriting-theoretic sense. All of that may appear later as specialization. At the first step, the important fact is only that we have objects, their interpretations, and admissible transitions between them.

Remark 2.2. Later the space Y may take very different forms. In some examples it will be an ordinary value space, in others a class of normal forms, and in yet others a more complicated observer-level quotient. What matters is not the exact nature of Y, but the fact that interpretation need not preserve the full structural information carried by the object.

2.2.1 A First Example

Example 2.3 (Bracketed words)**.** Let X consist of all well-formed binary bracketings over the alphabet $\{a, b, c\}$. Let $[\![\cdot]\!]$ forget all parentheses and return only the linear word, and let R consist of admissible reassociations of the form

$$((uv)w) \leftrightarrow (u(vw)).$$

Then $(X, [\![\cdot]\!], R)$ is a symbolic structure. The objects may differ at the structural level while having the same semantic image.

This example matters because it immediately illustrates the main methodological thesis of the chapter: semantic equivalence is, in general, too coarse. It recognizes agreement of value but not differences in compositional geometry.

2.2.2 Semantic Equivalence

Definition 2.4 (Semantic equivalence)**.** For a symbolic structure $S = (X, [\![\cdot]\!], R)$ define the relation

$$x \sim_{\text{sem}} y \quad \Longleftrightarrow \quad [\![x]\!] = [\![y]\!].$$

Proposition 2.5. *The relation $\sim_{\text{sem}}$ is an equivalence relation on X.*

Proof. Reflexivity, symmetry, and transitivity follow immediately from the corresponding properties of equality in the value space Y. □

At first sight this looks trivial. Its role is nevertheless fundamental. The semantic quotient $X/\sim_{\text{sem}}$ corresponds exactly to an operation that is too often performed implicitly in mathematics and physics: different realizations of an object are immediately identified as soon as they "mean the same thing". In this book such identification will be treated as the outermost and coarsest level of factorization.

Example 2.6 (Why the semantic quotient is coarse)**.** Suppose symbolic objects encode different computation histories of the same function. Their outputs may coincide, yet they may differ in computation depth, number of intermediate steps, reduction cost, internal symmetry, and stability under local changes. All these distinctions disappear after passing to $\sim_{\text{sem}}$.

2.3 Admissible Transformations and Their Compatibility with Semantics

Once semantic equivalence has been introduced, two questions must be separated.

The first is: which transformations are admissible at all for objects in X? The second is: how do those transformations relate to semantic interpretation? These questions must not be conflated. In some models only meaning-preserving steps are admissible; in others a wider class of transitions may be allowed, while only a special subclass is semantically sound for a given analysis.

Definition 2.7 (Semantically sound transformation)**.** Let xRy. The step $x \to y$ is semantically sound if

$$[\![x]\!] = [\![y]\!].$$

Definition 2.8 (Semantically sound symbolic structure)**.** A symbolic structure is called semantically sound if every admissible transformation preserves semantics, that is,

$$xRy \Longrightarrow [\![x]\!] = [\![y]\!].$$

Remark 2.9. Later we will consider more complex situations in which admissible transitions may change not the immediate value but only some observer-level coarse variable. At the level of the structural core, however, it is natural to begin with the semantically sound case.

2.4 Morphisms of Symbolic Structures

If we want to work not with one structure but with a whole class of them, we must introduce from the outset a language of maps between them.

Definition 2.10 (Morphism of symbolic structures)**.** Let

$$S_1 = (X_1, [\![\cdot]\!]_1, R_1), \qquad S_2 = (X_2, [\![\cdot]\!]_2, R_2).$$

A morphism $f : S_1 \to S_2$ is a map $f : X_1 \to X_2$ satisfying two conditions:

1. if xR_1y, then $f(x)R_2f(y)$;
2. there exists a map of semantic domains

   $$\bar{f} : Y_1 \to Y_2$$

   such that

   $$[\![f(x)]\!]_2 = \bar{f}([\![x]\!]_1)$$

   for all $x \in X_1$.

Remark 2.11. The second condition is what prevents a morphism from being merely a syntactic recoding. A valid morphism must transport both admissible structural transitions and the semantic interpretation attached to them.

2.5 Sequential Objects as the Next Structural Layer

One symbolic structure is not enough if the book is meant to reach assembly, kernels, and physical histories. For that we need a compositional layer on which objects can be combined into longer constructions, compared through normal forms, and analyzed piecewise.

Definition 2.12 (Sequential object)**.** A sequential object is a system of data

$$(W, D, [\![\cdot]\!], R, \otimes, I),$$

where:

- W is the space of composite symbolic states;
- D is the space of admissible data or decorations accompanying the states;
- $[\![\cdot]\!]$ is semantic interpretation;
- R is the relation of admissible local transformations;
- $\otimes$ is composition of states;
- I is the neutral element of composition.

2.6 The Minimal Axiomatic Package for Sequential Objects

1. Composition must be closed on W.
2. The neutral element I must be compatible with composition.
3. Admissible transformations must be local, meaning that they affect only a controlled part of a composite object.
4. Semantics must be compatible with composition and with local transformations.
5. There must exist at least a partial normalization procedure allowing one to compare different realizations of the same structural class.

2.7 A Second Example: Computation as a Sequential Object

Example 2.13 (Computation histories)**.** Let W consist of finite computation configurations, let R be a family of local reduction rules, and let $\otimes$ denote composition of independent computational subtasks. Then the same result may be obtained by several different reduction paths. At the semantic level this may yield one and the same answer. At the structural level the paths differ. At the sequential level they also differ in the way their local fragments are assembled and interact.

2.8 The Tower of Equivalences

Definition 2.14 (Normalization equivalence)**.** If a normalization map c is given, define the relation

$$x \sim_c y \quad \Longleftrightarrow \quad c(x) = c(y).$$

Definition 2.15 (Reduction equivalence)**.** Define the relation

$$x \sim_R y,$$

to mean that objects x and y are connected by an admissible chain of transformations or reduce to a common reachable representative in the chosen version of reduction theory.

2.9 A Guiding Proposition That Sets the Direction of the Chapter

Proposition 2.16 (Guiding proposition on the coarseness of the semantic quotient)**.** *In a sufficiently rich sequential system, semantic equivalence is too coarse to recover the compositional geometry of an object, its reduction history, and the data of its canonical assembly.*

2.10 Lean Orientation for the Next Chapter

1. Introduce the structure of a symbolic system with fields for object space, semantics, and the transformation relation.

2. Then introduce the equivalence relations induced by semantics, normalization, and reduction.

3. After that introduce the extended structure of a sequential object with composition and neutral element.

4. Finally formulate the basic lemmas about the correctness of semantic equivalence and the compatibility of composition with the chosen comparison relations.

2.11 Interim Conclusion

This chapter made two fundamental moves. First, symbolic structures were introduced as the initial class of objects, reducible neither to pure syntax nor to already completed semantics. Second, sequential objects were introduced, preparing the language of composition, locality, and normalization.

This yields the minimal framework needed to move to the next chapter, where h-patterns, laminarity, elementary functorial units, and canonical EFU decomposition will be introduced. That is where the intermediate structural layer first appears in an explicit form and makes the rest of the program mathematically substantive.

Chapter 3

h-Patterns, EFU, and Object Assembly

3.1 Why an Intermediate Layer Is Needed Between Reduction and Semantics

By the end of the previous chapter it is already clear that the semantic quotient alone is not enough, and that reduction reachability does not by itself provide a final picture either. Reduction tells us what can be obtained from what. Semantics tells us what has been obtained in the sense of value. But between those two levels one central question remains open: how is the geometry of admissible transformations itself organized, and what is the compositional architecture of an object in the part that does not disappear under normalization yet is not reducible to mere reachability?

This is exactly where a special intermediate layer becomes necessary. In this book that layer is described in the language of h-patterns. An h-pattern is meant to retain not the concrete notation of an object and not only its final semantics, but the stable organization of its structural subblocks and the way they are nested or kept independent.

3.2 The Intuition Behind the h-Pattern

Example 3.1 (Same semantics, different pattern geometry)**.** Suppose two objects have the same semantic result and even reduce to the same normal form, but in the first case intermediate assembly passes through two independent large blocks, whereas in the second case it passes through one block nested inside another. Semantically there is no difference. Reduction may connect the two realizations. Yet the h-pattern must distinguish them, because their assembly geometry is different.

3.3 Definition of the h-Pattern

Definition 3.2 (Abstract h-pattern extractor)**.** Let $(W, D, [\![\cdot]\!], R, \otimes, I)$ be a sequential object. An abstract h-pattern extractor is a map

$$H : W \to \mathbf{HPat}$$

into a class **HPat** of structural pattern objects such that for every $x \in W$ the object $H(x)$ satisfies the following requirements:

1. $H(x)$ is invariant under admissible reductions;
2. $H(x)$ is compatible with composition;
3. $H(x)$ ignores purely trivial structural details;
4. $H(x)$ preserves information about nesting, independence, and the hierarchy of essential substructures.

Remark 3.3. The point of the extractor is that H is part of the data of the theory, not an informal summary chosen after the fact. Different choices of extractor may lead to different notions of essential structure, so later EFU statements must always be read relative to the chosen H.

3.4 h-Equivalence

Definition 3.4 (h-equivalence)**.** For objects $x, y \in W$ set

$$x \sim_h y \quad \Longleftrightarrow \quad H(x) \cong H(y),$$

where $\cong$ denotes isomorphism in the category of h-patterns.

Proposition 3.5. *If isomorphism in the category of h-patterns is an equivalence relation, then $\sim_h$ is also an equivalence relation on W.*

Proof. This follows from reflexivity, symmetry, and transitivity of isomorphism. □

3.5 Laminarity as a Condition of Canonicity

Definition 3.6 (Laminar family of blocks)**.** A family of essential subblocks of an object is called laminar if for any two blocks A and B one of the following holds:

1. $A \cap B = \varnothing$;

2. $A \subseteq B$;

3. $B \subseteq A$.

Example 3.7 (Why partial overlaps are dangerous)**.** Suppose an object contains two essential subblocks with a nontrivial overlap, while neither is contained in the other. Then decomposition loses canonicity. Laminarity excludes exactly this situation.

3.6 Elementary Functorial Units

Definition 3.8 (Elementary functorial unit)**.** Let $x \in W$. One says that x is an elementary functorial unit if $H(x)$ is nontrivial and does not admit a decomposition

$$H(x) \cong A \otimes B$$

in which both A and B are nontrivial h-patterns.

Definition 3.9 (The class of EFU sectors)**.** Let $\mathrm{EFU}(W)$ denote the class of all objects of the sequential system that are elementary functorial units with respect to the chosen extractor H.

3.7 Decomposition into EFUs

Definition 3.10 (EFU decomposition)**.** An EFU decomposition of an object $x \in W$ is a representation of the form

$$x \sim_h r \otimes e_1 \otimes \cdots \otimes e_n,$$

where r is an h-trivial part, each e_i belongs to $\mathrm{EFU}(W)$, and the assembly is compatible with the pattern $H(x)$.

Definition 3.11 (h-trivial part)**.** An object $r \in W$ is called h-trivial if its pattern is isomorphic to the distinguished trivial pattern unit in **HPat**.

3.8 Why Canonicity Cannot Mean Literal Uniqueness

Definition 3.12 (Canonicity up to local moves)**.** An EFU decomposition is said to be canonical up to local moves if any two EFU decompositions of the same object differ only by a finite sequence of the following operations:

1. permutation of independent factors;

2. reassociation of brackets in the associative composition;

3. insertion or removal of h-trivial background;

4. a local equivalent rebuilding of a nested shell that does not change the h-pattern.

3.9 An Example of EFU Decomposition

Example 3.13 (A toy decomposition)**.** Suppose an object x consists of three essential blocks: two are independent and lie in parallel, while the third is nested inside one of the first two. Then the natural EFU decomposition isolates two outer factors, one of which contains a nested essential subblock. If that nested block is irreducible, it becomes a separate EFU element.

3.10 Target Theorems on EFU Decomposition

Theorem 3.14 (Target theorem on the existence of EFU decomposition)**.** *Suppose a sequential system admits an extractor H satisfying invariance under reduction, compatibility with composition, and the laminarity condition for essential blocks. Then every object $x \in W$ admits an EFU decomposition up to h-equivalence.*

Theorem 3.15 (Target theorem on the canonicity of EFU decomposition)**.** *Under the same assumptions, any two EFU decompositions of the same object are equivalent up to a finite sequence of local moves.*

3.11 Why EFU Is Already More Than a Technique of Structural Analysis

Once EFU has been introduced, the picture of the whole book changes substantially. A natural notion of assembly word appears, the path toward kernel/resolvent language opens, and the future alphabet of composite sectors in the physical and later symbolic parts of the theory becomes visible.

3.12 Lean Orientation for the Next Chapter

1. Introduce the type of h-patterns and the relation of isomorphism between them.

2. Then define the extractor $H : W \to \mathbf{HPat}$ as a separate structure.

3. After that formalize laminarity.

4. Next introduce the predicate “isEFU” and define the class of elementary functorial units.

5. Finally formulate the theorems on the existence and canonicity of EFU decomposition.

3.13 Conclusion of the Chapter

This chapter constructed the intermediate layer missing between reduction and semantics. It introduced h-patterns, h-equivalence, laminarity, elementary functorial units, and the notion of EFU decomposition. As a result, structural theory acquired for the first time an internal alphabet of stable blocks and a controlled language for the assembly of complex objects.

Chapter 4

Transfer Kernels, Marked Kernels, Moments, and Spectral Control

4.1 Why a Kernel-Level Language Is Needed After EFU

By the end of the previous chapter, structural theory can already decompose complex objects into stable assembly blocks. However, the language of EFU decomposition alone is not enough for the later physical program. The reason is that the following chapters must speak not only about the existence of assembly, but also about quantitative control of assembly classes, about admissible transitions between them, about observables that mark atomic transitions, and about the singularities of the corresponding generating series.

At precisely this point a transfer-kernel description becomes necessary. It does three things at once. First, it replaces the unstructured sum over all assemblies by a resolvent expression. Second, it allows the combinatorial layer to be separated from the spectral layer. Third, it prepares the language in which amplitudes, masses, resonances, flavour channels, and the rules for extracting observables by marking will later be defined.

Methodologically, this chapter fixes one of the most important principles of the whole book: numerical quantities are not primary objects. The primary objects are structural nuclei, assembly words, and transfer kernels. Numerical series, coefficients, and observables arise only as decategorified shadows of a richer kernel-level formalism.

4.2 Pattern-Valued and Assembly-Valued Series

Let a sequential system $\mathbb{S}$ be given with class of objects $W_{\mathbb{S}}$, class of EFU elements $\mathcal{E}_{\mathbb{S}}$, extractor $\mathcal{H}$, grading map

$$\gamma : W_{\mathbb{S}} \to \mathbb{N}^d,$$

and weight function

$$\mu : \mathrm{Im}(\mathcal{H}) \to R,$$

where R is a fixed commutative semiring of coefficients.

Definition 4.1 (Pattern-valued generating series)**.** The pattern-valued generating series of the system $\mathbb{S}$ is the formal series

$$\mathfrak{Z}^{\mu}_{\mathbb{S}}(\mathbf{t}) := \sum_{x \in W_{\mathbb{S}}} \mu\big([\mathcal{H}(x)]\big)\, \mathbf{t}^{\gamma(x)}\, [\mathcal{H}(x)],$$

where $[\mathcal{H}(x)]$ denotes the h-class of the object and $\mathbf{t}^{\gamma(x)}$ is the multigraded monomial attached to its grading.

Definition 4.2 (Assembly-valued generating series)**.** The assembly-valued generating series of the same system is the formal series

$$\mathfrak{A}^{\mu}_{\mathbb{S}}(\mathbf{t}) := \sum_{x \in W_{\mathbb{S}}} \mu\big([\mathcal{H}(x)]\big)\, \mathbf{t}^{\gamma(x)}\, \mathrm{Asm}(x),$$

where $\mathrm{Asm}(x)$ denotes the assembly word associated with a chosen canonical EFU decomposition representative of the object x.

These two series must not be conflated. The first forgets the exact assembly path and retains only the pattern class. The second preserves the sequence of assembly atoms and is therefore substantially richer. Later it is precisely the assembly-valued series that will factor through the transfer kernel, while the pattern-valued series will be obtained from it by collapse.

Proposition 4.3 (Assembly collapse)**.** *There exists a natural collapse morphism*

$$\mathrm{Col}_{\mathbb{S}} : \mathrm{AsmSeries}(\mathbb{S}) \to \mathrm{PatSeries}(\mathbb{S})$$

such that

$$\mathrm{Col}_{\mathbb{S}}\big(\mathfrak{A}^{\mu}_{\mathbb{S}}(\mathbf{t})\big) = \mathfrak{Z}^{\mu}_{\mathbb{S}}(\mathbf{t}).$$

Proof. By definition, $\mathrm{Col}_{\mathbb{S}}$ acts on each assembly word by replacing it with the h-class of the assembled object. After linear extension to formal sums, the contribution of each object x to the assembly-valued series becomes exactly the contribution of the same object to the pattern-valued series. □

4.3 Free EFU Factorization and the Geometric Series

The simplest case arises when every ordered word in the EFU alphabet is admissible as a valid assembly. This case is too poor for physics, but it is important as a control model because it immediately exhibits the appearance of resolvent structure.

Definition 4.4 (Freely EFU-factorizable system)**.** A system $\mathbb{S}$ is called freely EFU-factorizable if every object has a canonical ordered EFU decomposition, every finite word in the alphabet $\mathcal{E}_{\mathbb{S}}$ is admissible, and the weight and grading are multiplicatively-additively compatible with concatenation of EFU factors.

Definition 4.5 (Atomic EFU series)**.** The atomic EFU series is the formal series

$$\mathfrak{E}^{\mu}_{\mathbb{S}}(\mathbf{t}) := \sum_{e \in \mathcal{E}_{\mathbb{S}}} \mu\big([\mathcal{H}(e)]\big)\, \mathbf{t}^{\gamma(e)}\, [e].$$

Theorem 4.6 (Free factorization formula)**.** *If the system $\mathbb{S}$ is freely EFU-factorizable, then*

$$\mathfrak{A}^{\mu}_{\mathbb{S}}(\mathbf{t}) = \mathbf{1} + \mathfrak{E}^{\mu}_{\mathbb{S}}(\mathbf{t}) + \big(\mathfrak{E}^{\mu}_{\mathbb{S}}(\mathbf{t})\big)^2 + \cdots = \big(\mathbf{1} - \mathfrak{E}^{\mu}_{\mathbb{S}}(\mathbf{t})\big)^{-1}.$$

Proof. Each object corresponds to a unique word in the free monoid on $\mathcal{E}_{\mathbb{S}}$. Since the grading adds and the weight multiplies across atoms, summation over all objects coincides with summation over all words of all lengths. This yields the noncommutative geometric series. □

Example 4.7 (A free two-atom model)**.** Let $\mathcal{E}_{\mathbb{S}} = \{a, b\}$, assume that all words are admissible, that $\gamma(a) = \gamma(b) = 1$, and that the weights are $\mu([\mathcal{H}(a)]) = \alpha$ and $\mu([\mathcal{H}(b)]) = \beta$. Then

$$\mathfrak{E}^{\mu}_{\mathbb{S}}(t) = t\,(\alpha[a] + \beta[b])$$

and therefore

$$\mathfrak{A}^{\mu}_{\mathbb{S}}(t) = \big(1 - t(\alpha[a] + \beta[b])\big)^{-1}.$$

This example matters not in itself but as the first case in which the assembly-level sum takes resolvent form.

4.4 Finite-State Factorization and the Transfer Kernel

Real systems are almost never free. The admissibility of the next atom depends on the boundary data of the current partial assembly, on the type of channel, on orientation, or on other local compatibility conditions. Therefore a finite graph of states is needed instead of a free monoid.

Definition 4.8 (Finite-state EFU factorization)**.** A system $\mathbb{S}$ is called finite-state EFU-factorizable if there exists a finite set of states

$$B_{\mathbb{S}} = \{1, \ldots, s\}$$

and maps

$$\mathrm{in}, \mathrm{out} : \mathcal{E}_{\mathbb{S}} \to B_{\mathbb{S}}$$

such that a word $e_1 \cdots e_n$ is admissible if and only if it forms a path

$$b_0 \xrightarrow{e_1} b_1 \xrightarrow{e_2} \cdots \xrightarrow{e_n} b_n$$

with the conditions $\mathrm{out}(e_i) = \mathrm{in}(e_{i+1})$, where b_0 belongs to the set of initial states and b_n belongs to the set of terminal states.

Definition 4.9 (Transfer kernel)**.** The transfer kernel of the system $\mathbb{S}$ is defined as the matrix

$$K^{\mu}_{\mathbb{S}}(\mathbf{t}) = \big(K^{\mu}_{ab}(\mathbf{t})\big)_{a,b\in B_{\mathbb{S}}},$$

where

$$K^{\mu}_{ab}(\mathbf{t}) = \sum_{\substack{e\in\mathcal{E}_{\mathbb{S}} \\ \mathrm{in}(e)=a,\ \mathrm{out}(e)=b}} \mu\big([\mathcal{H}(e)]\big)\, \mathbf{t}^{\gamma(e)}\, [e].$$

If $I_{\mathbb{S}} \subseteq B_{\mathbb{S}}$ and $T_{\mathbb{S}} \subseteq B_{\mathbb{S}}$ are the distinguished sets of initial and terminal states, then ι and ω denote their indicator vectors.

Theorem 4.10 (Resolvent representation)**.** *If the system $\mathbb{S}$ is finite-state EFU-factorizable, then its assembly-valued generating series has the form*

$$\mathfrak{A}^{\mu}_{\mathbb{S}}(\mathbf{t}) = \iota^{\top}\big(I - K^{\mu}_{\mathbb{S}}(\mathbf{t})\big)^{-1}\omega.$$

Proof. The power $(K^{\mu}_{\mathbb{S}})^n$ sums all admissible EFU words of length n between the corresponding boundary states. Multiplication on the left by $\iota^{\top}$ and on the right by ω selects only those paths that begin in an allowed initial state and end in an allowed terminal state. Summation over all $n \geq 0$ gives the geometric series in the transfer-kernel matrix. □

Corollary 4.11 (Decategorified resolvent)**.** *After any scalar decategorification, the corresponding numerical generating series also has a matrix-resolvent representation.*

Proof. Decategorification is a semiring-type morphism and therefore transports the matrix formula without changing its algebraic form. □

Example 4.12 (A two-state system)**.** Suppose there are two states 0 and 1 and three atoms: $a : 0 \to 0$, $b : 0 \to 1$, and $c : 1 \to 1$. Then

$$K(t) = \begin{pmatrix} t\alpha[a] & t\beta[b] \\ 0 & t\gamma[c] \end{pmatrix},$$

if all gradings are equal to one. With initial vector $\iota = (1,0)^{\top}$ and terminal vector $\omega = (1,1)^{\top}$ we obtain

$$\mathfrak{A}(t) = \iota^{\top}(I - K(t))^{-1}\omega.$$

In this example one already sees that word admissibility is controlled not by the dictionary as a whole but by transitions along a finite graph of states.

4.5 Strict Transfer and Simulation Invariance

After the kernel-level description has been built, one must separately fix what counts as a correct transfer from one system to another. This is critical for the rest of the book: without such a step, it will later be impossible to say honestly that different realizations have the same observable content.

Definition 4.13 (EFU-compatible transfer morphism)**.** Let $\mathbb{S}$ and $\mathbb{T}$ be finite-state EFU-factorizable systems. An EFU-compatible transfer morphism

$$F : \mathbb{S} \to \mathbb{T}$$

consists of:

1. a morphism of coefficients and grading data;
2. a map F_E from atoms of the system $\mathbb{S}$ to admissible EFU words of the system $\mathbb{T}$;
3. a state map $\phi : B_{\mathbb{S}} \to B_{\mathbb{T}}$,

such that each atom $e : a \to b$ is sent to an admissible word from $\phi(a)$ to $\phi(b)$.

Theorem 4.14 (Strict transfer invariance)**.** *Assume that the EFU-compatible transfer morphism satisfies the conditions*

$$F_{\#}(K_{\mathbb{S}}) = K_{\mathbb{T}}, \qquad F_{\#}(\iota_{\mathbb{S}}) = \iota_{\mathbb{T}}, \qquad F_{\#}(\omega_{\mathbb{S}}) = \omega_{\mathbb{T}}.$$

Then

$$\widehat{F}\big(\mathfrak{A}_{\mathbb{S}}\big) = \mathfrak{A}_{\mathbb{T}}.$$

After assembly collapse, the equality of pattern-valued series follows as well.

Proof. Since $\widehat{F}$ is a morphism at the level of formal series, it commutes with addition and multiplication. Therefore

$$\widehat{F}\big((I - K_{\mathbb{S}})^{-1}\big) = \widehat{F}\Big(\sum_{n\geq 0} K_{\mathbb{S}}^{n}\Big) = \sum_{n\geq 0} \widehat{F}(K_{\mathbb{S}})^{n} = (I - K_{\mathbb{T}})^{-1}.$$

Multiplication by the transported boundary vectors gives the claim. □

Theorem 4.15 (Transfer by simulation)**.** *Suppose there exists a matrix P for which the equalities*

$$PK_{\mathbb{S}} = K_{\mathbb{T}}P, \qquad P\omega_{\mathbb{S}} = \omega_{\mathbb{T}}, \qquad \iota_{\mathbb{S}}^{\top} = \iota_{\mathbb{T}}^{\top}P.$$

hold. Then the corresponding decategorified generating series coincide.

Proof. The intertwining relation immediately implies

$$P(I - K_{\mathbb{S}})^{-1} = (I - K_{\mathbb{T}})^{-1} P.$$

It remains only to multiply on the left and right by the boundary vectors and use the compatibility of the initial and terminal data. □

4.6 Marked Kernels and Moments of Observables

The next layer is essential for the future physics. Observables must not be attached post factum to an already summed series. The correct level for their introduction is the level of atoms or residue data, that is, precisely the level at which local transitions can be marked.

Definition 4.16 (Atom-level observable)**.** Let $\xi : \mathcal{E}_{\mathbb{S}} \to A$ be an observable at the atomic level. Then the marked transfer kernel is defined by

$$K_{\mathbb{S}}^{\xi}(\mathbf{t}, s) = \big(K_{ab}^{\xi}(\mathbf{t}, s)\big)_{a,b \in B_{\mathbb{S}}},$$

where

$$K_{ab}^{\xi}(\mathbf{t}, s) = \sum_{\substack{e \in \mathcal{E}_{\mathbb{S}} \\ \mathrm{in}(e)=a,\ \mathrm{out}(e)=b}} \mu\big([\mathcal{H}(e)]\big)\, \mathbf{t}^{\gamma(e)}\, s^{\xi(e)}\, [e].$$

Definition 4.17 (Marked assembly series and moments)**.** The marked assembly series is defined by

$$\mathfrak{A}_{\mathbb{S}}^{\mu,\xi}(\mathbf{t}, s) = \iota^{\top} \big(I - K_{\mathbb{S}}^{\xi}(\mathbf{t}, s)\big)^{-1} \omega.$$

After scalar decategorification, the numerical marked series is denoted by

$$Z_{\mathbb{S}}^{\chi,\mu,\xi}(\mathbf{t}, s).$$

Its k-th marked moment is defined by

$$\mathcal{M}_{\mathbb{S}}^{(k)}(\mathbf{t}) := \left(s\frac{\partial}{\partial s}\right)^{k} Z_{\mathbb{S}}^{\chi,\mu,\xi}(\mathbf{t}, s)\bigg|_{s=1}.$$

Proposition 4.18 (Meaning of marked differentiation)**.** *Marked moments extract from the transfer formalism not merely the total weight of objects but the statistics of the distribution of the chosen atomic observable across all admissible assembly paths.*

Proof. Differentiation with respect to the parameter s isolates the powers $s^{\xi(e)}$ and therefore aggregates the contributions of words while taking into account the total value of the observable across atoms. Repeated application of the operator $s\partial_s$ extracts the ordinary power moments of that statistic. □

Theorem 4.19 (Moment stability)**.** *Let $F : \mathbb{S} \to \mathbb{T}$ be a strict transfer morphism compatible with the chosen atom-level observable ξ, with decategorification, and with the residue data. Then the marked assembly series, the pattern-valued marked series, and all their numerical moments are preserved under the transfer F.*

Proof. Compatibility of the observable means that the marked kernels are sent to one another under the action of F. Consequently, the proof of the theorem on strict transfer invariance carries over without change to marked kernels. After that, differentiation with respect to s is applied to already identical series, so all moments coincide as well. □

Example 4.20 (Counting atoms of a given type)**.** Let $\xi(e) = 1$ if the atom e belongs to a distinguished class, and $\xi(e) = 0$ otherwise. Then the first marked moment measures the average total weight of occurrences of that class of atoms across all admissible assemblies of a given grading degree. In later chapters such a mechanism will be used to extract charge-like, flavour-like, and channel-like statistics.

4.7 Singular Profiles of Coefficient Series

After the marked formalism, the next question is this: what can be said about the singularities of the coefficient series arising from the transfer formalism? It is exactly here that structural theory begins to pass from pure combinatorics to controlled analytic behavior.

Definition 4.21 (Ordinary and Dirichlet series)**.** Let $M_{\mathbb{S}}(N)$ be a coefficient sequence extracted from a marked or residue-weighted series. Then one defines the series

$$\mathfrak{M}_{\mathbb{S}}(z) = \sum_{N \geq 0} M_{\mathbb{S}}(N) z^N, \qquad \mathfrak{D}_{\mathbb{S}}(s) = \sum_{N \geq 1} \frac{M_{\mathbb{S}}(N)}{N^s}.$$

Definition 4.22 (Singular profile)**.** If in a neighborhood of a point z_0 the ordinary generating series has a local expansion

$$\mathfrak{M}(z) = \sum_{j=1}^{J} c_j (1 - z/z_0)^{-\alpha_j} + H(z),$$

where H is analytic and $\alpha_1 > \cdots > \alpha_J$, then the singular profile at the point z_0 is defined as the set

$$\mathrm{SP}_{z_0}(\mathfrak{M}) = \{(\alpha_j, c_j) : 1 \leq j \leq J\}.$$

The singular profile for a Dirichlet series is defined analogously from the principal part in a neighborhood of a point s_0.

Theorem 4.23 (Exact stability of the singular profile)**.** *If two systems have the same coefficient sequence, then their ordinary and Dirichlet series coincide as formal series. Consequently, whenever meromorphic continuation exists, their singular profiles also coincide.*

Proof. Equality of coefficients identically implies equality of formal series. If both sides are then continued meromorphically, their local Laurent or Laurent–Puiseux expansions coincide as well, and therefore so do their singular profiles. □

Theorem 4.24 (Stability of the leading singularity under lower-order perturbation)**.** *Suppose the difference of two coefficient sequences has lower singular order than the dominant term of the first system. Then the leading singular profile is preserved.*

Proof. The difference of the series produces only a less singular correction. Therefore the dominant exponent and its coefficient cannot change. □

4.8 Spectral Control of Transfer Kernels

After scalar decategorification, the transfer kernel becomes a matrix-valued analytic function

$$\mathbf{K}_{\mathbb{S}}(z) = \chi\big(K_{\mathbb{S}}(z)\big),$$

and the corresponding generating function takes the form

$$G_{\mathbb{S}}(z) = \iota^{\top}(I - \mathbf{K}_{\mathbb{S}}(z))^{-1}\omega.$$

At this level structural theory meets the classical language of resolvents, spectral radius, and the analytic theory of transfer operators.

Definition 4.25 (Singular spectrum and spectral profile)**.** The singular spectrum of the kernel $\mathbf{K}$ is defined as the set

$$\Sigma(\mathbf{K}) = \{z : \det(I - \mathbf{K}(z)) = 0\}.$$

The spectral profile at a point $z_0 \in \Sigma(\mathbf{K})$ consists of the local eigenvalue branches $\lambda_j(z)$ satisfying $\lambda_j(z_0) = 1$, together with their algebraic multiplicities.

Theorem 4.26 (Singularities come from the kernel spectrum)**.** *If $\mathbf{K}(z)$ is analytic in a domain U, then*

$$G(z) = \iota^{\top}(I - \mathbf{K}(z))^{-1}\omega$$

is meromorphic on $U \setminus \Sigma(\mathbf{K})$, and all its singularities inside U lie in $\Sigma(\mathbf{K})$.

Proof. On the set where the matrix $I - \mathbf{K}(z)$ is invertible, Cramer's rule expresses the resolvent through the adjugate matrix and the determinant. Therefore each component of the resolvent is meromorphic and can have singularities only at zeros of the determinant. □

Theorem 4.27 (Dominant eigenbranch control)**.** *Assume that* $\mathbf{K}(z)$ *is analytic on the real interval* $0 \le z < R_0$*, has nonnegative coefficients, is irreducible for* $z \in (0, R_0)$*, and that there exists* $r \in (0, R_0)$ *such that*

$$\rho\big(\mathbf{K}(r)\big) = 1, \qquad \rho\big(\mathbf{K}(z)\big) < 1 \text{ for } 0 \le z < r,$$

where ρ *denotes the spectral radius. If the dominant eigenvalue branch* $\lambda_*(z)$ *is simple at the point* r *and* $\lambda_*'(r) \neq 0$*, then* r *is the dominant singularity of the function* $G(z)$ *and*

$$G(z) = \frac{(\iota^\top v_r)(u_r^\top \omega)}{1 - \lambda_*(z)} + H(z),$$

where u_r *and* v_r *are normalized left and right eigenvectors and* H *is analytic in a neighborhood of* r*.*

Proof. By the Perron–Frobenius theorem, the dominant eigenvalue of a nonnegative irreducible kernel is simple and positive. For $z < r$ the resolvent exists because the spectral radius is strictly smaller than one. At the point r the first singularity appears exactly when the dominant branch crosses the value 1. Decomposing the space into the dominant eigendirection and an invariant complement gives the stated formula, because on the complement the spectrum remains strictly inside the unit circle. □

Corollary 4.28 (Leading singular profile from eigenbranch data)**.** *Under the assumptions of the previous theorem, the leading singular profile of the function* G *at the point* r *is determined completely by the dominant eigenbranch and the boundary vectors. In the case of a simple pole it is*

$$\mathrm{LSP}_r(G) = \left(1, \frac{(\iota^\top v_r)(u_r^\top \omega)}{-\lambda_*'(r)}\right).$$

Proof. It is enough to expand $1 - \lambda_*(z)$ linearly in a neighborhood of the point r. □

Theorem 4.29 (Spectral invariance under strict transfer)**.** *If after scalar decategorification two transfer-equivalent systems are related by a similarity matrix compatible with the boundary vectors, then their singular spectra, spectral profiles, and corresponding generating functions coincide.*

Proof. Similarity preserves the determinant and local spectral data, and hence the singular spectrum. The resolvent formula under similarity yields equality of the generating functions after multiplication by the compatible boundary vectors. □

4.9 Why This Chapter Is a Methodological Turning Point for the Whole Book

After this chapter, the theory acquires for the first time a full bridge between the three levels of description.

First, the assembly-level world remains rich enough to remember the order and compatibility of local blocks. Second, the decategorified world begins to produce numerical series, moments, and coefficients. Third, the spectral level shows exactly where singularities come from and why they are not external decorations of the series but internal consequences of kernel geometry.

This is exactly why all later physical quantities must be defined not directly on numbers but through the corresponding marked kernels, residue weights, or eigenbranch data. Mass, amplitude, resonance lifetime, effective charge, and statistical observables will be interpreted as derived constructions of the transfer formalism rather than as independent primitives.

4.10 Lean Orientation for the Next Chapter

For later formalization, the chapter should be split into at least three independent files.

1. The algebraic layer: assembly-valued series, free and finite-state factorization, resolvent representation.

2. The marked layer: marked kernels, formal differentiation, moments, transfer invariance of observables.

3. The analytic layer: singular profiles, dominant eigenbranch, spectral control, invariance under similarity.

It is especially important not to mix within a single formal entity the algebraic resolvent identity and analytic statements about meromorphic continuation. For Lean these must be separate dependency chains.

4.11 Conclusion of the Chapter

This chapter built the kernel-level apparatus of structural theory. It introduced pattern-valued and assembly-valued series, the transfer kernel, the resolvent representation, marked kernels, moments, singular profiles, and spectral control through the dominant eigenbranch. In this way the structural core of the book reached for the first time a form in which one can pass to the observer, to structural dynamics, and then to the quantum-information and microphysical layers without loss of mathematical discipline.

Chapter 5

Causal Rewriting and the Observer

5.1 Why the Physical Layer Begins with Causality Rather Than Space

After the structural core has been constructed, it is natural to expect that the next chapter should immediately introduce geometry, metric structure, and effective fields. Such a move would, however, be premature. The kernel-level apparatus of the previous chapters has already shown that numerical and even spectral observables are not primary. Still less can space, time, and velocity be primary, since all of them require an additional regime for reading structural history.

For that reason, the physical layer in this book begins with a weaker but more fundamental object: the causal organization of rewrite events. At this level there is still no coordinate grid, no global time parameter, and no universal propagation speed. There are only states, local admissible rewritings, and the relation of structural dependence between them.

This order of introducing concepts is important for future formalization as well. If one first postulates spacetime and only afterward tries to derive the observer, then the observer becomes an internal object on an already prepared stage. In our theory the direction must be the opposite: the observer is a mechanism of projection that produces space and time as its images.

5.2 Rewrite Events and Causal Order

Definition 5.1 (Rewrite event)**.** A rewrite event of the system $\mathbb{S}$ is an elementary admissible transition

$$e : x \to_R y,$$

where $x, y \in W$ and the transition rule belongs to a fixed class of local rewrite rules.

Definition 5.2 (Causal dependence of rewrite events)**.** For rewrite events e_1 and e_2 we write

$$e_1 \prec e_2,$$

if executing e_2 requires that the structural effect of the event e_1 has already been realized. In other words, e_2 cannot be applied correctly within the same structural history without the prior execution of e_1.

Definition 5.3 (Causal rewriting system)**.** A rewriting system is called causal if the reflexive closure of the relation $\prec$ is a partial order on the set of rewrite events, that is, it is reflexive, antisymmetric, and transitive on the class of actually realizable dependencies.

Proposition 5.4 (Absence of fundamental time)**.** *The partial order $\prec$ by itself determines neither a global time coordinate nor a unique numbering of events. It determines only a relation of dependence and compatibility.*

Proof. If the order is only partial, then incomparable pairs of events exist. Consequently, no unique linear parameter is part of the structure itself. A linear ordering can appear only after an additional slicing rule has been chosen. □

This result is methodologically important. It means that a fundamental history is not a sequence of frames. It is a causal object for which a time line arises only after observer-level linearization.

5.3 History as a Causal Object

Definition 5.5 (Structural history)**.** A structural history is a pair

$$\Gamma = (E_\Gamma, \prec_\Gamma),$$

where E_Γ is the set of rewriting events realized in the given history and $\prec_\Gamma$ is the restriction of the causal order to that set.

Definition 5.6 (Compatible events)**.** Events $e_1, e_2 \in E_\Gamma$ are called compatible if they are incomparable in the order $\prec_\Gamma$ and their realization does not violate each other's local admissibility conditions.

Remark 5.7. Incomparability in the causal order is not sufficient for full compatibility, since two rewrite events may be logically independent and yet compete for the same local region of support. Therefore a history must store not only a partial order but also implicit local compatibility constraints.

Example 5.8 (Minimal fork)**.** Suppose the event e_0 creates a configuration after which two local rules e_1 and e_2 may act independently on disjoint subblocks. Then

$$e_0 \prec e_1, \qquad e_0 \prec e_2, \qquad e_1 \not\prec e_2, \qquad e_2 \not\prec e_1.$$

For one observer such a history may look like two simultaneous processes, while for another it may look like two successive stages of coarse-grained evolution. At the fundamental level both interpretations are merely different readings of the same causal object.

Theorem 5.9 (History is prior to the time line)**.** *For any causal rewrite system, every admissible time line is a linear extension of some structural history, but in general one and the same structural history admits many different time lines.*

Proof. Any admissible time line must respect the causal order and hence is a linear extension of the partial order $\prec_\Gamma$. If the history contains incomparable events, then more than one linear extension exists. Therefore the time line is not fixed uniquely by the fundamental structure. □

5.4 The Observer as a Projection Mechanism

We can now introduce the central notion of the chapter. In this theory the observer is not an object located inside an already prepared space. It is a way of reading structural history and associated states in the form of geometry, time, and physically distinguishable observables.

Definition 5.10 (Observer)**.** An observer is a triple

$$\mathcal{O} = (\Pi_\mathcal{O}, \Theta_\mathcal{O}, \chi_\mathcal{O}),$$

where:

1. $\Pi_\mathcal{O}$ is a rule of spatial projection from states or histories into some class of geometric objects $\mathcal{G}_\mathcal{O}$;
2. $\Theta_\mathcal{O}$ is a temporal slicing rule that turns a causal history into an observer-time organization;
3. $\chi_\mathcal{O}$ is a decategorification/coarse-graining functor that determines which distinctions the observer regards as physically distinguishable.

Definition 5.11 (Observer time)**.** Let $\Gamma = (E_\Gamma, \prec_\Gamma)$ be a structural history. The observer time for the observer $\mathcal{O}$ is given by a map

$$t_\mathcal{O} : E_\Gamma \to T_\mathcal{O},$$

that is monotone with respect to the causal order and is induced by the rule $\Theta_{\mathcal{O}}$.

Definition 5.12 (Observer space)**.** The observer space for the observer $\mathcal{O}$ is given by a map

$$\Pi_{\mathcal{O}} : W \to \mathcal{G}_{\mathcal{O}},$$

that assigns to each structural configuration its geometric image in the chosen coarse-grained model of space.

Remark 5.13. Nothing in these definitions requires all observers to use the same space, the same history-slicing rule, or the same resolution scale. This is exactly why space and time are derivative rather than fundamental entities.

5.5 Why Rejecting a Fundamental Limiting Speed Clarifies the Theory

Ordinary physical intuition pushes one toward postulating a universal maximal speed already at the level of local rewrite rules. Within the present program that would be a logical mistake. Velocity is a ratio of two observer-level quantities: spatial displacement and temporal increment. But neither of those concepts exists before the observer is fixed.

Proposition 5.14 (Velocity is not a fundamental quantity)**.** *At the level of structural histories velocity is undefined. It arises only after the simultaneous choice of* $\Pi_{\mathcal{O}}$ *and* $\Theta_{\mathcal{O}}$*.*

Proof. Without spatial projection it is impossible to speak about displacement, and without temporal slicing it is impossible to speak about duration. Consequently, without an observer the ratio

$$\frac{\Delta\text{space}}{\Delta\text{time}}$$

has no mathematical meaning. □

Definition 5.15 (Observer velocity)**.** Let $\mathfrak{p}$ be a stable object or a stable evolution channel. Then its velocity relative to the observer $\mathcal{O}$ is defined by

$$v_{\mathcal{O}}(\mathfrak{p}) = \frac{\Delta_{\mathcal{G}_{\mathcal{O}}}\big(\Pi_{\mathcal{O}}(\mathfrak{p})\big)}{\Delta t_{\mathcal{O}}},$$

provided both the numerator and the denominator are defined in the chosen coarse-grained regime.

Corollary 5.16 (Absence of a fundamental meaning for the term "superluminal")**.** *If the theory does not fix a fundamental observer-independent speed, then the term "superluminal" can mean only a comparison of two channels inside one and the same observer regime, not a property of reality at the fundamental level.*

Proof. Velocity is defined only after an observer has been chosen. Therefore comparison with a distinguished reference channel is also an observer-dependent statement. □

5.6 Two Observers of the Same History

Example 5.17 (Different slicings of one causal history)**.** Consider the structural history from the minimal fork example. Let the observer $\mathcal{O}_1$ choose a temporal slicing in which the events e_1 and e_2 lie in one layer, while the observer $\mathcal{O}_2$ chooses a coarse-graining in which the effect of e_1 is aggregated first and the effect of e_2 afterward.

Then $\mathcal{O}_1$ sees one stage of splitting into two parallel channels, whereas $\mathcal{O}_2$ sees a two-step sequential process. If, in addition, their spatial projections differ, then one and the same deep process may acquire different observer trajectories and different velocities. This is not a contradiction but a direct consequence of the fact that observer space and observer time are not fundamental.

Example 5.18 (Coarse and fine spatial projection)**.** Suppose $\Pi_{\mathcal{O}_1}$ distinguishes two distant subsystems as separate graph nodes, while $\Pi_{\mathcal{O}_2}$ collapses them into one coarse-grained effective site. Then the same rewrite channel has nonzero spatial displacement for $\mathcal{O}_1$ and zero displacement for $\mathcal{O}_2$. Consequently, even the fact of whether "the object moves or not" may depend on observer resolution.

5.7 Admissible Change of Observer and Invariants

After introducing observer-dependent quantities, it is necessary to fix which properties have a chance to be fundamental. Otherwise the theory would decompose into disconnected descriptions without a common physical core.

Definition 5.19 (Admissible observer change)**.** An admissible change of observer is a transition

$$\mathcal{O} \rightsquigarrow \mathcal{O}'$$

between two observers satisfying the following conditions:

1. the causal order is not violated: if $e_1 \prec e_2$, then the observer $\mathcal{O}'$ cannot invert that dependence;
2. the classes of admissible histories are preserved;
3. the observer functor $\chi_{\mathcal{O}'}$ does not destroy those structural observables that have been declared fundamental in advance;
4. the change can be realized as a compatible coarse-graining, refinement, or transfer-compatible reparametrization.

Definition 5.20 (Observer-invariant observable)**.** An observable $\mathcal{I}$ is called observer-invariant if for every admissible observer change

$$\mathcal{O} \rightsquigarrow \mathcal{O}'$$

one has

$$\mathcal{I}(\mathcal{O}) = \mathcal{I}(\mathcal{O}').$$

Theorem 5.21 (Causal structure is invariant under admissible observer change)**.** *If the transition $\mathcal{O} \rightsquigarrow \mathcal{O}'$ is admissible, then the causal order of a structural history is preserved as a fundamental invariant. In particular, only its observer images may differ, but not the underlying dependence relation.*

Proof. This is built into the definition of admissible observer change itself. An observer may alter the representation of the history, its layering, its coarse-grained geometry, and the list of visible observables, but cannot turn a dependent event into an independent one or vice versa without violating admissibility of the description. □

Theorem 5.22 (Transfer-stable observables are natural candidates for observer invariants)**.** *Suppose an observable is defined through transfer-compatible data: marked kernels, residue observables, or the spectral profile of a decategorified resolvent. Then under admissible observer changes induced by compatible transfers or simulations, such an observable is preserved.*

Proof. The previous chapter already established that strict transfer and simulation preserve assembly series, pattern-valued series, marked moments, and spectral profiles under the corresponding compatibility hypotheses. If an admissible observer change is realized through such a transfer, then the observable survives the change of observer. □

5.8 The Role of This Chapter in the General Architecture of the Book

This chapter completes the necessary conceptual inversion without which all later physics would be written in the old externally physical style. After it, one can no longer think as if space, time, and velocity were given prior to the theory. They must be treated as observer images of a deeper structural-causal dynamics.

This changes the status of all subsequent quantities. The geometric graph of the next chapter will not be a fundamental stage but a geometric projection. Structural action, energy, and momentum will not be defined through already prepared spacetime, but will be extracted from causal histories and marked kernels. Even effective field-like equations will turn out not to be initial axioms, but secondary organized descriptions of an observer-level regime.

5.9 Lean Orientation for the Next Chapter

For Lean formalization, this part should be decomposed into three independent layers.

1. The type of rewrite events and a partial-order structure for causal histories.
2. The structure of an observer with fields $\Pi_{\mathcal{O}}$, $\Theta_{\mathcal{O}}$, and $\chi_{\mathcal{O}}$.
3. A separate namespace for admissible observer changes and lemmas about observer invariants.

It is especially important not to mix in one formal construction the fundamental causal order and the observer-induced time function. In the prose of the book this may look obvious, but in Lean they must be distinct entities with an explicit theorem stating the monotonicity of $t_{\mathcal{O}}$ with respect to $\prec$.

5.10 Conclusion of the Chapter

This chapter introduced the physical layer beginning not with geometry but with causality. Rewrite events, structural histories, and the observer as a triple of spatial projection, temporal slicing, and decategorification functor were defined. It was shown that space, time, and velocity are not fundamental entities but arise as observer-dependent images of a deeper causal structure. In this way the theory is prepared for the next step: the introduction of geometric projection and a gravity-like channel without returning to the naive assumption that spacetime is primary.

Chapter 6

Geometric Projection and the Gravity-Like Channel

6.1 Why Geometry Appears Only After the Causal Layer

The previous chapter fixed the crucial turn of the whole book: space and time are not original entities but observer-dependent images of a deeper structural-causal dynamics. One may now introduce the geometric layer, but only with the correct status. Geometry must not be a fundamental arena on which rewritings then take place. On the contrary, it must be a special projection of the same states and histories that already carry causal structure.

For that reason, the goal of this chapter is twofold. First, one must introduce a graph-geometric substrate as an observer image of structural states. Second, one must show how a gravity-like channel naturally arises on that graph image through the Laplacian/Poisson formalism. This is what connects the early structural apparatus with the first genuinely geometric regime of the book.

At the same time, the levels must not be confused. The geometric graph, the potential, and the effective force are not autonomous primary objects, but coarse-grained observables extracted from the projection layer. This is precisely what later makes it possible to speak coherently about geometry, spectral observables, and subsequent low-energy physics without destroying the already established hierarchy of descriptions.

6.2 Physical Graph Projection

Definition 6.1 (Physical graph projection)**.** A physical graph projection of the system is given by a map

$$\Pi : W \to \mathfrak{G},$$

which assigns to each coarse-grained structural state $x \in W$ a graph

$$\Pi(x) = G_\alpha(x) = (V_x, E_{\mathrm{loc}}(x) \cup E_{\mathrm{long}}(x)).$$

Here $E_{\mathrm{loc}}(x)$ forms the local backbone, while $E_{\mathrm{long}}(x)$ consists of long-range connections distributed according to a power-law with respect to observer-level distance.

Definition 6.2 (Power-law regime)**.** One says that the projection Π realizes a power-law regime with parameter $\alpha > 0$ if, for coarse-grained distance d, the expected number of long-range links of length d obeys the asymptotic law

$$\mathbb{E}[N_d] \asymp c_\alpha N d^{-\alpha}$$

in the corresponding scaling window.

Remark 6.3. In this definition the parameter d is not a fundamental length in an ontological sense. It already belongs to the chosen geometric projection. Thus a power-law graph does not describe deep reality "as it is," but fixes a stable observer-readable geometric image of a structural state.

Proposition 6.4 (Separation of geometry and dynamics)**.** *If a system is described by the data*

$$(W, R, \Pi),$$

then rewrite dynamics acts on structural states, whereas geometric observables are computed from $\Pi(x)$*. Consequently, two states may have close geometric images and differ substantially in causal-rewrite content, and conversely.*

Proof. By definition, the map Π is a projection from the structural layer to the geometric layer. It need not be injective, nor complete with respect to information. Therefore different structural configurations may have the same or similar geometric image, while similar rewrite dynamics may project to different graph-level pictures. □

6.3 Graph Laplacian and the Geometric Channel

After the graph layer has been introduced, one needs a minimal geometric operator that makes it possible to extract observer-level fields from distributed sources. The natural such operator is the graph Laplacian.

Definition 6.5 (Graph Laplacian)**.** Let $G = (V, E)$ be a finite or locally finite weighted graph with adjacency matrix A and diagonal degree matrix D. The graph Laplacian is defined by

$$L = D - A.$$

If the graph carries edge weights, then A and D are understood as the weighted adjacency and weighted degree matrices.

Definition 6.6 (Source density)**.** A source density on the graph image is a map

$$\rho : V \to \mathbb{R}$$

or, more generally, a signed measure on V, encoding the coarse-grained distribution of sources for the observer-level geometric channel.

Definition 6.7 (Poisson/Green channel)**.** Let $\varepsilon > 0$ be a regularizing parameter. Then the geometric Poisson channel is defined by the equation

$$(L + \varepsilon I)\phi = \rho,$$

where ϕ is a potential on the vertices. Equivalently,

$$\phi = (L + \varepsilon I)^{-1}\rho.$$

The corresponding operator

$$\mathcal{G}_\varepsilon := (L + \varepsilon I)^{-1}$$

is called the Green channel of the given observer-level geometric regime.

Remark 6.8. The regularizer εI is important not only technically. It separates the strict kernel of the Laplacian from the actually observed coarse-grained channel, in which zero modes are no longer read as literal symmetries of an infinitely precise system.

Theorem 6.9 (Observer-level potential)**.** *Let $x \in W$ and let its geometric image be $G_\alpha(x)$. Then the solution of the Poisson/Green channel equation defines an observer-level scalar potential*

$$V_{\mathcal{O}}(x) := \chi_{\mathcal{O}}(\phi_x),$$

where $\phi_x = (L_x + \varepsilon I)^{-1}\rho_x$ and $\chi_{\mathcal{O}}$ is the chosen decategorification/coarse-graining functor of the observer.

Proof. The graph operator $L_x + \varepsilon I$ is defined on the geometric image of the state x. Solving the linear equation on that graph yields the potential ϕ_x. Applying the observer functor $\chi_{\mathcal{O}}$ converts this graph-level object into a collection of observer numerical quantities or effective fields. Therefore $V_{\mathcal{O}}$ is not a fundamental object but a decategorified image of the geometric channel. □

6.4 Effective Force and the Status of the Newtonian Limit

Once the potential has been introduced, it is natural to define an effective force. But this is exactly the point where the theory must be especially careful. At this stage of the book Newton's law must not be declared a proved result. It may only be a target theorem asserting the existence of a certain scaling window.

Definition 6.10 (Observer-level effective force)**.** Suppose the observer space of the observer $\mathcal{O}$ admits a discrete or coarse-grained gradient operator $\nabla_{\mathcal{O}}$. Then the effective force is defined by

$$F_{\mathcal{O}} := -\nabla_{\mathcal{O}} V_{\mathcal{O}}.$$

If the potential channel also depends on a distinguished probe object, then $F_{\mathcal{O}}$ is read as the effective response of that object to the observer-level geometric landscape.

Theorem 6.11 (Target theorem: Newtonian scaling window)**.** *Suppose the projection* Π *maps states into a family of power-law graphs* G_α*, and suppose the Green channel exists and is stable in the corresponding coarse-grained ensemble. Assume that for sources from some class of localized structural sectors there exists an intermediate scaling window in which:*

1. *local geometric discreteness is averaged out;*
2. *the influence of long-range links remains substantial but does not destroy effective isotropy;*
3. *the observer-level potential admits radial asymptotics.*

Then in that window the effective force admits a low-energy approximation of Newtonian type

$$F_{\mathcal{O}}(r) \approx -\frac{C_{\mathcal{O}} m_1 m_2}{r^2}$$

or its corresponding dimension-dependent analog.

Remark 6.12. This theorem is deliberately formulated as a target theorem. At the present stage of the manuscript it fixes the exact status of the desired conclusion, but does not pretend that the conclusion has already been proved. Later the book must either provide a strict derivation under explicit hypotheses on the graph ensemble or honestly record a failure mode: the absence of a stable scaling window.

6.5 Local and Global Dimensions

Power-law graphs are important for this program not merely because they carry a Laplacian. Their essential physical value lies in the possibility that different effective notions of dimension may diverge.

Definition 6.13 (Local dimension)**.** The local dimension of an observer-level geometric regime is the dimensional characteristic read off from local neighborhood growth, local degrees, or short-scale transport behavior.

Definition 6.14 (Global or gravitational dimension)**.** The global, or gravitational, dimension is the effective dimension seen by the long-range Green channel or by the large-scale asymptotics of the potential.

Proposition 6.15 (Possible divergence of dimensions)**.** *For power-law graph projections, the local and global dimensions need not coincide. In particular, the geometry may look almost one-dimensional locally, while the Green channel exhibits large-scale behavior characteristic of a higher effective dimension.*

Proof. Local dimension is controlled by nearest-neighbor edges and by the short-scale structure of the backbone. Global dimension is also determined by the statistics of long-range links and by their contribution to the Green operator. Under power-law tails, these two scaling mechanisms are generally different, so equality of the dimensions is not mandatory. □

Remark 6.16. This difference is not a defect of the model. On the contrary, it explains why one and the same system may be geometrically poor at the local level and yet possess a much richer long-range effective channel.

6.6 Examples

Example 6.17 (Minimal backbone with one long-range link)**.** Consider a chain of vertices $1, 2, 3, 4, 5$ with local edges $(1, 2), (2, 3), (3, 4), (4, 5)$ and one additional long-range link $(1, 5)$. Let a unit source be placed at the vertex 3.

Locally, this geometry differs only slightly from a one-dimensional chain. But the Green channel is already sensitive to the additional bypass, because the matrix $L + \varepsilon I$ changes not only at the endpoints but in the global structure of available paths. In the observer-level potential this appears as a deviation from a purely one-dimensional decay profile.

Example 6.18 (Empirically testable part of the program)**.** In this chapter the empirically substantive content lies not in the abstract definitions themselves, but in the existence and

stability of the scaling window. What is testable is whether the family of graph projections extracted from the theory actually yields:

1. a stable Green channel;
2. a non-accidental effective potential;
3. an observer-reproducible approximate force law on intermediate scales.

If such a window does not exist, the gravity-like sector should be regarded as unsuccessful, not merely as "awaiting a better interpretation."

6.7 How This Chapter Connects to the Subsequent Dynamics

The present chapter does not yet introduce energy, momentum, and force at the fundamental level. It does something more modest but necessary: it fixes the geometric channel on which observer-level forces can later be read at all. This distinction is fundamental. The basic dynamics still remains in causal histories and rewrite events. The graph potential and the effective force are only images of that dynamics after geometric projection and decategorification.

Therefore the next chapter must translate energy-like and inertia-like quantities into the language of structural action and marked kernels, rather than deriving them directly from graph potential. Otherwise the theory would quietly return to the old scheme in which geometry is primary and structure is secondary.

6.8 Lean Orientation for the Next Chapter

For formalization of this material in Lean, it is reasonable to divide the code into at least three blocks.

1. The graph projection layer: graph types, the projection map Π, local edges, and long-range edges.
2. The operator layer: the Laplacian, the regularized Green operator, and basic lemmas about their existence in the finite case.
3. The observer layer: the decategorified potential, the effective force, and the formulation of the target theorem as a proposition with explicit hypotheses.

It is especially important that the target theorem on the Newtonian window be separated in Lean from the already proved linear-algebra facts about $L + \varepsilon I$. Otherwise

the status of conjectural or target content will again be mixed with the basic mathematical infrastructure.

6.9 Conclusion of the Chapter

In this chapter the geometric projection was introduced as an observer image of structural states rather than as the fundamental arena of the theory. Power-law graph projection, the graph Laplacian, the Poisson/Green channel, the observer-level potential, and the effective force were defined. It was also fixed that the Newtonian limit at the present stage has the status of a target theorem rather than a closed result. Finally, the possible divergence of local and global dimensions was isolated, which makes the gravity-like sector substantively different from the simple one-dimensional geometry of the backbone. In this way the manuscript is prepared for the next step: translating energy, momentum, inertia, and force into the language of structural action and kernel-level observables.

Chapter 7

Structural Action, Energy, Momentum, Inertia, and Force

7.1 Why a Language of Action Is Needed After the Geometric Channel

The previous chapter introduced the observer-level geometric channel: power-law graph projection, the Laplacian, the Green operator, and the effective potential. But that geometric part still does not provide a fundamental definition of energy, momentum, and inertia. If one simply declares them to be standard quantities on the resulting graph, the theory immediately loses its central idea: geometry becomes primary again, while structural-causal dynamics becomes secondary.

For that reason, the next step must go in the opposite direction. Energy, momentum, and inertia are not fundamental; a deeper quantity is fundamental, from which they arise after an observer has been chosen. In this book that quantity will be the structural action of a history. It measures the cost of causal-rewrite evolution in terms of local events, and only then gives rise to energy as action density per observer time, momentum as conjugacy to observer-space shift, inertia as stiffness of the transport branch, and force as the gradient of observer energy.

In this way the chapter plays the role of a conceptual bridge between causal histories and effective equations of motion. Without it, the later dynamics would turn out either purely phenomenological or secretly dependent on classical intuitions not derived from the structural program itself.

7.2 The Structural Action of a History

Definition 7.1 (Local weight of a rewrite event)**.** To each rewrite event e one assigns a positive local weight

$$w(e) > 0,$$

interpreted as the elementary admissibility or cost of the given structural transition at the multiplicative level.

Definition 7.2 (Weight of a history)**.** Let $\Gamma = (E_\Gamma, \prec_\Gamma)$ be a structural history. The weight of the history is the product of the local weights of its elementary events:

$$W(\Gamma) := \prod_{e \in E_\Gamma} w(e).$$

If the history admits a decomposition into sequentially or in parallel organized layers, then the product is understood as multiplicative aggregation over all realized elementary transitions.

Definition 7.3 (Structural action)**.** The structural action of a history is the quantity

$$\mathcal{S}(\Gamma) := -\log W(\Gamma).$$

Equivalently,

$$\mathcal{S}(\Gamma) = \sum_{e \in E_\Gamma} a(e), \qquad a(e) := -\log w(e).$$

The quantity $a(e)$ is called the local cost of the rewrite event.

Remark 7.4. The logarithmic definition is not a decorative choice. It translates multiplicative assembly/transfer data into an additive functional on histories. This is exactly the transition that makes it possible to speak about action, and then about energy and momentum, without breaking with the transfer-kernel formalism.

Definition 7.5 (Class of admissible local action functionals)**.** The local cost $a(e)$ may depend on structural markers of the event, for example on changes of h-geometry, shell/depth complexity, EFU rank, residue data, and influence on observer-level projection. Schematically, one allows a class of functionals of the form

$$a(e) = \beta_h \Delta h(e) + \beta_s \Delta \delta(e) + \beta_r \Delta \mathrm{rk}_{\mathrm{EFU}}(e) + \beta_\rho \Xi(\rho_e) + \beta_g \Delta g_\Pi(e) + \beta_0,$$

where the coefficients β_* are fixed by the chosen physical regime.

Proposition 7.6 (Additivity of the action)**.** *If a history Γ decomposes into a composition of two compatible subsystem histories Γ_1 and Γ_2 with multiplicative rule of weights*

$$W(\Gamma) = W(\Gamma_1) W(\Gamma_2),$$

then the structural action is additive:

$$\mathcal{S}(\Gamma) = \mathcal{S}(\Gamma_1) + \mathcal{S}(\Gamma_2).$$

Proof. This follows immediately from the definition through the negative logarithm of the product. □

7.3 Energy as Action Density per Observer Time

Fundamentally, the theory provides action, not energy. Energy appears only after an observer has been chosen and hence a time calibration of the history has been fixed.

Definition 7.7 (Observer energy of a process)**.** Let $\mathcal{O}$ be an observer with time function $t_{\mathcal{O}}$. The energy of a history Γ relative to $\mathcal{O}$ is the ratio

$$E_{\mathcal{O}}(\Gamma) := \frac{\Delta\mathcal{S}(\Gamma)}{\Delta t_{\mathcal{O}}(\Gamma)},$$

provided the corresponding coarse-grained time increment is defined.

Remark 7.8. This definition shows that energy is not a stock of "substance" or a primary material quantity. It is the intensity of expenditure of structural action in the chosen temporal calibration of the observer.

Definition 7.9 (Rest energy)**.** Let $\mathfrak{p}$ be a stable EFU class or a stable assembly class. The rest energy relative to the observer $\mathcal{O}$ is the quantity

$$E_{\mathcal{O}}^{\mathrm{rest}}(\mathfrak{p}) := \inf_{\Gamma \in \mathrm{Stat}(\mathfrak{p},\mathcal{O})} \frac{\mathcal{S}(\Gamma)}{t_{\mathcal{O}}(\Gamma)},$$

where $\mathrm{Stat}(\mathfrak{p}, \mathcal{O})$ consists of histories preserving the structural class of the object and carrying no directed observer-level transport.

Proposition 7.10 (Rest energy as the cost of stability)**.** *If the stable class* $\mathfrak{p}$ *requires nonzero self-maintenance at the level of histories, then*

$$E_{\mathcal{O}}^{\mathrm{rest}}(\mathfrak{p}) > 0.$$

Proof. If every stationary history realizing the class $\mathfrak{p}$ contains at least one nontrivial set of rewrite events with positive local cost, then the total action is positive. Division by finite positive observer time preserves the sign. □

7.4 Momentum as Conjugacy to Observer-Space Transport

If space is a derived projection, then momentum cannot be a fundamental invariant either. It appears only after observer space has been chosen and only in regimes where directed transport makes sense.

Definition 7.11 (Observer displacement)**.** Let $\mathcal{O}$ be fixed. For a rewrite event e, its observer displacement is defined as

$$\Delta x_{\mathcal{O}}(e),$$

that is, as the change of the chosen spatial projection induced by the event e.

Definition 7.12 (Momentum through action)**.** If a family of histories admits variation with respect to an observer-level transport parameter, then momentum is the quantity conjugate to that parameter:

$$p_{\mathcal{O}} := \frac{\partial \mathcal{S}}{\partial(\Delta x_{\mathcal{O}})}.$$

Definition 7.13 (Marked transport kernel)**.** Suppose a finite-state kernel is given for a particle sector $\mathfrak{p}$. Its marked transport kernel is defined by

$$K^{(\mathfrak{p})}_{\sigma\tau}(q,\eta) = \sum_{e:\sigma\to\tau} \exp\big(iq\,\Delta x_{\mathcal{O}}(e) - \eta a(e)\big),$$

where q is an observer quasi-momentum marker and η marks the action/energy contribution.

Definition 7.14 (Dominant transport branch)**.** Let

$$\lambda_{\mathfrak{p}}(q,\eta)$$

denote the dominant eigenvalue of the marked transport kernel $K^{(\mathfrak{p})}(q,\eta)$, or the corresponding simple dominant eigenbranch in a regime where spectral control applies.

Proposition 7.15 (Momentum as a spectral transport observable)**.** *In marked-kernel language, momentum may be read as the observable conjugate to the parameter q, for example through the derivative*

$$p_{\mathcal{O}} \sim \partial_q \log \lambda_{\mathfrak{p}}(q,\eta).$$

Proof. The marker q encodes directed transport in observer space. Therefore differentiation of the dominant spectral branch with respect to q extracts the sensitivity of the regime to directed displacement, which is exactly the quantity interpreted as momentum. □

7.5 On-Shell Energy and the Effective Dispersion Branch

The advantage of the marked transport kernel is that energy is no longer introduced through a separate phenomenological formula, but is instead defined through a spectral condition.

Definition 7.16 (On-shell energy branch)**.** The energy of the particle sector $\mathfrak{p}$ as a function of quasi-momentum q is determined by the condition

$$\lambda_{\mathfrak{p}}\big(q, E_{\mathfrak{p}}(q)\big) = 1.$$

The resulting function $E_{\mathfrak{p}}(q)$ is called the effective dispersion branch of that sector.

Corollary 7.17 (Rest energy as an on-shell value)**.** *The rest energy of the sector* $\mathfrak{p}$ *is given by*

$$E_{\mathfrak{p}}^{\mathrm{rest}} = E_{\mathfrak{p}}(0),$$

that is, by the condition

$$\lambda_{\mathfrak{p}}(0, E_{\mathfrak{p}}^{\mathrm{rest}}) = 1.$$

Remark 7.18. Thus mass-like and energy-like observables are extracted from the same kernel-level mechanism that was already used for assembly series, marked moments, and spectral control. This eliminates the need to introduce a separate ad hoc mass formula.

7.6 Inertia as Stiffness of the Transport Branch

Definition 7.19 (Fundamental inertia)**.** For a stable class $\mathfrak{p}$, inertia is the minimal additional action required to move it from one observer transport sector to another:

$$\mathcal{I}(\mathfrak{p}) := \inf_{\Gamma\colon (\mathfrak{p},u_1)\to(\mathfrak{p},u_2)} \big(\mathcal{S}(\Gamma) - \mathcal{S}_{\min}(\mathfrak{p})\big).$$

Remark 7.20. In this definition inertia is not "mass," but structural rigidity with respect to a change of transport regime. An object with large inertia requires a substantial action cost to retune its transport.

Definition 7.21 (Spectral inertial mass)**.** If the effective dispersion branch admits, for small q, an expansion of the form

$$E_{\mathfrak{p}}(q) = E_{\mathfrak{p}}^{\mathrm{rest}} + v_{0,\mathfrak{p}} q + \frac{q^2}{2M_{\mathfrak{p}}} + O(q^3),$$

then the quantity $M_{\mathfrak{p}}$ is called the effective inertial mass of the corresponding observer regime.

Theorem 7.22 (Formula for the effective inertial mass)**.** *Assume that the branch $\lambda_{\mathfrak{p}}(q,\eta)$ is smooth in a neighborhood of the point*

$$(0, E_{\mathfrak{p}}^{\mathrm{rest}})$$

and that

$$\partial_\eta \lambda_{\mathfrak{p}}(0, E_{\mathfrak{p}}^{\mathrm{rest}}) \neq 0.$$

If the linear term in q vanishes by symmetry, then

$$\frac{1}{M_{\mathfrak{p}}} = -\frac{\partial_q^2 \lambda_{\mathfrak{p}}(0, E_{\mathfrak{p}}^{\mathrm{rest}})}{\partial_\eta \lambda_{\mathfrak{p}}(0, E_{\mathfrak{p}}^{\mathrm{rest}})}.$$

Proof. Differentiate the on-shell condition

$$\lambda_{\mathfrak{p}}(q, E_{\mathfrak{p}}(q)) = 1$$

with respect to q. At first differentiation one gets a relation between $\partial_q \lambda$ and $E_{\mathfrak{p}}'(q)$. At second differentiation, after setting $q = 0$ and using the vanishing of the linear term by symmetry, one obtains the formula for the quadratic coefficient of the expansion, that is, for $M_{\mathfrak{p}}$. □

Proposition 7.23 (Separation of mass, rest energy, and inertia)**.** *In general, the mass-like spectral invariant, the resting energy density, and inertial rigidity need not coincide. Their identification may arise only as a special emergent regime.*

Proof. Rest energy is determined by the minimum of action density in the stationary sector, inertia by curvature or transport rigidity, and the mass-like invariant by a spectral parameter of the stable particle-sector kernel. These three extraction procedures are different, so their coincidence is not a formal consequence of the definitions. □

7.7 Force as the Gradient of Observer Energy

After the energy branch and the geometric potential have been introduced, force no longer looks like an independent agent. It arises as a derivative effect of already constructed observer-dependent quantities.

Definition 7.24 (Observer force)**.** Suppose the observer space admits a gradient operator $\nabla_{\mathcal{G}_{\mathcal{O}}}$. Then force is defined by

$$F_{\mathcal{O}} := -\nabla_{\mathcal{G}_{\mathcal{O}}} E_{\mathcal{O}}^{\mathrm{int}},$$

where $E_{\mathcal{O}}^{\mathrm{int}}$ is the interaction part of observer energy.

Definition 7.25 (Total energy in a geometric field)**.** Let ϕ_i be the observer-level potential from the previous chapter, and let $g_{\mathfrak{p}}$ be the coupling of the given particle sector to the geometric channel. Then the total observer energy is defined by

$$E_{\mathfrak{p}}(q,i) = E_{\mathfrak{p}}^{\mathrm{int}}(q) + g_{\mathfrak{p}}\phi_i.$$

Proposition 7.26 (Observer equation of motion)**.** *If an observer regime is chosen in which an effective Hamiltonian-like description applies, then the equations of motion take the form*

$$\dot{x} = \partial_q E_{\mathfrak{p}}(q,i), \qquad \dot{q} = -\nabla_i E_{\mathfrak{p}}(q,i).$$

In the low-momentum regime this becomes

$$M_{\mathfrak{p}}\ddot{x} \approx -g_{\mathfrak{p}}\nabla_i\phi_i.$$

Proof. The first pair of formulas is the standard effective Hamiltonian reading of the chosen energy branch. Substituting the decomposition

$$E_{\mathfrak{p}}(q,i) = E_{\mathfrak{p}}^{\mathrm{int}}(q) + g_{\mathfrak{p}}\phi_i$$

separates transport and potential contributions. The low-momentum expansion of the energy then gives

$$\partial_q E_{\mathfrak{p}}^{\mathrm{int}}(q) \approx q/M_{\mathfrak{p}},$$

from which the Newton-like form of the second equation follows. □

7.8 Bound States and Negative Binding Energy

Definition 7.27 (Binding energy)**.** Let two stable objects $\mathfrak{p}_1$ and $\mathfrak{p}_2$ separately have rest energies E_1 and E_2, and suppose their joint assembly configuration has resting energy

$$E_{12}^{\mathrm{rest}} < E_1 + E_2.$$

Then the binding energy is defined by

$$E_{\mathrm{bind}} := E_{12}^{\mathrm{rest}} - (E_1 + E_2) < 0.$$

Remark 7.28. Negative binding energy means that the joint EFU/assembly organization uses a more favorable structural regime: shared shell layers, reduced rewrite tension, or a transition to a more favorable spectral branch. Those are precisely the mechanisms that should later generate chemistry-like and many-body binding sectors.

Example 7.29 (Free regime)**.** Suppose the particle sector $\mathfrak{p}$ has a symmetric dispersion branch

$$E_{\mathfrak{p}}(q) = E_{\mathfrak{p}}^{\mathrm{rest}} + \frac{q^2}{2M_{\mathfrak{p}}} + O(q^4)$$

and does not feel an external geometric channel. Then

$$\dot{q} = 0, \qquad \dot{x} = \frac{q}{M_{\mathfrak{p}}} + O(q^3).$$

This corresponds to free observer-level transport without external force.

Example 7.30 (Regime with an external potential)**.** Suppose the same sector is placed in a geometric field with potential ϕ_i. Then

$$E_{\mathfrak{p}}(q, i) = E_{\mathfrak{p}}^{\mathrm{int}}(q) + g_{\mathfrak{p}}\phi_i,$$

and in the low-momentum approximation the motion obeys the equation

$$M_{\mathfrak{p}}\ddot{x} \approx -g_{\mathfrak{p}}\nabla_i\phi_i.$$

This example shows that observer force arises not as a fundamental agent but as the gradient of already derived observer energy.

Example 7.31 (A simplest bound state)**.** Suppose two assembly sectors separately have resting energies E_1 and E_2, but there exists a joint sector in which part of the rewrite events becomes shared and the dominant spectral branch shifts into a more favorable position. If the resulting resting energy is

$$E_{12}^{\mathrm{rest}} = E_1 + E_2 - \Delta, \qquad \Delta > 0,$$

then

$$E_{\mathrm{bind}} = -\Delta < 0.$$

This is exactly what should be interpreted as structural-spectral binding energy. This is exactly the structural-spectral binding energy of the joint sector.

7.9 How This Chapter Prepares Explicit Equations of Motion

This chapter has not yet built concrete toy kernels for individual particles and has not yet carried out explicit low-energy calculations. But it is here that the entire vocabulary needed for the next step has appeared. One can now write not merely general principles

but explicit marked transport kernels, on-shell conditions, effective dispersion laws, and toy equations of motion.

It also closes an important conceptual gap. After the causal chapter and the geometric chapter, one might still have the impression that geometry almost fully determines physics. The present chapter shows that even energy and force must be extracted not from geometry as such, but from the interplay between structural action, transfer kernels, and observer projection.

7.10 Lean Orientation for the Next Chapter

For formalization of this material in Lean, it is useful to separate four independent layers.

1. The history-action layer: local weights, the action functional, and stationary histories.
2. The observer-energy layer: definitions of energy, rest energy, and observer force.
3. The marked-kernel layer: transport markers, the on-shell branch, and derivatives of the dominant eigenbranch.
4. The low-energy layer: extraction of inertial mass and toy Hamiltonian-like equations.

It is especially important to separate definitional content from target or model-dependent content. The definitions of action, energy, and inertia must be formalized independently of any concrete graph model or any concrete particle kernel. By contrast, low-energy expansions and explicit equations should live in additional sections with explicit hypotheses.

7.11 Conclusion of the Chapter

In this chapter the fundamental mechanical quantities were translated into the language of structural-causal theory. Structural action of a history was introduced as the primitive quantity, from which observer energy, rest energy, momentum as a transport observable, inertia as stiffness of a spectral branch, and force as the gradient of observer energy were then derived. On-shell energy branch, effective inertial mass, and binding energy were also introduced. In this way the book acquired a general dynamic vocabulary sufficient for the next step: constructing explicit toy models and effective equations of motion based on concrete marked transport kernels.

Chapter 8

Effective Equations of Motion and Explicit Toy Models

8.1 Why Explicit Models Are Needed After the General Dynamical Vocabulary

By the previous chapter, the theory had already acquired a general vocabulary: structural action, observer energy, transport-marked kernels, the on-shell dispersion branch, effective inertial mass, and observer force. But this is still not enough for a working physical program. If one stops at the level of definitions, it remains unclear whether the apparatus actually produces computable formulas rather than merely a well-organized language.

Therefore the goal of the present chapter is to pass from general structure to explicit toy dynamics. We do not try here to build a final microphysics. The task is more modest and more important: to show that once one chooses a concrete class of local events and a concrete finite-state transport model, the theory really does produce dispersion, velocity, inertial mass, and observer-level equations of motion in explicit form.

Such toy models will then play a triple role. First, they provide testable computational examples. Second, they show which elements of the model are actually essential for the low-energy limit. Third, they are natural candidates for a first Lean formalization of the dynamical layer.

8.2 Minimal Geometry and the Particle Sector

We begin with the simplest observer geometry already compatible with the previous chapters.

Definition 8.1 (Minimal projected graph)**.** Let $G_{\alpha,N}$ be a finite cycle-like backbone with vertices

$$V = \mathbb{Z}/N\mathbb{Z},$$

local edges

$$E_{\mathrm{loc}} = \{(i, i+1) : i \in V\},$$

and, possibly, additional long-range edges distributed according to a power-law rule. This graph is treated as the minimal observer-level geometry of the toy model.

Definition 8.2 (Internal state space of a toy particle)**.** A toy particle sector $\mathfrak{p}$ is specified by a finite set of internal states

$$\Sigma_{\mathfrak{p}} = \{\sigma_1, \ldots, \sigma_m\}.$$

These states encode the internal transport regime of the particle at the level of coarse-grained rewriting dynamics.

Remark 8.3. Even in the simplest toy model it is important to keep the internal state space nontrivial. Otherwise the marked kernel immediately degenerates into a scalar model and the ability to distinguish transport channels, metastable internal states, and spectral mixing is lost.

8.3 Elementary Rewrite Events and Their Cost

Suppose elementary events of the form

$$e : (i, \sigma) \to (j, \tau),$$

are now allowed, where $i, j \in V$ and $\sigma, \tau \in \Sigma_{\mathfrak{p}}$. Each such event simultaneously changes observer-level position and internal transport state.

Definition 8.4 (Toy action cost)**.** For an elementary event $e : (i, \sigma) \to (j, \tau)$ define the local cost

$$a(e) = \kappa_0 + \kappa_h \Delta h(e) + \kappa_s \Delta s(e) + \kappa_r \Delta r(e) + \kappa_\ell d_G(i, j),$$

where:

1. $\Delta h(e)$ measures the change in internal structural complexity;
2. $\Delta s(e)$ measures the change in a shell/depth marker;
3. $\Delta r(e)$ measures a residue-sensitive transport correction;
4. $d_G(i, j)$ is the graph distance in the projected geometry.

Remark 8.5. In the toy chapter, the coefficients κ_* are not interpreted as fundamental constants of nature. They play the role of effective model parameters on which one can test stability of low-energy outputs.

8.4 An Explicit Marked Transport Kernel

One can now build a kernel that is actually computable.

Definition 8.6 (Toy marked transport kernel)**.** The marked transport kernel of the toy sector $\mathfrak{p}$ is the matrix

$$K^{(\mathfrak{p})}(q,\eta) = \big(K^{(\mathfrak{p})}_{\sigma\tau}(q,\eta)\big)_{\sigma,\tau\in\Sigma_{\mathfrak{p}}},$$

where

$$K^{(\mathfrak{p})}_{\sigma\tau}(q,\eta) = \sum_{e:\sigma\to\tau} \exp\big(iq\Delta x(e) - \eta a(e)\big).$$

Here q marks observer-level transport along the backbone, and η marks the action or energy contribution.

Definition 8.7 (Nearest-neighbour symmetric toy kernel)**.** The simplest nontrivial case occurs for two internal states

$$\Sigma_{\mathfrak{p}} = \{A, B\}$$

and four types of elementary events:

1. $A \to A$ with shift $+1$;
2. $A \to B$ with no shift;
3. $B \to A$ with no shift;
4. $B \to B$ with shift -1.

Then the kernel takes the form

$$K^{(\mathfrak{p})}(q,\eta) = \begin{pmatrix} e^{-\eta a_{AA}} e^{iq} & e^{-\eta a_{AB}} \\ e^{-\eta a_{BA}} & e^{-\eta a_{BB}} e^{-iq} \end{pmatrix}.$$

Example 8.8 (A symmetric two-state toy model)**.** If additionally

$$a_{AA} = a_{BB} = a, \qquad a_{AB} = a_{BA} = b,$$

then

$$K(q,\eta) = \begin{pmatrix} e^{-\eta a} e^{iq} & e^{-\eta b} \\ e^{-\eta b} & e^{-\eta a} e^{-iq} \end{pmatrix}.$$

Its eigenvalues are

$$\lambda_{\pm}(q,\eta) = e^{-\eta a}\cos q \pm \sqrt{e^{-2\eta b} - e^{-2\eta a}\sin^2 q}.$$

In this model the dominant branch is already explicitly computable, so the on-shell dispersion relation can be studied analytically.

8.5 On-Shell Dispersion and Rest Energy

Definition 8.9 (Toy dispersion branch)**.** The effective dispersion branch of the toy sector is defined by the condition

$$\lambda_*(q, E_{\mathfrak{p}}(q)) = 1,$$

where λ_* is the dominant eigenbranch of the marked kernel.

Proposition 8.10 (Rest energy of the toy sector)**.** *The rest energy is given by*

$$E_{\mathfrak{p}}^{\mathrm{rest}} = E_{\mathfrak{p}}(0),$$

that is, it solves the equation

$$\lambda_*(0, E_{\mathfrak{p}}^{\mathrm{rest}}) = 1.$$

For the symmetric two-state model this yields

$$e^{-E_{\mathfrak{p}}^{\mathrm{rest}} a} + e^{-E_{\mathfrak{p}}^{\mathrm{rest}} b} = 1$$

for the dominant branch.

Proof. At $q = 0$ both transport directions coincide at the level of the observer momentum marker, and the on-shell condition reduces to an algebraic equation for the dominant eigenvalue. For the indicated symmetric matrix, the dominant eigenvalue is the sum of the diagonal and off-diagonal contributions. □

8.6 Low-Energy Expansion and Inertial Mass

Theorem 8.11 (Quadratic low-energy regime)**.** *Assume that the toy model is symmetric under $q \mapsto -q$, and that the dominant branch $\lambda_*(q, \eta)$ is analytic in a neighborhood of the point*

$$(0, E_{\mathfrak{p}}^{\mathrm{rest}}).$$

Then the dispersion branch admits an expansion

$$E_{\mathfrak{p}}(q) = E_{\mathfrak{p}}^{\mathrm{rest}} + \frac{q^2}{2M_{\mathfrak{p}}} + O(q^4),$$

where

$$\frac{1}{M_{\mathfrak{p}}} = -\frac{\partial_q^2 \lambda_*(0, E_{\mathfrak{p}}^{\mathrm{rest}})}{\partial_\eta \lambda_*(0, E_{\mathfrak{p}}^{\mathrm{rest}})}.$$

Proof. The symmetry $q \mapsto -q$ kills the linear term. The remaining claim follows from implicit differentiation of the on-shell equation

$$\lambda_*(q, E_{\mathfrak{p}}(q)) = 1.$$

□

Corollary 8.12 (Group velocity)**.** *In the low-energy regime the observer-level velocity is*

$$v_{\mathfrak{p}}(q) = \partial_q E_{\mathfrak{p}}(q) = \frac{q}{M_{\mathfrak{p}}} + O(q^3).$$

Proof. It is enough to differentiate the quadratic expansion of the dispersion branch. □

8.7 External Geometric Field and Effective Equations of Motion

Now the geometric potential from Chapter 5 is brought back in.

Definition 8.13 (Toy energy in an external field)**.** Let ϕ_i be an observer-level potential on the projected graph, and let $g_{\mathfrak{p}}$ be the coupling of the particle sector to that field. Then the total energy of the toy particle is

$$E_{\mathfrak{p}}(q, i) = E_{\mathfrak{p}}^{\mathrm{int}}(q) + g_{\mathfrak{p}}\phi_i.$$

Definition 8.14 (Discrete effective equations)**.** The effective motion of the toy particle is given by

$$\dot{x} = \partial_q E_{\mathfrak{p}}(q, i), \qquad \dot{q} = -\nabla_i E_{\mathfrak{p}}(q, i).$$

For cycle geometry one may take the symmetric discrete gradient

$$\nabla_i \phi_i := \frac{\phi_{i+1} - \phi_{i-1}}{2}.$$

Theorem 8.15 (Low-energy Newton-like equation)**.** *In the low-momentum regime and for a slowly varying potential one has the approximation*

$$M_{\mathfrak{p}}\ddot{x} \approx -g_{\mathfrak{p}}\nabla_i \phi_i.$$

Proof. From the previous corollary we have

$$\dot{x} \approx q/M_{\mathfrak{p}}.$$

On the other hand, the gradient acts only on the potential contribution, so

$$\dot{q} = -g_{\mathfrak{p}}\nabla_i \phi_i.$$

Differentiating the first relation in time gives the stated approximation. □

8.8 A Fully Worked Toy Example

Example 8.16 (A two-state transport particle on a ring)**.** Consider a ring of size N with no long edges and a two-state particle with symmetric kernel

$$K(q,\eta) = \begin{pmatrix} e^{-\eta a}e^{iq} & e^{-\eta b} \\ e^{-\eta b} & e^{-\eta a}e^{-iq} \end{pmatrix}, \qquad a, b > 0.$$

Suppose additionally that on the ring a potential ϕ_i is given as the solution of the discrete Poisson equation with a single source. Then the computational scheme is fully explicit:

1. for a given q, solve the algebraic on-shell equation $\lambda_*(q, E(q)) = 1$;
2. extract E^{rest}, M, and $v(q)$ from that solution;
3. then integrate the system

$$\dot{x} = \partial_q E(q), \qquad \dot{q} = -g\nabla_i \phi_i.$$

This example can already be implemented either as a numerical notebook or as a Lean-compatible finite model.

Example 8.17 (A free toy regime)**.** If $\phi_i \equiv 0$, then

$$\dot{q} = 0, \qquad \dot{x} = v_{\mathfrak{p}}(q).$$

Consequently, motion is fully controlled by the intrinsic dispersion branch. In the low-energy regime one gets almost uniform transport with velocity proportional to q.

Example 8.18 (Toy bound tendency)**.** If a potential well is created by two sources or by a locally modified graph sector, and the resulting total energy admits a local minimum in position, then the particle possesses a bounded observer regime. At the toy level this is not yet a full theory of bound states, but it already demonstrates how binding-like behavior emerges from the same action-plus-kernel machinery.

8.9 A Practical Computational Pipeline

For this chapter it is especially important to fix not only formulas but also a real order of computations.

1. Specify the projected graph $G_{\alpha,N}$ and source ρ.
2. Solve the equation

$$(L + \varepsilon I)\phi = \rho.$$

3. Choose the particle sector $\mathfrak{p}$ and its internal state space $\Sigma_{\mathfrak{p}}$.

4. List the admissible rewrite events and compute their costs $a(e)$.

5. Assemble the marked transport kernel $K^{(\mathfrak{p})}(q, \eta)$.

6. Find the dominant eigenbranch $\lambda_*(q, \eta)$.

7. Solve the on-shell equation for $E_{\mathfrak{p}}(q)$.

8. Extract rest energy, inertial mass, and group velocity.

9. Integrate the effective equations of motion.

Remark 8.19. This pipeline shows that once the local action cost is chosen, most of the remaining dynamics is forced by the formalism. For that reason, the weak point of the program lies not in the algebra after the kernel has been built, but in the physically meaningful choice of event costs and admissible microscopic transitions.

8.10 What Exactly Is Testable Here

The toy chapter is useful only if it clearly separates those consequences that are mathematically derived from the model from those places where choices are inserted into the model.

1. Formally derived consequences: the on-shell equation, the inertial mass formula, and the low-energy motion law.

2. Model-dependent input: the internal state space, the list of admissible events, the local action cost, and the coupling to the geometric field.

3. Empirically meaningful outputs: the dispersion profile, transport asymmetry, effective mass, and the existence or absence of bounded regimes.

This distinction must be preserved later as well. Otherwise the toy model would be credited with explanatory force that properly belongs only to the explicitly chosen input assumptions.

8.11 Lean Orientation for the Next Chapter

For formalization of the explicit toy dynamics in Lean, it is useful to split the code into four parts.

1. The finite graph model and a discrete Poisson solver in the finite case.

2. Finite internal-state kernels and their characteristic-value and eigenvalue computations.

3. The implicit definition of the on-shell energy branch together with theorems on the small-q expansion.

4. A finite-dimensional derivation of effective equations of motion.

It is especially helpful that in the toy model most objects are finite. This makes the chapter a natural first target for formal verification: there are fewer analytic obstacles here than in later continuum and quantum-field chapters.

8.12 Conclusion of the Chapter

In this chapter the structural-dynamical program was carried all the way to explicit computable formulas. Finite toy particle sectors, explicit marked transport kernels, on-shell dispersion branches, low-energy expansions, and Newton-like effective equations of motion were constructed. In this way the book demonstrated that its dynamical language does not reduce to general declarations: for a concrete choice of microscopic event costs it produces real computational mechanics. This creates the correct launching point for the next step: the passage to complex amplitudes, quantum states, and the resolvent-amplitude formalism.

Chapter 9

Complex Amplitudes, Quantum States, and the Resolvent-Amplitude Formalism

9.1 Why the Quantum Layer Must Begin with Amplitudes Rather Than an External Wave Function

At this point in the book, the structural program has already built a complete path from symbolic structures to transport kernels, geometric channels, and effective equations of motion. The next step is often taken from the outside: one postulates a Hilbert space, a state vector, and independent quantum axioms. Within the present book such a move would be too abrupt and logically uneconomical.

Here the quantum layer must grow out of the already existing kernel-level formalism. If in the previous chapters observables were introduced through real marked kernels and their spectral branches, then one must now allow complex weights of histories and complex-valued transport amplitudes. That is the first meaning of "quantumness" in the present theory. Not a separate external entity called a wave function, but a coherent sum over structural histories with phases.

For that reason, this chapter does four things. First, it introduces the complex amplitude kernel as a continuation of the marked transport kernel. Second, it defines history amplitudes and coarse-grained amplitudes for classes of histories. Third, it assembles these sums into a resolvent-amplitude form compatible with the entire earlier style of the book. Fourth, it explains why a "quantum state" is treated here as a derived object extracted from amplitude sectors rather than as a primitive building block of the theory.

9.2 The Complex Amplitude Kernel

Definition 9.1 (Complex event amplitude)**.** To each elementary rewrite event e one assigns a complex event amplitude

$$\alpha(e) = r(e)e^{i\phi(e)}, \qquad 0 \leq r(e) \leq 1,$$

where $r(e)$ is the modulus part of the amplitudinal weight and $\phi(e)$ is the phase of the event.

Remark 9.2. The real action weights of the previous chapters do not disappear. They enter the modulus part of the amplitude through a damping factor, whereas the phase part encodes coherent transport and interference information.

Definition 9.3 (Amplitude-marked kernel)**.** Let $\mathcal{O}$ be a fixed observer. Then the amplitude-marked kernel is defined by

$$K_{\mathcal{O}}^{\mathrm{amp}}(z, \theta, \beta, \varphi) = \sum_{e} \alpha(e) z^{\gamma(e)} e^{i\theta \cdot \Delta x_{\mathcal{O}}(e)} e^{-\beta a(e)} e^{i\varphi \xi_D(e)} [e],$$

where:

1. $z^{\gamma(e)}$ encodes grading data;
2. θ marks observer-level displacement;
3. β marks the action weight;
4. φ marks a detector or record channel;
5. $\xi_D(e)$ is the detector-mark observable of the event e.

Proposition 9.4 (The amplitude kernel extends the marked transport kernel)**.** *If all phases vanish, the detector mark is switched off, and the event amplitudes have the form*

$$\alpha(e) = 1,$$

then the amplitude-marked kernel reduces to the usual marked transport formalism with real action weight.

Proof. For $\phi(e) = 0$ and $\varphi = 0$, the complex factors reduce to one. What remains is only the grading, the transport marker, and damping by action cost, that is, exactly the previously introduced marked-kernel structure. □

9.3 History Amplitude and Coarse-Grained Classes

One can now pass from kernel-level description to histories themselves.

Definition 9.5 (History amplitude)**.** Let

$$\Gamma = (e_1, \ldots, e_n)$$

be an admissible structural history. Its amplitude relative to the observer $\mathcal{O}$ is defined by

$$\mathcal{A}_{\mathcal{O}}(\Gamma) := \prod_{k=1}^{n} \alpha(e_k) e^{i\theta \cdot \Delta x_{\mathcal{O}}(e_k)} e^{-\beta a(e_k)} e^{i\varphi \xi_D(e_k)}.$$

Definition 9.6 (Coarse-grained amplitude of a history class)**.** Let $\mathfrak{C}$ be a class of histories selected by a coarse-graining rule or observer partition. Then its amplitude is defined by the sum

$$\Psi_{\mathcal{O}}(\mathfrak{C}) := \sum_{\Gamma \in \mathfrak{C}} \mathcal{A}_{\mathcal{O}}(\Gamma).$$

Remark 9.7. It is exactly in this sum that true interference behavior first appears. Different structural histories may belong to one and the same coarse-grained class and therefore contribute coherently before any probabilistic interpretation is imposed.

Definition 9.8 (Amplitude sector)**.** An amplitude sector is a family of coarse-grained history classes, closed under the chosen observer partition and equipped with their complex amplitudes

$$\{\Psi_{\mathcal{O}}(\mathfrak{C}_a)\}_a.$$

9.4 The Resolvent-Amplitude Representation

One of the strengths of the entire book is that sums over structured objects are not rewritten by hand each time, but assembled into resolvent form. The quantum layer must preserve exactly this style.

Definition 9.9 (Resolvent amplitude)**.** The full resolvent amplitude for a chosen observer regime is defined by

$$\Psi_{\mathcal{O}}(z, \theta, \beta, \varphi) = \iota^{\top} \big(I - K_{\mathcal{O}}^{\mathrm{amp}}(z, \theta, \beta, \varphi)\big)^{-1} \omega.$$

Theorem 9.10 (The resolvent amplitude sums admissible histories)**.** *If the observer-amplitude system is finite-state in the same sense as the earlier transfer systems, then the formal expansion*

$$\big(I - K_{\mathcal{O}}^{\mathrm{amp}}\big)^{-1} = \sum_{n \geq 0} (K_{\mathcal{O}}^{\mathrm{amp}})^n$$

sums all admissible histories with their complex amplitudes.

Proof. Each power of the kernel lists all admissible paths of a fixed length in the finite-state observer-amplitude automaton. Since the entries already contain complex phase, action damping, and record marks, summation over all lengths yields exactly the total coherent sum over histories. □

Corollary 9.11 (Coarse-grained amplitudes arise by projection)**.** *The amplitude of any coarse-grained history class is obtained from the full resolvent amplitude by applying the corresponding projection or coarse-graining operator to the sum over admissible histories.*

Proof. The full resolvent amplitude already contains all admissible histories. Coarse-graining merely groups them according to the chosen partition and sums the corresponding terms. □

9.5 Quantum State as a Derived Object

At this point one can answer the main methodological question of the chapter: what exactly plays the role of the "quantum state" in this theory?

Definition 9.12 (Amplitude state)**.** An amplitude state of the observer is an organized family of coarse-grained amplitudes

$$\Psi_{\mathcal{O}} = \big(\Psi_{\mathcal{O}}(\mathfrak{C}_a)\big)_a$$

for the chosen partition into classes of histories.

Remark 9.13. This is not an external object added to the theory. It is built from amplitude sums over histories and therefore inherits all dependencies on action, observer, transport markers, and record structure. In other words, the "state" here is a compact encoding of amplitude sectors, not a primitive ontological datum.

Definition 9.14 (Normalized amplitude state)**.** If an inner-product-like observer pairing is chosen and the norm is finite, then the normalized amplitude state is

$$\widehat{\Psi}_{\mathcal{O}} := \frac{\Psi_{\mathcal{O}}}{\|\Psi_{\mathcal{O}}\|}.$$

Proposition 9.15 (State reduction to transport sectors)**.** *If the coarse-graining chosen by the observer ignores detector marks and groups histories only by transport sector, then the amplitude state reduces to a transport-sector superposition built from the same kernel-resolvent data.*

Proof. Detector-label dependence is entirely encoded in the phase factor $e^{i\varphi\xi_D(e)}$. Removing that marker and regrouping by transport sectors leaves only transport-labeled coherent sums. □

9.6 Two-Channel Interference as a Minimal Example

Example 9.16 (Two-channel interference)**.** Suppose that a fixed coarse-grained result class can be realized by two competing history channels Γ_1 and Γ_2 with amplitudes

$$\mathcal{A}_1 = r_1 e^{i\phi_1}, \qquad \mathcal{A}_2 = r_2 e^{i\phi_2}.$$

Then the total amplitude is

$$\Psi = \mathcal{A}_1 + \mathcal{A}_2,$$

and the squared modulus is

$$|\Psi|^2 = r_1^2 + r_2^2 + 2r_1 r_2 \cos(\phi_1 - \phi_2).$$

This is the minimal interference law arising purely from coherent summation of histories.

Example 9.17 (Phase-mark dependence)**.** If one family of histories acquires an additional detector or environmental phase shift δ, then the interference term becomes

$$2r_1 r_2 \cos\big((\phi_1 - \phi_2) + \delta\big).$$

Thus even simple phase marks already control the passage between constructive and destructive interference. This makes phase-labeled kernels a natural prelude to the measurement and decoherence theory of the next chapter.

9.7 Connection with the Subsequent Measurement Block

The present chapter deliberately stops short of an autonomous measurement theory. It introduces the detector-mark parameter φ and record-sensitive amplitudes, but it does not yet build a full decoherence functional and probability calculus. This separation is principled. It allows one first to separate the coherent amplitude layer from the later mechanisms of record selection and classicalization.

In this way the book avoids a common logical mistake: mixing the "quantum state" with the "measurement result." In the present architecture one first has amplitudes of histories, then amplitude sectors, then a record-sensitive partition, and only after that a Born-like rule, a decoherence functional, and effective classical readout.

Remark 9.18. The Born-like probability rule is not introduced in this chapter as a derived theorem. If a minimal probability postulate is needed, it must be stated clearly only in the subsequent measurement chapter.

9.8 How This Chapter Connects to the Earlier Layers

The amplitude chapter does not break its connection either to transport mechanics or to structural action. On the contrary, it shows that the quantum-like layer naturally grows on top of already existing objects.

1. From Chapter 6 the action language is retained: the factor $e^{-\beta a(e)}$ continues to play the role of event cost.
2. From Chapter 7 the transport marking is retained: the factor $e^{i\theta\cdot\Delta x_{\mathcal{O}}(e)}$ continues to encode observer displacement.
3. From the structural core the resolvent architecture is retained: sums over histories again assemble into $(I-K)^{-1}$.

For that reason, the quantum layer here is not opposed either to the classical or to the structural layers. It merely adds complex coherence to the same kernel-resolvent machinery.

9.9 Lean Orientation for the Next Chapter

For formalization of this material in Lean, it is reasonable to divide the code into four blocks.

1. The complex-kernel layer: finite-state kernels with complex coefficients.
2. The history-amplitude layer: the multiplicative amplitude of a history and coarse-grained sums over history classes.
3. The resolvent layer: the theorem that the resolvent expansion enumerates admissible histories with complex weights.
4. The state layer: amplitude sectors, normalization, and projection to detector-marked partitions.

It is especially important that the definition of "state" in Lean depend explicitly on the chosen coarse-graining. Otherwise one gets the false impression that a state vector is an absolute object independent of the observer partition.

9.10 Chapter Audit Summary

- Definitions introduced: complex amplitude sectors, amplitude-marked kernels, coarse-grained amplitude classes, and the derived observer-level state language.

- Standing assumptions: transfer-kernel closure, observer-level access to coherent history sums, and admissible coarse-graining of amplitude sectors.
- Internally established results: the resolvent-amplitude representation and the interpretation of the quantum state as a derived organization of amplitude sectors rather than primitive ontology.
- Imported identifications used: ordinary Hilbert-space and interference language as observer-level compression of the internal amplitude layer.
- Open targets deferred: a full probability calculus and complete derivation of the later measurement block.
- Downstream dependence: the measurement, decoherence, and information chapters use this amplitude layer as their coherent precursor.

9.11 Conclusion of the Chapter

In this chapter the minimal quantum layer of structural-physical theory was built. Complex event amplitudes, amplitude-marked kernels, history amplitudes, coarse-grained amplitude classes, and the resolvent-amplitude representation were introduced. It was shown that in this architecture the quantum state is a derived object, namely a compact organization of amplitude sectors, rather than an external primitive. Finally, a minimal two-channel interference example showed that coherent phase-dependent behavior arises naturally from summation over histories. In this way the manuscript is prepared for the next step: the strict observer–measurement–decoherence block.

Chapter 10

Measurement, Record Space, and Decoherence

10.1 Why the Measurement Block Must Follow the Amplitude Chapter

The previous chapter stopped at exactly the right point: complex amplitudes of histories, amplitude sectors, and the resolvent-amplitude representation were introduced, but there was still neither a probabilistic rule, nor an autonomous theory of records, nor a mechanism of classicalization. This separation must now be preserved. Otherwise the quantum layer and the measurement layer would be logically conflated.

The present chapter constructs exactly the observer–measurement–decoherence block that already existed in the initial sketches as the OMD program. Its task is not to "derive collapse" immediately, but to define carefully the space of records, stable record sectors, the decoherence functional, and a minimal Born-like rule. Statements that are taken in the book as axioms must be marked separately and explicitly from those that still remain target theorems.

The internal logic of the chapter is as follows. First, micro-histories and stable record basins are introduced. Then channel amplitudes and detector projectors are defined. After that, the decoherence functional and a precise criterion of effective classicality are introduced. Only then are the reduced detector state, the conditional post-measurement state, and the emergent Lueders form formulated. Exactly in this order the measurement layer remains a continuation of kernel-resolvent theory rather than an external insertion.

10.2 Micro-Histories and Observer Record Labels

Definition 10.1 (Micro-history)**.** A micro-history is a finite causal-compatible chain of microstates

$$\Gamma : \quad x_0 \xrightarrow{e_1} x_1 \xrightarrow{e_2} \cdots \xrightarrow{e_N} x_N, \qquad e_k \in R.$$

An observer

$$\mathcal{O} = (\Pi_{\mathcal{O}}, \Theta_{\mathcal{O}}, \chi_{\mathcal{O}})$$

induces observer times

$$0 = t_0 < t_1 < \cdots < t_N =: T_{\mathrm{read}}$$

and coarse-grained detector labels

$$r_k(\Gamma) := \chi_D(x_k) \in \{0, 1, \ldots, M\}.$$

Here the label 0 means that no stable record has yet formed.

Definition 10.2 (Stable record basin)**.** For each outcome $m \in \{1, \ldots, M\}$ one specifies a stable basin

$$\mathcal{B}_m \subset W$$

with the properties:

1. $\mathcal{B}_m \cap \mathcal{B}_n = \varnothing$ for $m \neq n$;
2. if $x \in \mathcal{B}_m$, then $\chi_D(x) = m$;
3. small local fluctuations do not eject the state from the basin.

Definition 10.3 (Stabilization time)**.** For a history Γ and an outcome m, define the stabilization time by

$$\tau_{\mathrm{st}}^{(m)}(\Gamma) := \inf\Big\{ t_k : \ x_k \in \mathcal{B}_m \text{ and } \forall \ell \geq k, \ x_\ell \in \mathcal{B}_m \Big\}.$$

If no such k exists, set $\tau_{\mathrm{st}}^{(m)}(\Gamma) = +\infty$.

Definition 10.4 (Outcome history class)**.** The histories realizing the outcome m form the class

$$\mathcal{C}_m := \Big\{ \Gamma : \ \tau_{\mathrm{st}}^{(m)}(\Gamma) < \infty \Big\}.$$

10.3 The Axiomatic Layer of Records

One now fixes the minimal axioms of the measurement layer. These points must not be presented as derived results; their role is to make explicit exactly which block of assumptions is used later.

Definition 10.5 (Detector projectors)**.** Let

$$\mathcal{R}_{\mathcal{O}} = \{r_1, \dots, r_M\}$$

be the space of stable records of the observer. For each m introduce a projector

$$P_m$$

onto the m-th record sector, such that

$$P_m P_n = \delta_{mn} P_m, \qquad \sum_{m=1}^{M} P_m = I.$$

Definition 10.6 (Axiom M1: locality of recording)**.** There exists a distinguished subset of detector events

$$R_D \subset R$$

such that only events from R_D can change the detector label. If $e \notin R_D$, then the corresponding record mark is zero.

Definition 10.7 (Axiom M2: stability of a record)**.** If a coarse-grained state enters the basin $\mathcal{B}_m$, then sufficiently small local perturbations preserve the same record label m.

Definition 10.8 (Axiom M3: record amplification)**.** If after the first formation event a history switches between different formed records, the total action acquires an additional penalty

$$\Delta\mathcal{S}_{\text{flip}}(\Gamma) = \lambda N_{\text{flip}}(\Gamma), \qquad \lambda > 0,$$

where $N_{\text{flip}}(\Gamma)$ counts inter-record flips.

Definition 10.9 (Axiom M4: environmental dephasing)**.** The total phase of a history decomposes as

$$\Phi(\Gamma) = \Phi_{\text{sys}}(\Gamma) + \Phi_E(\Gamma),$$

and for histories ending in distinct record sectors one has a dephasing bound of the form

$$\left| \mathbb{E}\, e^{i(\Phi_E(\Gamma) - \Phi_E(\Gamma'))} \right| \leq e^{-\sigma^2 \tau_{\text{sep}}(\Gamma,\Gamma')/2}.$$

Definition 10.10 (Axiom M5: completeness of record channels)**.** The projectors $\{P_m\}$ form a complete coarse-grained decomposition of the detector record space.

10.4 Channel Amplitudes and Decomposition of the Total State

Definition 10.11 (Channel amplitude)**.** The amplitude of the m-th outcome is defined by

$$|\Psi_m\rangle := \sum_{\Gamma \in \mathcal{C}_m} \mathcal{A}(\Gamma)|r_m\rangle.$$

Proposition 10.12 (Channel decomposition)**.** *The full amplitude state decomposes as*

$$|\Psi\rangle = \sum_{m=1}^{M} |\Psi_m\rangle, \qquad |\Psi_m\rangle = P_m|\Psi\rangle.$$

Proof. The history classes $\mathcal{C}_m$ partition the set of histories according to the final stable record. Since the projectors P_m form a complete decomposition, projection onto the m-th record sector extracts exactly the corresponding channel amplitude. □

Definition 10.13 (Total action with record penalty)**.** The measurement-aware total action is defined by

$$\mathcal{S}_{\text{tot}}(\Gamma) := \mathcal{S}(\Gamma) + \lambda N_{\text{flip}}(\Gamma).$$

Definition 10.14 (Complex history amplitude with record structure)**.** The history amplitude entering the measurement block has the form

$$\mathcal{A}(\Gamma) = \mathfrak{a}(\Gamma) e^{-\beta \mathcal{S}_{\text{tot}}(\Gamma)} e^{i\Phi_{\text{sys}}(\Gamma)} e^{i\Phi_E(\Gamma)}, \qquad |\mathfrak{a}(\Gamma)| \leq 1.$$

10.5 The Decoherence Functional and the Classicality Criterion

Definition 10.15 (Decoherence functional)**.** For two record sectors m, n define

$$D_{mn} := \langle \Psi_m, \Psi_n \rangle.$$

Definition 10.16 (Born-like rule)**.** At this stage of the book, probabilities of readout outcomes are introduced by the minimal rule

$$p_m := \frac{|\Psi_m|^2}{\sum_{n=1}^{M} |\Psi_n|^2}.$$

Remark 10.17. The Born-like rule in this chapter should be treated as an explicit minimal axiom of probabilistic interpretation, not as an already derived theorem. This distinction must be preserved later as well.

Proposition 10.18 (Total probability with interference)**.** *For any union of outcomes $A \subseteq \{1, \dots, M\}$ one has*

$$\left| \sum_{m \in A} \Psi_m \right|^2 = \sum_{m \in A} |\Psi_m|^2 + \sum_{\substack{m,n \in A \\ m \neq n}} D_{mn}.$$

Proof. This is the direct expansion of the squared norm of a sum of channel amplitudes. □

Corollary 10.19 (Classical readout regime)**.** *If the off-diagonal elements satisfy*

$$D_{mn} = 0, \qquad m \neq n,$$

then probabilities become classically additive over mutually exclusive record sectors.

Proof. When all off-diagonal contributions vanish, the interference correction vanishes identically. □

Definition 10.20 (Effective classicality condition)**.** The observer is in a classical readout regime if

$$|D_{mn}| \ll |\Psi_m|\,|\Psi_n|, \qquad m \neq n.$$

10.6 Weak and Strong Decoherence

Lemma 10.21 (Weak off-diagonal bound)**.** *If the channel partition sums are finite and the environmental dephasing bound holds, then there exists a constant $C_{mn} > 0$ such that*

$$|D_{mn}| \leq C_{mn} e^{-\sigma^2 \tau_{mn}^{\min}/2}, \qquad m \neq n,$$

where $\tau_{mn}^{\min}$ is the minimal separation time between histories contributing to channels m and n.

Proof. Expand D_{mn} as a double sum over histories in $\mathcal{C}_m$ and $\mathcal{C}_n$, bound each term by the modulus of the amplitude, and use the environmental dephasing estimate. The remaining absolute sums are finite by hypothesis and are absorbed into C_{mn}. □

Theorem 10.22 (Weak decoherence theorem)**.** *Under axioms M1–M5 and finite channel partition sums, off-diagonal decoherence terms satisfy an exponential suppression bound of the form*

$$|D_{mn}| \leq C_{mn} e^{-\sigma^2 \tau_{mn}^{\min}/2}, \qquad m \neq n.$$

Proof. This is the previous lemma together with the decomposition of amplitudes into record channels. □

Definition 10.23 (Strong amplification hypothesis)**.** Assume that for histories with unstable record switching the number of flips grows at least linearly with instability duration:

$$N_{\mathrm{flip}}(\Gamma) \geq \nu\, \tau_{\mathrm{unst}}(\Gamma), \qquad \nu > 0.$$

Theorem 10.24 (Target theorem: Strong decoherence theorem)**.** *If, in addition to the weak decoherence hypotheses, the strong amplification hypothesis holds, then there exists a constant* $\widetilde{C}_{mn} > 0$ *such that*

$$|D_{mn}(\tau)| \leq \widetilde{C}_{mn} e^{-(\beta\lambda\nu + \sigma^2/2)\tau}, \qquad m \neq n.$$

Remark 10.25. This theorem should still be regarded as a target theorem unless all microscopic assumptions needed to justify the linear growth of flip count are proved inside the theory. The statement is included here because it is the natural quantitative goal of the OMD program.

10.7 Reduced States and the Emergent Classical Detector

Definition 10.26 (Normalized total state)**.** The normalized total post-amplitude state is

$$\rho := \frac{|\Psi\rangle\langle\Psi|}{\langle\Psi, \Psi\rangle}.$$

Definition 10.27 (Detector reduced state)**.** The detector reduced state is defined by

$$\rho_D := \mathrm{Tr}_{S,E}\rho.$$

Proposition 10.28 (Detector state formula)**.** *Writing*

$$|\Psi\rangle = \sum_{m=1}^{M} |\Psi_m\rangle,$$

the reduced detector state takes the form

$$\rho_D = \frac{\sum_m |\Psi_m|^2 |r_m\rangle\langle r_m| + \sum_{m\neq n} D_{mn} |r_m\rangle\langle r_n|}{\sum_k |\Psi_k|^2 + \sum_{i\neq j} D_{ij}}.$$

Proof. Expand $|\Psi\rangle\langle\Psi|$ in the record basis and then trace out the system and environment degrees of freedom. □

Corollary 10.29 (Diagonal detector limit)**.** *If the effective classicality condition holds,*

then

$$\rho_D \approx \sum_{m=1}^{M} p_m |r_m\rangle\langle r_m|.$$

Proof. The off-diagonal terms are negligible compared to the diagonal weights and therefore disappear in the coarse-grained limit. □

Definition 10.30 (Conditional post-measurement state)**.** Conditioned on outcome m, the measured subsystem is assigned the state

$$\rho_{S|m} := \frac{\mathrm{Tr}_{D,E}(P_m \rho P_m)}{\mathrm{Tr}(P_m \rho)}.$$

Proposition 10.31 (Emergent Lueders form)**.** *In the decoherence regime the update*

$$\rho \longmapsto \frac{P_m \rho P_m}{\mathrm{Tr}(P_m \rho)}$$

arises as the effective conditional state-update rule.

Remark 10.32. This proposition should be read as derived effective dynamics in the decoherence regime, not as a primitive collapse postulate.

10.8 Examples

Example 10.33 (Two stable record sectors)**.** Suppose the detector has two stable basins $\mathcal{B}_1$ and $\mathcal{B}_2$. Histories leading to these basins define channel amplitudes Ψ_1 and Ψ_2. Then

$$D_{12} = \langle \Psi_1, \Psi_2 \rangle$$

measures the residual coherence between the two readout outcomes. If detector amplification is weak, this term may remain visible; if amplification and dephasing are strong, it becomes exponentially suppressed.

Example 10.34 (Record stabilization versus flip penalty)**.** Suppose a history enters the basin $\mathcal{B}_1$ but later leaves it and switches to $\mathcal{B}_2$. Such a branch carries a positive flip count and therefore an additional penalty $\lambda N_{\mathrm{flip}}(\Gamma)$. In this sense stable readout is not only environmentally preferred but also action-favored inside structural theory itself.

10.9 How This Chapter Prepares the Information Layer

This chapter stops at the first point where information-theoretic language becomes mathematically legitimate. Once stable record channels, reduced detector states, and an effective classical readout regime are available, entropy and mutual information can be defined without ambiguity. Before that point they would be only heuristic labels.

10.10 Lean Orientation for the Next Chapter

For Lean formalization, the OMD block should be split into four namespaces.

1. Record geometry: basins, stabilization time, outcome history classes.
2. Channel amplitudes: projectors, channel decomposition, decoherence functional.
3. Reduced states: detector state, conditional state, diagonal detector limit.
4. Target theorems: weak and strong decoherence, emergent Lueders form.

It is especially important that the Born-like rule and the strong decoherence statements be marked with explicit theorem-status metadata in comments or naming conventions. Otherwise the formal library will quickly lose the distinction between assumptions and proved consequences.

10.11 Conclusion of the Chapter

The observer–measurement–decoherence block is now in place: stable record sectors, channel amplitudes, decoherence bounds, reduced detector states, and the effective Lueders update have all been formulated in one consistent order. The next chapter can therefore work directly with entropy and reduced-state functionals instead of returning to the measurement axioms themselves.

Chapter 11

Information, Entropy, and Density Operators

11.1 Why the Informational Layer Must Follow Measurement

The previous chapter introduced record sectors, detector projectors, channel amplitudes, the decoherence functional, and the effective classical-readout regime. That was the correct stopping point for the measurement block, because information-theoretic quantities cannot be introduced responsibly before one has specified what counts as a subsystem, what counts as a stable record, and under which conditions amplitudes can be interpreted through effectively classical outcome weights. Without that preparation, entropy language would float free of the structural and observer-level constructions developed earlier.

This chapter completes the first informational layer of the book. Its task is fourfold. First, it introduces the density-operator language as a derived reorganization of the amplitude description rather than as a new primitive ontology. Second, it defines reduced states of detector, system, and environment sectors, thereby giving formal meaning to the notion of accessible and inaccessible information. Third, it introduces record entropy, structural entropy, subsystem entropy, and mutual information. Fourth, it formulates the basic positivity, normalization, and monotonicity facts that will later be required when statistical mechanics and thermodynamics are built over history ensembles.

The chapter remains deliberately conservative. No thermodynamic interpretation is yet imposed. In particular, entropy here is still an information-theoretic and structural invariant attached to a chosen coarse partition or reduced state. Temperature, free energy, and entropy production belong to the next chapter, where positive weights on ensembles of histories are introduced explicitly.

11.2 Normalized Total State and Density Operators

Definition 11.1 (Normalized amplitude state)**.** Let $|\Psi\rangle$ be the total amplitude state constructed from coarse-grained history classes. If

$$0 < \langle \Psi, \Psi \rangle < \infty,$$

the associated normalized amplitude state is

$$|\widehat{\Psi}\rangle := \frac{|\Psi\rangle}{\sqrt{\langle \Psi, \Psi \rangle}}.$$

Definition 11.2 (Density operator)**.** The density operator associated with the total amplitude state is

$$\rho := |\widehat{\Psi}\rangle\langle\widehat{\Psi}| = \frac{|\Psi\rangle\langle\Psi|}{\langle \Psi, \Psi \rangle}.$$

Proposition 11.3 (Positivity and normalization)**.** *The operator ρ is positive and has unit trace:*

$$\rho \geq 0, \qquad \mathrm{Tr}(\rho) = 1.$$

Proof. For any vector $|\varphi\rangle$ one has

$$\langle \varphi, \rho\varphi \rangle = |\langle \widehat{\Psi}, \varphi \rangle|^2 \geq 0,$$

so ρ is positive. Since $|\widehat{\Psi}\rangle$ is normalized, its rank-one projector has trace one. □

Remark 11.4. At this stage the density operator is not a new fundamental object. It is a derived reorganization of the amplitude-level description in a form suited for subsystem reduction, entropy functionals, and later coarse-grained statistical constructions.

11.3 Subsystem Splittings and Reduced States

Definition 11.5 (Admissible subsystem factorization)**.** An admissible subsystem factorization is a decomposition of the relevant observer-level Hilbert space into a tensor product

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_D \otimes \mathcal{H}_E,$$

where S denotes the measured system, D the detector-record sector, and E the inaccessible or environmental sector. Such a factorization is admissible if the detector projectors P_m from the previous chapter act nontrivially only on $\mathcal{H}_D$.

Definition 11.6 (Reduced state)**.** For an admissible subsystem factorization, the reduced

state of a subsystem A is obtained by tracing out its complement:

$$\rho_A := \mathrm{Tr}_{A^c}(\rho).$$

In particular,

$$\rho_D := \mathrm{Tr}_{S,E}(\rho), \qquad \rho_S := \mathrm{Tr}_{D,E}(\rho), \qquad \rho_E := \mathrm{Tr}_{S,D}(\rho).$$

Proposition 11.7 (Reduced states are normalized positive operators)**.** *For every subsystem A, the reduced state ρ_A is positive and has unit trace.*

Proof. Partial trace preserves positivity and total trace. Therefore every ρ_A inherits these properties from ρ. □

Definition 11.8 (Detector basis)**.** Let

$$\{|r_m\rangle\}_{m=1}^{M}$$

be the stable detector basis associated with the record projectors $P_m = |r_m\rangle\langle r_m|$ of the classical readout regime.

Proposition 11.9 (Explicit detector reduced state)**.** *If the total amplitude decomposes into channel amplitudes*

$$|\Psi\rangle = \sum_{m=1}^{M} |\Psi_m\rangle, \qquad |\Psi_m\rangle = P_m|\Psi\rangle,$$

then the detector reduced state has the form

$$\rho_D = \frac{\sum_{m=1}^{M} |\Psi_m|^2 |r_m\rangle\langle r_m| + \sum_{m\neq n} D_{mn}|r_m\rangle\langle r_n|}{\sum_{k=1}^{M} |\Psi_k|^2 + \sum_{i\neq j} D_{ij}},$$

where $D_{mn} = \langle \Psi_m, \Psi_n\rangle$ is the decoherence functional introduced in the previous chapter.

Proof. Expand $\rho = |\Psi\rangle\langle\Psi|/\langle\Psi,\Psi\rangle$ in the channel decomposition and trace out the system and environmental factors. The diagonal terms contribute the classical record weights, and the off-diagonal terms contribute the decoherence matrix elements D_{mn}. □

Corollary 11.10 (Diagonal detector limit)**.** *If*

$$|D_{mn}| \ll |\Psi_m|\,|\Psi_n|, \qquad m \neq n,$$

then

$$\rho_D \approx \sum_{m=1}^{M} p_m |r_m\rangle\langle r_m|, \qquad p_m := \frac{|\Psi_m|^2}{\sum_{n=1}^{M} |\Psi_n|^2}.$$

Proof. In the effective classicality regime the off-diagonal entries are negligible relative to the diagonal weights, so the reduced detector state becomes approximately diagonal in the record basis. □

11.4 Conditional States and Effective State Update

Definition 11.11 (Conditional post-measurement state)**.** For a fixed outcome m, the conditional post-measurement state of the measured subsystem is

$$\rho_{S|m} := \frac{\mathrm{Tr}_{D,E}(P_m \rho P_m)}{\mathrm{Tr}(P_m \rho)}.$$

The unconditional subsystem state remains

$$\rho_S = \mathrm{Tr}_{D,E}(\rho).$$

Proposition 11.12 (Effective Lueders update in the decoherence regime)**.** *Assume that the detector basis is stable and the off-diagonal decoherence terms are negligible. Then the observer-level update associated with a readout m takes the effective form*

$$\rho \longmapsto \frac{P_m \rho P_m}{\mathrm{Tr}(P_m \rho)}.$$

Proof. In the strong-record regime, histories that end in distinct detector basins do not interfere at observable scale. Conditioning on the readout therefore amounts to restricting the state to the corresponding detector sector and renormalizing. □

Remark 11.13. This statement remains an effective theorem of the reduced observer description, not a fundamental collapse postulate. The underlying structural dynamics still resides at the level of histories, actions, amplitudes, and environmental dephasing.

11.5 Record Entropy and Structural Entropy

Definition 11.14 (Record entropy)**.** If the detector is in the classical readout regime with outcome probabilities $\{p_m\}_{m=1}^{M}$, define the record entropy by

$$S_{\mathrm{rec}} := -\sum_{m=1}^{M} p_m \log p_m.$$

Definition 11.15 (Structural partition)**.** A structural partition is a finite or countable family

$$\mathcal{P} = \{\mathcal{C}_\alpha\}_{\alpha \in A}$$

of pairwise disjoint history classes or assembly sectors whose union covers the domain under study.

Definition 11.16 (Structural entropy)**.** Let $q_\alpha \geq 0$ with $\sum_{\alpha \in A} q_\alpha = 1$ be the normalized weights of a chosen structural partition $\mathcal{P}$. The structural entropy relative to $\mathcal{P}$ is

$$S_{\mathrm{str}}(\mathcal{P}) := -\sum_{\alpha \in A} q_\alpha \log q_\alpha.$$

Remark 11.17. Structural entropy is partition-dependent. This dependence is not a defect: it records precisely which level of coarse-graining is being used. Later thermodynamic entropy will arise only after the relevant ensemble and admissible coarse level have been fixed.

Example 11.18 (Pure record)**.** If one outcome occurs with probability one, say $p_1 = 1$ and $p_m = 0$ for $m > 1$, then

$$S_{\mathrm{rec}} = 0.$$

This is the minimal entropy case: the observer has a perfectly definite record.

Example 11.19 (Equiprobable readout)**.** If all M outcomes occur with equal probability $p_m = 1/M$, then

$$S_{\mathrm{rec}} = \log M.$$

This is the maximal record entropy for a detector with M readout channels.

11.6 Von Neumann Entropy and Basic Identities

Definition 11.20 (Von Neumann entropy)**.** For any reduced state ρ_A, define

$$S(\rho_A) := -\operatorname{Tr}(\rho_A \log \rho_A).$$

In particular,

$$S_S := S(\rho_S), \qquad S_D := S(\rho_D), \qquad S_E := S(\rho_E).$$

Proposition 11.21 (Classical detector entropy identity)**.** *If the detector reduced state is diagonal in the stable record basis,*

$$\rho_D = \sum_{m=1}^{M} p_m |r_m\rangle\langle r_m|,$$

then

$$S_D = S_{\mathrm{rec}} = -\sum_{m=1}^{M} p_m \log p_m.$$

Proof. For a diagonal density matrix the von Neumann entropy is the Shannon entropy of its eigenvalue list. Here the eigenvalues are exactly the outcome probabilities p_m. □

Lemma 11.22 (Entropy is nonnegative)**.** *For every reduced state ρ_A one has*

$$S(\rho_A) \geq 0.$$

Proof. The eigenvalues of a density operator lie in $[0, 1]$ and sum to one. Therefore the sum $-\sum_j \lambda_j \log \lambda_j$ is nonnegative. □

11.7 Mutual Information and Correlation Structure

Definition 11.23 (Mutual information)**.** For two subsystems A and B, define

$$I(A : B) := S(\rho_A) + S(\rho_B) - S(\rho_{AB}).$$

In particular,

$$I(S : D) = S(\rho_S) + S(\rho_D) - S(\rho_{SD}).$$

Proposition 11.24 (Nonnegativity of mutual information)**.** *For every bipartite reduced state ρ_{AB} one has*

$$I(A : B) \geq 0.$$

Proof. This is the standard subadditivity consequence for von Neumann entropy. In the present book it functions as the minimal consistency statement that the information shared between two subsystems cannot be negative. □

Theorem 11.25 (Entanglement entropy identity for pure global state)**.** *If the total state ρ is pure, then complementary reduced entropies coincide:*

$$S(\rho_D) = S(\rho_{SE}), \qquad S(\rho_S) = S(\rho_{DE}), \qquad S(\rho_E) = S(\rho_{SD}).$$

Proof. For a pure global state, the nonzero spectra of reduced density operators of complementary subsystems coincide. The entropy identities follow immediately. □

Example 11.26 (Simple bipartite decomposition)**.** Consider the pure two-sector state

$$|\widehat{\Psi}\rangle = \sqrt{p}\,|0\rangle_S \otimes |0\rangle_D + \sqrt{1-p}\,|1\rangle_S \otimes |1\rangle_D, \qquad 0 < p < 1.$$

Then

$$\rho_S = p|0\rangle\langle 0| + (1-p)|1\rangle\langle 1|,$$

and likewise for ρ_D. Therefore

$$S(\rho_S) = S(\rho_D) = -p \log p - (1-p) \log(1-p),$$

while

$$I(S:D) = 2[-p\log p - (1-p)\log(1-p)].$$

This is the minimal explicit model in which measurement-generated correlation becomes visible at the level of reduced states.

11.8 Information Gain and Coarse-Grained Monotonicity

Definition 11.27 (Information gain of measurement)**.** The information gain of measurement is defined by

$$\Delta I_{\text{meas}} := I(S:D)_{\text{after}} - I(S:D)_{\text{before}}.$$

Proposition 11.28 (Positive information gain in a successful readout)**.** *Suppose that before interaction the system and detector are uncorrelated,*

$$\rho_{SD}^{\text{before}} = \rho_S^{\text{before}} \otimes \rho_D^{\text{before}},$$

and after interaction the detector enters a stable classical readout regime with nontrivial correlation to the measured sector. Then

$$\Delta I_{\text{meas}} > 0.$$

Proof. Initially, the product form implies $I(S:D)_{\text{before}} = 0$. A successful readout creates nontrivial correlation between detector records and system sectors, hence $I(S:D)_{\text{after}} > 0$. □

Remark 11.29 (Failure mode)**.** If reduced states depend too strongly on an arbitrarily chosen cut between accessible and inaccessible sectors, then the information-theoretic language loses physical meaning. For that reason, every later use of entropy or mutual information must specify the structural basis of the subsystem split and the observer access assumptions under which the reduction is performed.

11.9 Lean Orientation for the Next Chapter

For later formalization, the chapter should be split into four layers.

1. A basic file defining normalized states, density operators, and partial trace on admissible tensor factorizations.

2. A subsystem file proving positivity, normalization, and the explicit reduced-state formulas for detector and measured subsystems.

3. An entropy file defining Shannon-type and von-Neumann-type entropy functionals, together with the nonnegativity and diagonalization lemmas.

4. An information file defining mutual information and measurement information gain, with separate lemmas for product states, classical readout states, and pure-state bipartite symmetry.

This separation matters. The partial trace and density-operator infrastructure belong to the general mathematical library of the project, whereas record entropy and measurement information gain depend on the specific observer-measurement architecture built in the preceding chapters.

11.10 Conclusion of the Chapter

The first observer-accessible informational layer is complete. Density operators, reduced states, entropy functionals, and mutual information now exist as derived constructions over the amplitude and measurement blocks. The next chapter will not repeat this information theory; it will use it as input when building ensembles of histories, partition resolvents, and coarse-grained thermodynamic quantities.

Chapter 12

Partition Resolvent, Statistical Mechanics, and Thermodynamics

12.1 Why Statistical Mechanics Must Be Built on Histories Rather Than Ready-Made Phase Space

The previous chapter completed the first informational layer of the theory. The manuscript now has density operators, reduced states, record entropy, structural entropy, and mutual information. That does not yet amount to statistical mechanics. Entropy has been defined so far either for reduced observer-accessible states or for explicitly chosen structural partitions. To pass from these quantities to thermodynamics, the book must now introduce genuine ensembles.

Within the present program, such ensembles cannot be postulated on an external continuous phase space. The entire architecture of the book requires that coarse-grained thermodynamics arise from weighted families of structural histories. For that reason the basic object of the chapter is not a Gibbs distribution on an a priori phase manifold, but a partition sum over admissible histories or over finitely factorized assembly sectors. The action language from the dynamical chapters and the resolvent language from the structural chapters must therefore meet in one place.

The goal of the chapter is fourfold. First, it defines microcanonical and canonical ensembles of histories. Second, it rewrites the canonical partition function in kernel-resolvent form. Third, it introduces thermodynamic potentials and fluctuation formulas. Fourth, it records exactly which thermodynamic statements are formal identities of the ensemble and which remain regime-dependent target theorems.

12.2 Admissible Histories and History Observables

Definition 12.1 (Admissible history space)**.** Let

$$\mathfrak{H}$$

denote the class of all admissible observer-compatible histories

$$\Gamma: \quad x_0 \xrightarrow{e_1} x_1 \xrightarrow{e_2} \cdots \xrightarrow{e_N} x_N, \qquad e_k \in R,$$

such that causal compatibility with the observer structure and coarse-grained record evolution are well defined.

Definition 12.2 (Total history cost)**.** The total history cost entering the statistical layer is

$$\mathcal{S}_{\mathrm{tot}}(\Gamma) := \mathcal{S}(\Gamma) + \lambda N_{\mathrm{flip}}(\Gamma), \qquad \lambda > 0,$$

where $\mathcal{S}(\Gamma)$ is the structural action and $N_{\mathrm{flip}}(\Gamma)$ is the inter-record flip count introduced in the measurement chapter.

Remark 12.3. The statistical layer is therefore built not on a purely kinematic counting of histories but on the same physically relevant quantity that already governed transport, measurement stability, and decoherence.

Definition 12.4 (Additive history observable)**.** An additive history observable is a map

$$\Xi_a : \mathfrak{H} \to \mathbb{R}$$

such that for every concatenation-compatible decomposition of a history into fragments one has additive composition of the corresponding observable values. Typical examples are observer time, total displacement, EFU count, shell complexity, and detector record count.

12.3 Microcanonical and Canonical Ensembles of Histories

Definition 12.5 (Microcanonical history sector)**.** For fixed values S_0 and $\Xi^0 = (\Xi_1^0, \ldots, \Xi_r^0)$, define the microcanonical history sector by

$$\mathfrak{H}_{\mathrm{mc}}(S_0, \Xi^0) := \big\{\Gamma \in \mathfrak{H} : \ \mathcal{S}_{\mathrm{tot}}(\Gamma) = S_0, \ \Xi_a(\Gamma) = \Xi_a^0 \ \forall a\big\}.$$

Definition 12.6 (History density of states)**.** The density of states of the microcanonical sector is

$$\Omega(S_0, \Xi^0) := \#\mathfrak{H}_{\mathrm{mc}}(S_0, \Xi^0),$$

or its measure-valued analogue when the coarse variables are effectively continuous.

Definition 12.7 (Microcanonical entropy)**.** The microcanonical entropy is

$$S_{\mathrm{mc}}(S_0, \Xi^0) := \log \Omega(S_0, \Xi^0).$$

Definition 12.8 (Canonical history weight)**.** Let β and $h = (h_1, \dots, h_r)$ be conjugate parameters to the total history cost and the additive observables. The canonical weight of a history is

$$w_{\beta,h}(\Gamma) := \exp\Big(-\beta \mathcal{S}_{\mathrm{tot}}(\Gamma) + \sum_{a=1}^{r} h_a \Xi_a(\Gamma)\Big).$$

Definition 12.9 (Canonical partition function)**.** The canonical partition function of the history ensemble is

$$Z(\beta, h) := \sum_{\Gamma \in \mathfrak{H}} w_{\beta,h}(\Gamma).$$

Whenever this sum is finite, the corresponding canonical probability measure is

$$\mathbb{P}_{\beta,h}(\Gamma) := \frac{w_{\beta,h}(\Gamma)}{Z(\beta, h)}.$$

12.4 Statistical Transfer Kernel and Partition Resolvent

Definition 12.10 (Statistical transfer kernel)**.** Suppose each elementary event e carries a local action cost $a(e)$ and local additive marks $\xi_a(e)$ such that

$$\Xi_a(\Gamma) = \sum_{e \in \Gamma} \xi_a(e).$$

Then the statistical transfer kernel is

$$K_{\mathcal{O}}^{\mathrm{stat}}(\beta, h) := \sum_{e} \exp\Big(-\beta a(e) + \sum_{a=1}^{r} h_a \xi_a(e)\Big)[e].$$

Remark 12.11. This is the exact point at which statistical mechanics becomes internal to the structural core. The same kernel-resolvent architecture used for transfer, amplitude, and marked transport is now used to build thermodynamic quantities.

Definition 12.12 (Partition resolvent)**.** If the history system admits a finite-state transfer factorization, define the partition resolvent by

$$Z(\beta, h) = \iota^{\top} \big(I - K_{\mathcal{O}}^{\mathrm{stat}}(\beta, h)\big)^{-1} \omega.$$

Proposition 12.13 (Equivalence of path sum and resolvent form)**.** *If admissible histories are enumerated by the finite-state transfer factorization generated by $K_{\mathcal{O}}^{\mathrm{stat}}$, then the*

canonical partition function agrees with the partition resolvent:

$$\sum_{\Gamma\in\mathfrak{H}} \exp\Big(-\beta\mathcal{S}_{\mathrm{tot}}(\Gamma) + \sum_{a=1}^{r} h_a \Xi_a(\Gamma)\Big) = \iota^{\top}\big(I - K_{\mathcal{O}}^{\mathrm{stat}}(\beta,h)\big)^{-1}\omega.$$

Proof. Expand the resolvent as a geometric series. Each power of the statistical kernel lists admissible concatenations of local events, and the exponential event factors multiply into the total cost and total additive observables of the resulting history. □

12.5 Thermodynamic Potentials of the History Ensemble

Definition 12.14 (Free energy)**.** The free energy of the history ensemble is

$$F(\beta,h) := -\frac{1}{\beta}\log Z(\beta,h).$$

Definition 12.15 (Internal action-energy)**.** The internal action-energy is

$$U(\beta,h) := -\partial_\beta \log Z(\beta,h).$$

Definition 12.16 (Generalized expectation values)**.** For each additive observable Ξ_a, define

$$\langle \Xi_a \rangle_{\beta,h} := \partial_{h_a} \log Z(\beta,h).$$

Definition 12.17 (Heat capacity and susceptibility matrix)**.** Define

$$C(\beta,h) := \beta^2 \partial_\beta^2 \log Z(\beta,h)$$

and

$$\chi_{ab}(\beta,h) := \partial_{h_a}\partial_{h_b} \log Z(\beta,h).$$

Definition 12.18 (Thermodynamic entropy)**.** The thermodynamic entropy of the history ensemble is

$$S_{\mathrm{th}}(\beta,h) := \beta U(\beta,h) + \log Z(\beta,h) - \sum_{a=1}^{r} h_a \langle \Xi_a \rangle_{\beta,h}.$$

Proposition 12.19 (Fluctuation identities)**.** *Whenever differentiation under the partition sum is justified,*

$$U = \langle \mathcal{S}_{\mathrm{tot}} \rangle_{\beta,h}, \qquad \partial_\beta^2 \log Z = \mathrm{Var}_{\beta,h}(\mathcal{S}_{\mathrm{tot}}),$$

and

$$\chi_{ab} = \mathrm{Cov}_{\beta,h}(\Xi_a, \Xi_b).$$

Proof. These are direct derivative identities for $\log Z$. Differentiation with respect to β produces insertions of $\mathcal{S}_{\mathrm{tot}}$, and differentiation with respect to the source parameters h_a produces insertions of the additive observables. □

Corollary 12.20 (Nonnegativity of heat capacity and susceptibility)**.** *Whenever the canonical ensemble is well defined,*

$$C(\beta, h) \geq 0,$$

and the matrix $\chi_{ab}(\beta, h)$ is positive semidefinite.

Proof. Heat capacity is proportional to a variance, and the susceptibility matrix is a covariance matrix. □

Proposition 12.21 (Legendre relation)**.** *The thermodynamic potentials satisfy*

$$F = U - \frac{1}{\beta} S_{\mathrm{th}} - \frac{1}{\beta} \sum_{a=1}^{r} h_a \langle \Xi_a \rangle.$$

Proof. Substitute the definition of S_{th} and solve for F. □

12.6 Examples

Example 12.22 (Minimal finite-state history system)**.** Assume there are only two admissible history classes Γ_1 and Γ_2 with costs S_1, S_2 and one additive observable values X_1, X_2. Then

$$Z(\beta, h) = e^{-\beta S_1 + hX_1} + e^{-\beta S_2 + hX_2}.$$

Hence

$$U = \frac{S_1 e^{-\beta S_1 + hX_1} + S_2 e^{-\beta S_2 + hX_2}}{e^{-\beta S_1 + hX_1} + e^{-\beta S_2 + hX_2}},$$

and

$$\langle X \rangle = \frac{X_1 e^{-\beta S_1 + hX_1} + X_2 e^{-\beta S_2 + hX_2}}{e^{-\beta S_1 + hX_1} + e^{-\beta S_2 + hX_2}}.$$

This is the simplest explicit model in which the formal definitions of the chapter become computable.

Example 12.23 (Detector readout as a coarse-grained subsystem)**.** Suppose a detector has classicalized outcome probabilities p_m and that each outcome sector contains many microscopic histories. Then record entropy

$$S_{\mathrm{rec}} = -\sum_m p_m \log p_m$$

and thermodynamic entropy S_{th} need not coincide. They coincide only under additional assumptions relating the chosen coarse record partition to the dominant macrosector decomposition of the history ensemble.

12.7 Microcanonical–Canonical Bridge and the Role of Temperature

Proposition 12.24 (Canonical–microcanonical bridge)**.** *Assume there exists a large-history limit with concentration on one dominant macrosector. Then, at the corresponding saddle point,*

$$S_{\mathrm{mc}}(S_0, \Xi^0) \approx S_{\mathrm{th}}(\beta, h)$$

for parameters (β, h) *satisfying*

$$U(\beta, h) = S_0, \qquad \langle \Xi_a \rangle_{\beta,h} = \Xi_a^0.$$

Remark 12.25. This is an ensemble-equivalence statement on spaces of histories, not on pre-given microstates. For that reason it must remain marked as a regime-dependent proposition rather than a universally closed theorem.

Definition 12.26 (History temperature)**.** Whenever the microcanonical entropy is differentiable with respect to the total history cost, define

$$\beta = \frac{\partial S_{\mathrm{mc}}}{\partial S_0}.$$

Thus temperature is introduced as the conjugate parameter to structural action-cost at the coarse-grained ensemble level.

12.8 Open Systems, Irreversibility, and Entropy Production

Definition 12.27 (Environment cost term)**.** Suppose the total history cost admits a coarse decomposition

$$\mathcal{S}_{\mathrm{tot}} = \mathcal{S}_S + \mathcal{S}_D + Q_E,$$

where $Q_E(\Gamma)$ measures the environment-coupled remainder not retained in the explicit system and detector sectors.

Definition 12.28 (Entropy production)**.** For an open observer-dependent subsystem,

define the coarse entropy production by

$$\Sigma := \Delta S_D + \beta \langle Q_E \rangle.$$

Theorem 12.29 (Target theorem: Nonnegativity of entropy production)**.** *Under Markovian coarse-graining, stable record basins, and sufficiently strong environmental decoherence, one expects*

$$\Sigma \geq 0.$$

Remark 12.30. This statement is intentionally left as a target theorem. It requires a monotonicity argument for the chosen coarse dynamics and cannot be presented honestly as already established merely from the present definitions.

Proposition 12.31 (First-law form for controlled history ensembles)**.** *Suppose external control parameters $\lambda^1, \dots, \lambda^k$ enter the local event costs. Then the differential of the internal action-energy takes the form*

$$dU = \delta Q + \sum_{i=1}^{k} Y_i \, d\lambda^i, \qquad Y_i := \left\langle \frac{\partial \mathcal{S}_{\mathrm{tot}}}{\partial \lambda^i} \right\rangle.$$

Remark 12.32. This is the history-level analogue of the first law: change of average action-cost splits into a heat-like part and a controlled work-like part. Its physical interpretation remains observer-level and coarse-grained.

12.9 Large Deviations and Preparation for Transport Theory

Definition 12.33 (Large-deviation functional)**.** Let $\Xi(\Gamma)$ be an extensive observable and let $T_{\mathcal{O}}(\Gamma)$ be observer time. If

$$\mathbb{P}_\beta\left(\frac{\Xi(\Gamma)}{T_{\mathcal{O}}(\Gamma)} \approx \xi \right) \asymp e^{-T_{\mathcal{O}} I(\xi)},$$

then $I(\xi)$ is the large-deviation rate function of the normalized observable.

Proposition 12.34 (Generating functional and rate function)**.** *If the scaled cumulant generating function*

$$\Lambda(h) := \lim_{T \to \infty} \frac{1}{T} \log Z(\beta, h)$$

exists and is differentiable, then the large-deviation rate function is given by the Legendre transform

$$I(\xi) = \sup_h \big(h\xi - \Lambda(h) \big).$$

Definition 12.35 (History current and effective diffusion coefficient)**.** For a transport observable $\Xi_Q(\Gamma)$ define the average current by

$$J_Q := \frac{\langle \Xi_Q \rangle}{\langle T_{\mathcal{O}} \rangle}.$$

If $X_{\mathcal{O}}(\Gamma)$ is total observer displacement, define the effective diffusion coefficient by

$$D_Q := \lim_{T \to \infty} \frac{\mathrm{Var}(X_Q)}{2T_{\mathcal{O}}}.$$

Theorem 12.36 (Target theorem: Green–Kubo-type history formula)**.** *Under stationarity and sufficient mixing, one expects a correlation formula of the form*

$$D_Q = \int_0^\infty \langle J_Q(0) J_Q(t) \rangle \, dt.$$

12.10 Chapter Audit Summary

- Definitions introduced: partition resolvents, history ensembles, thermodynamic potentials, history temperature, and open-system entropy-production quantities.
- Standing assumptions: coarse ensemble closure on spaces of histories, differentiability of thermodynamic potentials, and the saddle-point regime used for the ensemble bridge.
- Internally established results: the partition-resolvent formalism, Legendre relations, and the distinction between record entropy and thermodynamic entropy.
- Imported identifications used: ordinary statistical-mechanical language for canonical, microcanonical, and transport-preparation regimes.
- Open targets deferred: complete transport closure and fully proved Green–Kubo-type formulas beyond the current target-theorem status.
- Downstream dependence: the continuum transport and hydrodynamic chapters build on this thermodynamic layer.

12.11 Lean Orientation for the Next Chapter

For formalization, the statistical-mechanical layer should be separated into five files.

1. A history-ensemble file defining admissible histories, total history cost, additive observables, and canonical weights.

2. A partition-resolvent file proving equality between the path-sum and resolvent formulations under finite-state factorization assumptions.

3. A thermodynamic-potentials file defining F, U, S_{th}, susceptibilities, and fluctuation identities.

4. An open-systems file defining coarse entropy production and recording explicitly which monotonicity statements remain targets rather than derived theorems.

5. A transport-preparation file defining large-deviation functionals, history currents, and the hypotheses under which Green–Kubo-type limits should later be proved.

This decomposition preserves the central discipline of the project: formal identities of the partition function must not be mixed with regime-dependent physical interpretations.

12.12 Conclusion of the Chapter

This chapter turned the earlier action and information layers into a genuine ensemble theory of structural histories. The next step is no longer general thermodynamic bookkeeping, but continuum transport: continuity equations, constitutive laws, diffusion, and hydrodynamic closure must now be derived as large-scale limits of the history ensemble.

Chapter 13

Continuum, Transport, and the Hydrodynamic Limit

13.1 Why Continuum Physics Must Arise as a Large-Scale Limit of History Ensembles

The previous chapter introduced the partition resolvent, history ensembles, fluctuation identities, and the first thermodynamic potentials of the theory. That is exactly the point at which continuum physics can be addressed responsibly. Before ensembles, currents and transport coefficients would be only kinematic labels. After ensembles, they become coarse observables derived from weighted classes of histories.

The central principle of the present chapter is therefore fixed in advance: continuum mechanics is not fundamental ontology. It is an observer-level regime in which the coarse-graining map becomes stable enough that density fields, currents, constitutive laws, and hydrodynamic closures exist as effective descriptions of history ensembles. Space and time remain observer images, and the continuum is only the large-scale organization of those images.

This chapter has three aims. First, it defines the continuum window and the basic coarse fields. Second, it derives continuity equations, linear-response laws, diffusion, conductivity, and wave or telegraph limits. Third, it organizes these results into hydrodynamic closure and into a graph-to-continuum bridge compatible with the earlier geometric projection and Poisson channel.

13.2 Observer Continuum Window and Coarse Fields

Definition 13.1 (Observer continuum window)**.** Let $\mathcal{O}$ be a fixed observer. One says that $\mathcal{O}$ admits a continuum window at scale ℓ if the coarse-graining map partitions microstates

into cells

$$\mathcal{C}_i^{(\ell)}$$

such that local differences inside a cell are operationally unresolved, neighboring cells carry effective coordinates, and fluctuations inside a cell are small relative to inter-cell gradients for the observables of interest.

Remark 13.2. The continuum window is therefore not a statement about an absolute background manifold. It is a regime of observer coarse-graining.

Definition 13.3 (Coarse-grained density field)**.** Let Q be a transfer-stable observable. For a cell $\mathcal{C}_i^{(\ell)}$, define the coarse-grained density by

$$\rho_Q(x_i, t) := \frac{1}{|\mathcal{C}_i^{(\ell)}|} \sum_{x \in \mathcal{C}_i^{(\ell)}} Q(x, t).$$

Typical cases are mass-like density, charge density, number density, action-energy density, and entropy density.

Definition 13.4 (Coarse-grained current)**.** Let $\Xi_Q(\Gamma; \partial\mathcal{C}_i^{(\ell)}, [t, t+\Delta t])$ count the transport of Q through the boundary of a cell during the observer-time interval $[t, t+\Delta t]$. The effective current is

$$J_Q(x_i, t) := \frac{1}{\Delta t\, |\partial\mathcal{C}_i^{(\ell)}|} \mathbb{E}\big[\Xi_Q(\Gamma; \partial\mathcal{C}_i^{(\ell)}, [t, t+\Delta t])\big].$$

13.3 Continuity Equations

Definition 13.5 (Local balance defect)**.** Let $Q_i(t)$ denote the total amount of Q in the cell $\mathcal{C}_i^{(\ell)}$. Define the balance defect by

$$\Delta_Q(i; t, \Delta t) := Q_i(t+\Delta t) - Q_i(t) + \Delta t\ \mathrm{div}_\ell\, J_Q(i, t) - \Delta t\, \Sigma_Q(i, t),$$

where $\Sigma_Q(i, t)$ is the source term.

Definition 13.6 (Mesoscopic conservation hypothesis)**.** For a conserved transfer-stable observable Q, the mesoscopic conservation hypothesis is

$$\mathbb{E}\big[\Delta_Q(i; t, \Delta t)\big] = o(\Delta t)$$

in the continuum window.

Theorem 13.7 (Continuity equation)**.** *If the mesoscopic conservation hypothesis holds, then in the continuum window the density field satisfies*

$$\partial_t \rho_Q + \nabla \cdot J_Q = \sigma_Q,$$

where σ_Q is the continuum source density. In the strictly conserved case,

$$\partial_t \rho_Q + \nabla \cdot J_Q = 0.$$

Proof. Rewrite the balance defect identity at cell level, divide by cell volume and by Δt, and pass to the continuum-window limit in which the discrete divergence and coarse averages converge to observer-level continuum fields. □

Corollary 13.8 (Charge and number continuity)**.** *For charge-like and number-like observables one obtains the observer-level equations*

$$\partial_t \rho_q + \nabla \cdot J_q = \sigma_q, \qquad \partial_t \rho_n + \nabla \cdot J_n = \sigma_n,$$

with vanishing source terms when creation and annihilation are absent.

13.4 Constitutive Laws and Linear Response

Definition 13.9 (Local-equilibrium closure)**.** Local-equilibrium closure holds if, in the continuum window, the current of each observable depends only on a finite list of local fields and their first gradients:

$$J_Q = \mathcal{J}_Q\big(\rho_1, \dots, \rho_s, \nabla\rho_1, \dots, \nabla\rho_s\big) + o(\nabla).$$

Proposition 13.10 (Linear-response constitutive law)**.** *Assume stationarity of the history ensemble, sufficient mixing, local-equilibrium closure, and a small perturbation around a homogeneous background. Then the current admits the leading-order form*

$$J_Q = -\sum_{a=1}^{s} D_{Qa} \nabla \rho_a + \sum_{b=1}^{r} \mu_{Qb} F_b,$$

where D_{Qa} is the transport matrix, F_b are observer-level external fields, and μ_{Qb} are mobility coefficients.

Remark 13.11. This law is not inserted phenomenologically. It is the continuum re-expression of the response coefficients already extracted from the partition resolvent.

13.5 Diffusion, Conductivity, and the Einstein Relation

Definition 13.12 (Diffusive regime)**.** An observable Q is in the diffusive regime if, under diffusive scaling of observer coordinates and times, the coarse current remains finite and no ballistic drift survives at leading order.

Proposition 13.13 (Diffusion equation)**.** *If the continuity equation holds and the constitutive law reduces to*

$$J_Q = -D_Q \nabla \rho_Q + o(\nabla), \qquad D_Q > 0,$$

then

$$\partial_t \rho_Q = D_Q \Delta \rho_Q + \sigma_Q.$$

In the absence of sources this becomes the standard diffusion equation.

Proof. Substitute the diffusive constitutive law into the continuity equation and collect the leading gradient terms. □

Definition 13.14 (Effective diffusion coefficient)**.** If $X_Q(T)$ denotes the total observer-level displacement associated with the transport of Q over time T, define

$$D_Q := \lim_{T \to \infty} \frac{\mathrm{Var}(X_Q(T))}{2T}$$

whenever the limit exists.

Theorem 13.15 (Target theorem: Green–Kubo formula)**.** *Under stationarity, sufficient mixing, and finite correlation time, one expects*

$$D_Q = \int_0^\infty \langle J_Q(0) J_Q(t) \rangle \, dt.$$

Definition 13.16 (Mobility)**.** For a charge-like observable in a weak external field E, define the mobility by

$$\mu_q := \left. \frac{\partial \langle v_q \rangle}{\partial E} \right|_{E=0}.$$

Proposition 13.17 (Ohmic linear-response law)**.** *In the weak-field linear regime one has*

$$J_q = \sigma E - D_q \nabla \rho_q,$$

where σ is the effective conductivity. If density gradients are negligible, this reduces to the Ohmic form

$$J_q \approx \sigma E.$$

Theorem 13.18 (Target theorem: Einstein relation)**.** *In a thermalized diffusive regime one expects*

$$D_q = \mu_q T_{\mathrm{eff}}, \qquad \textit{equivalently} \qquad D_q = \mu_q / \beta.$$

13.6 Wave and Telegraph Limits

Definition 13.19 (Inertial transport regime)**.** An observable Q is in an inertial transport regime if the current satisfies a relaxation law of the form

$$\tau_Q \partial_t J_Q + J_Q = -D_Q \nabla \rho_Q.$$

Proposition 13.20 (Telegraph equation)**.** *Combining the inertial transport regime with the continuity equation yields*

$$\tau_Q \partial_t^2 \rho_Q + \partial_t \rho_Q = D_Q \Delta \rho_Q + \sigma_Q.$$

Proof. Differentiate the continuity equation in time and eliminate the current using the relaxation law. □

Proposition 13.21 (Wave-regime limit)**.** *If dissipation is negligible and the restoring response is linear, then an effective wave equation may emerge:*

$$\partial_t^2 u = c_Q^2 \Delta u.$$

Here c_Q is the emergent observer-level transport speed associated with the observable sector Q, not a fundamental constant of the ontology.

13.7 Hydrodynamic Variables and Closure

Definition 13.22 (Hydrodynamic state)**.** The hydrodynamic state is the collection of coarse fields

$$\mathfrak{h}(x,t) = \big(\rho_n, \rho_m, \rho_q, u, e, s\big),$$

where these denote number density, mass-like density, charge density, coarse velocity, action-energy density, and entropy density.

Definition 13.23 (Momentum density)**.** Using the earlier marked-kernel conception of momentum, define the coarse momentum density by

$$\pi(x,t) = \rho_m(x,t) u(x,t).$$

Proposition 13.24 (Euler-type hydrodynamic closure)**.** *Assume local equilibrium, negligible viscosity corrections, and the existence of a pressure functional $P = P(\rho_m, s, \dots)$. Then the continuum fields satisfy*

$$\partial_t \rho_m + \nabla \cdot (\rho_m u) = 0,$$

and

$$\partial_t(\rho_m u) + \nabla \cdot (\rho_m u \otimes u) + \nabla P = f_{\text{ext}}.$$

Theorem 13.25 (Target theorem: Navier–Stokes-type closure)**.** *If first-order dissipative corrections are retained, then one expects*

$$\partial_t(\rho_m u) + \nabla \cdot (\rho_m u \otimes u) + \nabla P = \eta \Delta u + \Big(\zeta + \frac{\eta}{3}\Big)\nabla(\nabla \cdot u) + f_{\text{ext}},$$

where η and ζ are history-derived shear and bulk viscosities.

Remark 13.26. The notation in this chapter is aligned with the statistical chapter: J_Q denotes the current associated with the observable Q, while D_Q denotes the corresponding transport coefficient extracted either from constitutive closure or from long-time fluctuations.

13.8 Entropy Balance in the Continuum Window

Definition 13.27 (Local entropy current)**.** Let J_E be the action-energy current and J_a the conserved currents with corresponding potentials μ_a. The local entropy current is

$$J_s := \frac{J_E - \sum_a \mu_a J_a}{T_{\text{eff}}}.$$

Proposition 13.28 (Local entropy balance)**.** *In the continuum window one has the local balance law*

$$\partial_t s + \nabla \cdot J_s = \sigma_s.$$

Under a dissipative closure compatible with the coarse-graining, one expects

$$\sigma_s \geq 0.$$

Corollary 13.29 (Global entropy production)**.** *For a spatial region Ω one obtains*

$$\frac{d}{dt}\int_\Omega s\, dx = -\int_{\partial\Omega} J_s \cdot n\, dS + \int_\Omega \sigma_s\, dx,$$

with nonnegative bulk entropy production in the irreversible regime.

13.9 Graph-to-Continuum Passage

Definition 13.30 (Coarse graph Laplacian)**.** Let the observer-level geometric projection induce a coarse graph

$$G_\alpha^{(\ell)}$$

with Laplacian

$$L^{(\ell)} = D^{(\ell)} - A^{(\ell)}.$$

Definition 13.31 (Continuum Laplacian convergence)**.** Continuum Laplacian convergence means that in the continuum window one has

$$L^{(\ell)} f \longrightarrow \mathcal{L}_{\text{eff}} f$$

in a suitable weak sense. In a local isotropic regime the limiting operator is typically of the form

$$\mathcal{L}_{\text{eff}} = \kappa \Delta.$$

Proposition 13.32 (Poisson-to-continuum limit)**.** *If the coarse graph Laplacian converges as above, then the observer-level Poisson channel*

$$(L^{(\ell)} + \varepsilon I)\phi^{(\ell)} = \rho^{(\ell)}$$

passes in the continuum limit to

$$(\mathcal{L}_{\text{eff}} + \varepsilon)\phi = \rho.$$

In the local isotropic regime this becomes

$$(\kappa\Delta + \varepsilon)\phi = \rho.$$

13.10 Large Deviations and Macroscopic Fluctuations

Definition 13.33 (Macroscopic fluctuation field)**.** If $\rho_Q^\varepsilon(x,t)$ is the coarse density at scale ε, define the fluctuation field by

$$\delta\rho_Q^\varepsilon(x,t) := \varepsilon^{-d/2}\big(\rho_Q^\varepsilon(x,t) - \bar{\rho}_Q(x,t)\big).$$

Theorem 13.34 (Target theorem: Gaussian fluctuation limit)**.** *In a central-limit regime one expects*

$$\delta\rho_Q^\varepsilon \Longrightarrow \mathcal{G}_Q,$$

where $\mathcal{G}_Q$ is a Gaussian fluctuation field whose covariance is determined by susceptibilities and transport coefficients.

Theorem 13.35 (Target theorem: Macroscopic fluctuation principle)**.** *For a macroscopic profile (ρ, J) deviating from the hydrodynamic law, one expects an action principle of the form*

$$\mathbb{P}[(\rho, J)] \asymp \exp\big(-\mathcal{I}[\rho, J]\big),$$

where $\mathcal{I}[\rho, J]$ is constrained by the continuity equation and the transport matrix.

13.11 Lean Orientation for the Next Chapter

For formalization, the continuum layer should be split into five files.

1. A continuum-window file defining coarse cells, density fields, currents, and the balance defect.

2. A continuity-and-response file proving continuity equations and recording the assumptions behind linear-response closure.

3. A transport file defining diffusion coefficients, mobility, telegraph limits, and the target Green–Kubo and Einstein relations.

4. A hydrodynamics file defining the hydrodynamic state and separating Euler-type identities from Navier–Stokes-type target closures.

5. A graph-limit file formalizing coarse graph Laplacians, continuum convergence, and the Poisson-to-continuum bridge.

This split keeps the continuum chapter compatible with the earlier discipline of the book: observer-level fields, transport coefficients, and hydrodynamic closures must remain visibly derived from history ensembles and transfer kernels. In particular, the notation J_Q, D_Q, and c_Q is reserved for transport data attached to a specific observable sector.

13.12 Conclusion of the Chapter

This chapter derived continuum physics as a large-scale observer regime of the history ensemble. Coarse density fields and currents were defined, continuity equations and constitutive laws were formulated, diffusion and conductivity were connected to response coefficients, telegraph and wave limits were isolated, hydrodynamic closure was organized, and the graph-based geometric channel was linked to continuum field equations.

The next chapter can therefore ask a sharper question: under which transport-sector conditions do these continuum fields synchronize into relativistic and field-theoretic effective symmetries?

Chapter 14

Relativistic and Field Limit

14.1 Why Relativistic Symmetry Must Appear as an Effective Transport Regime

The previous chapter established continuum fields, transport laws, diffusion, telegraph limits, and hydrodynamic closure as large-scale observer descriptions of history ensembles. That still does not amount to a relativistic theory. Continuum equations by themselves do not select a distinguished cone speed, do not enforce Klein–Gordon-type or Dirac-type structure, and do not explain why local field equations should organize themselves into Lorentz-like families.

Within the present book, the relativistic layer cannot be introduced as a primitive spacetime axiom. The ontology of the project is already fixed: symbolic states, causal rewriting, transfer kernels, and observer projections are fundamental, whereas space, time, and characteristic speeds are derived. Therefore the only acceptable route is this one: identify transport sectors through kernel-resolvent data, extract their low-momentum spectral branches, and show that in certain windows these branches synchronize into quadratic dispersion laws with a common effective cone speed.

The chapter has four tasks. First, it defines transport sectors and effective dispersion laws. Second, it formulates the relativistic transport window and derives Klein–Gordon-type and Dirac-type continuum images. Third, it states the conditions under which Lorentz-like symmetry and a universal causal cone emerge in the low-energy limit. Fourth, it fixes only the part of the local field image needed for the later pole, residue, and scattering language.

14.2 Observer Coordinates and Transport Sectors

Definition 14.1 (Observer continuum coordinates)**.** Assume that the observer $\mathcal{O}$ admits a continuum window. The resulting observer coordinates are written as

$$x^\mu = (t_\mathcal{O}, x^1_\mathcal{O}, \dots, x^d_\mathcal{O}),$$

where $t_\mathcal{O}$ arises from the observer slicing and $x_\mathcal{O}$ from coarse geometric projection.

Definition 14.2 (Transport sector)**.** A transport sector $\mathfrak{T}$ is a stable class of excitations selected by a marked or reduced kernel block

$$K_\mathfrak{T}$$

whose observable dynamics is controlled by a dominant spectral branch or an equivalent dispersion relation of the form

$$\mathfrak{P}_\mathfrak{T}(\omega, q) = 0.$$

Definition 14.3 (Effective dispersion law)**.** The effective observer-level dispersion law of the sector $\mathfrak{T}$ is the relation

$$\omega = \omega_\mathfrak{T}(q)$$

or, equivalently, the zero set of $\mathfrak{P}_\mathfrak{T}(\omega, q)$. In a low-momentum window it is assumed to admit an expansion of the form

$$\mathfrak{P}_\mathfrak{T}(\omega, q) = A_\mathfrak{T}\omega^2 - B_\mathfrak{T}|q|^2 - M^2_\mathfrak{T} + O(|q|^4, \omega^4, \omega^2|q|^2).$$

Remark 14.4. From this point on, the notation $c_\mathfrak{T}$, $m_\mathfrak{T}$, and $\Box_\mathfrak{T}$ is reserved for quantities attached to a specific relativistic transport sector. This separates sector-dependent relativistic data from the transport notation J_Q, D_Q, and c_Q used in the previous chapter.

14.3 The Relativistic Transport Window

Definition 14.5 (Relativistic transport window)**.** One says that the transport sector $\mathfrak{T}$ lies in a relativistic transport window if, in a neighborhood of $(\omega, q) = (0, 0)$, its dispersion relation takes the form

$$\omega^2 = c^2_\mathfrak{T}|q|^2 + m^2_\mathfrak{T} + O(|q|^4),$$

with

$$c^2_\mathfrak{T} = \frac{B_\mathfrak{T}}{A_\mathfrak{T}}, \qquad m^2_\mathfrak{T} = \frac{M^2_\mathfrak{T}}{A_\mathfrak{T}}.$$

Proposition 14.6 (Effective cone speed)**.** *If $A_\mathfrak{T} > 0$ and $B_\mathfrak{T} > 0$, then the transport sector*

determines a distinguished effective speed

$$c_{\mathfrak{T}} = \sqrt{B_{\mathfrak{T}}/A_{\mathfrak{T}}}.$$

Remark 14.7. This speed is not a universal constant of the theory. It is a derived parameter of a particular low-energy sector.

Definition 14.8 (Effective interval and d'Alembert operator)**.** For a relativistic transport sector define the effective interval

$$ds^2_{\mathfrak{T}} := c^2_{\mathfrak{T}} dt^2_{\mathcal{O}} - |dx_{\mathcal{O}}|^2$$

and the corresponding effective d'Alembert operator

$$\Box_{\mathfrak{T}} := \frac{1}{c^2_{\mathfrak{T}}}\partial_t^2 - \Delta.$$

Proposition 14.9 (Klein–Gordon-type limit)**.** *If the sector $\mathfrak{T}$ lies in a relativistic transport window and higher-order corrections are negligible, then the corresponding continuum field $\phi_{\mathfrak{T}}$ satisfies the effective field equation*

$$(\Box_{\mathfrak{T}} + m^2_{\mathfrak{T}})\phi_{\mathfrak{T}} = 0.$$

Proof. In the quadratic approximation, the dispersion relation is exactly the on-shell condition for the operator $\Box_{\mathfrak{T}} + m^2_{\mathfrak{T}}$. Passing from Fourier variables back to observer coordinates yields the stated continuum equation. □

Example 14.10 (Massive low-energy branch)**.** If a sector has low-momentum expansion

$$\omega^2 = c_*^2|q|^2 + m_*^2 + O(|q|^4),$$

then its continuum image is a Klein–Gordon-type mode with effective mass m_* and sector speed c_*. This is the minimal example of a relativistic transport sector that is not massless.

14.4 Massless Sectors and Effective Light Cones

Definition 14.11 (Massless transport sector)**.** A transport sector is massless if

$$m_{\mathfrak{T}} = 0.$$

In that case the low-energy dispersion law becomes

$$\omega^2 = c^2_{\mathfrak{T}}|q|^2 + O(|q|^4).$$

Proposition 14.12 (Effective light cone)**.** *For a massless transport sector, characteristic propagation surfaces satisfy*

$$ds_{\mathfrak{T}}^2 = 0.$$

Theorem 14.13 (Target theorem: Emergent light invariance)**.** *If there exists a distinguished massless sector $\mathfrak{L}$ that is stable under admissible observer change and under the relevant transfer maps, then its cone speed $c_{\mathfrak{L}}$ becomes the universal characteristic speed of the associated low-energy relativistic subsector.*

Example 14.14 (Massless synchronization candidate)**.** If several low-energy bosonic sectors all share the same leading coefficient ratio $B_{\mathfrak{T}}/A_{\mathfrak{T}}$ and one of them is massless, then they already furnish the minimal pattern from which a common low-energy cone can emerge.

14.5 Field-Theoretic Images of Kernel Blocks

Definition 14.15 (Bosonic quadratic kernel block)**.** Let a bosonic mode be governed by a kernel block whose low-momentum image has the form

$$\mathcal{D}_B(\omega, q) = Z_t\omega^2 - Z_x|q|^2 - m_B^2 + O(|q|^4, \omega^4).$$

Proposition 14.16 (Bosonic field image)**.** *After observer decategorification and continuum expansion, the corresponding quadratic action takes the form*

$$S_B^{(2)} \rightsquigarrow \int d^dx\, dt \left[Z_t(\partial_t X)^2 - Z_x|\nabla X|^2 - m_B^2 X^2\right].$$

Definition 14.17 (Fermionic first-order kernel block)**.** Let a fermionic sector have low-energy kernel image

$$\mathcal{D}_F(\omega, q) = \Gamma^0\omega - \sum_{i=1}^{d} \Gamma^i q_i - M + O(|q|^2, \omega^2).$$

Proposition 14.18 (Dirac-type continuum image)**.** *If the matrices Γ^μ satisfy an effective Clifford closure relation in the low-energy window, then the corresponding fermionic field obeys*

$$(i\Gamma^\mu\partial_\mu - M)\psi = 0.$$

Remark 14.19. The fermionic field equation is therefore not postulated independently. It is the first-order continuum image of an appropriate kernel block.

14.6 Lorentz-Like Symmetry and Sector Synchronization

Definition 14.20 (Lorentz-like sector)**.** A transport sector $\mathfrak{R}$ is Lorentz-like if its effective field equations are invariant under observer transformations preserving the interval

$$ds^2_{\mathfrak{R}} = c^2_{\mathfrak{R}} dt^2 - |dx|^2.$$

Proposition 14.21 (Effective boost form)**.** *In one spatial dimension, the effective boost associated with the sector speed $c_{\mathfrak{R}}$ is given by*

$$t' = \gamma\left(t - \frac{v}{c^2_{\mathfrak{R}}}x\right), \qquad x' = \gamma(x - vt), \qquad \gamma = (1 - v^2/c^2_{\mathfrak{R}})^{-1/2}.$$

Theorem 14.22 (Target theorem: Emergent Lorentz invariance)**.** *Assume that there exists a universal massless sector $\mathfrak{L}$, that the relevant low-energy bosonic and fermionic sectors share the same effective speed $c_{\mathfrak{L}}$, and that their interactions preserve the common cone up to infrared-irrelevant operators. Then the low-energy observer theory becomes Lorentz-like with shared cone speed $c_{\mathfrak{L}}$.*

14.7 Local Interactions and Effective Causality

Definition 14.23 (Interaction kernel expansion)**.** Suppose the full kernel action decomposes as

$$S_{\mathrm{ker}} = S_{\mathrm{quad}} + S_{\mathrm{int}}.$$

Its local low-energy field image has the form

$$S_{\mathrm{int}} \rightsquigarrow \int d^d x\, dt\, \mathcal{L}_{\mathrm{int}}(\phi_a, \psi_b, \partial\phi_a, \partial\psi_b).$$

Proposition 14.24 (Locality criterion)**.** *If interaction kernels decay sufficiently fast outside observer-local cells and ultraviolet corrections are irrelevant in the continuum window, then $\mathcal{L}_{\mathrm{int}}$ depends only on local fields and finitely many derivatives.*

Proposition 14.25 (Effective causal admissibility)**.** *Within a relativistic transport sector $\mathfrak{T}$, a process is effectively causally admissible if its support lies inside or on the sector cone:*

$$|dx_{\mathcal{O}}| \le c_{\mathfrak{T}}\, dt_{\mathcal{O}}.$$

Different sectors may carry different cone speeds before low-energy synchronization.

Theorem 14.26 (Target theorem: Universal low-energy cone)**.** *If all physically relevant low-energy sectors synchronize to the same massless branch $\mathfrak{L}$, then an effective universal*

causal cone emerges:

$$|dx_{\mathcal{O}}| \leq c_{\mathfrak{L}}\, dt_{\mathcal{O}}.$$

14.8 Graph Substrate, UV Corrections, and Infrared Robustness

Definition 14.27 (Graph origin of relativistic coefficients)**.** The coefficients $A_{\mathfrak{T}}$ and $B_{\mathfrak{T}}$ entering the low-momentum dispersion relation are read jointly from transport-kernel dynamics and from the observer-level coarse graph geometry induced by projection.

Remark 14.28. Relativistic data therefore come from the combined structural and geometric channel. They are not fixed by geometry alone.

Definition 14.29 (Lorentz-violating corrections)**.** Higher-order operators that deform the quadratic relativistic limit are collected into an ultraviolet correction term of the form

$$\delta\mathcal{L} = \sum_{n\geq 2} \frac{1}{\Lambda^{2n-2}} \mathcal{O}_n,$$

where the operators $\mathcal{O}_n$ contain higher spatial or temporal derivatives.

Theorem 14.30 (Target theorem: Infrared irrelevance of Lorentz-violating operators)**.** *If singular-profile stability and ultraviolet robustness hold in the sense already adopted in the structural chapters, then Lorentz-violating higher-order operators become infrared-irrelevant and the relativistic field image is the universal low-energy fixed description.*

14.9 Lean Orientation for the Next Chapter

For later formalization, the relativistic layer should be split into five files.

1. A transport-sector file defining effective dispersion relations, low-momentum coefficients, and relativistic transport windows.

2. A bosonic-image file proving the Klein–Gordon-type continuum limit from quadratic kernel blocks.

3. A fermionic-image file proving the Dirac-type continuum limit under effective Clifford closure assumptions.

4. A synchronization file recording the target theorems on common cone speed, Lorentz-like symmetry, and universal causal admissibility.

5. An infrared-robustness file separating local interaction criteria and ultraviolet irrelevance assumptions from the already proved continuum identities.

This split preserves the main methodological point of the chapter: relativistic and field-theoretic language is derived from transport-sector kernel data and is not an independent primitive layer of the book.

14.10 Conclusion of the Chapter

This chapter built the relativistic and field-theoretic limit as a derived low-energy regime of transport sectors. Effective dispersion laws, cone speeds, intervals, Klein–Gordon-type and Dirac-type images, Lorentz-like synchronization, local interaction criteria, and infrared robustness conditions were all formulated in a common kernel-based language.

The next chapter can therefore proceed to quantum field, interaction, and scattering theory in the proper sense: poles, residues, asymptotic sectors, bound states, resonances, and observer-level field quantization over the sectors isolated here.

Chapter 15

Quantum Field, Interaction, and Scattering Limit

15.1 Why Quantum Field Theory Must Be Read as the Pole-Residue Language of Transport Sectors

The previous chapter isolated the relativistic and field-theoretic limit of transport sectors. Effective dispersion laws, cone speeds, Klein–Gordon-type images, Dirac-type images, and local interaction criteria are now available. This is the natural point within the present framework at which observer-level quantum field theory is introduced. Earlier than that, one would only be naming formal symbols without the sector structure that gives them physical meaning.

The methodological rule of the chapter is strict. Fields, quanta, propagators, and scattering states are not to be postulated as an independent ontology layered on top of the theory. They are only the observer-decategorified language in which one describes poles, residues, interaction kernels, asymptotic sectors, bound-state channels, and resonances of the full kernel-resolvent dynamics.

This chapter therefore does five things. First, it introduces on-shell pole conditions, residue factors, and one-pole propagators. Second, it states the effective quantization rules and the associated Fock-like sectors. Third, it builds interaction kernels, Dyson-type expansions, generating functionals, and correlators. Fourth, it introduces asymptotic sectors, the S-matrix, the T-matrix, and an LSZ-like target theorem. Fifth, it describes bound states, resonances, and Euclidean continuation as the bridge to statistical mechanics.

15.2 Pole Conditions, Residues, and One-Particle Propagators

Definition 15.1 (Field sector)**.** A field sector $\mathfrak{F}$ is a transport sector whose low-energy observer image is stable enough to admit a continuum field description and an associated kernel block

$$\mathcal{D}_{\mathfrak{F}} = I - K_{\mathfrak{F}}.$$

It may be bosonic, fermionic, gauge-like, or composite.

Definition 15.2 (On-shell pole condition)**.** Let $\lambda_*^{(\mathfrak{F})}(\eta, q)$ be the dominant spectral branch of the field sector. The on-shell pole condition is

$$\lambda_*^{(\mathfrak{F})}(\eta, q) = 1.$$

In the rest frame this reduces to

$$\lambda_*^{(\mathfrak{F})}(m_{\mathfrak{F}}, 0) = 1.$$

Definition 15.3 (Residue factor)**.** The residue factor of the field sector is

$$Z_{\mathfrak{F}}(q) := \left[\partial_\eta\big(1 - \lambda_*^{(\mathfrak{F})}(\eta, q)\big)\right]^{-1}_{\eta=\eta_*(q)}.$$

Definition 15.4 (One-pole propagator)**.** In the one-pole approximation, the observer-level propagator of the field sector has the form

$$G_{\mathfrak{F}}(\eta, q) \approx \frac{Z_{\mathfrak{F}}(q)}{\eta - \eta_*(q) + i0}.$$

Proposition 15.5 (Particle interpretation of isolated poles)**.** *If the propagator of a field sector has an isolated simple pole, then the observer-level theory admits a one-particle interpretation with dispersion $\eta_*(q)$, residue factor $Z_{\mathfrak{F}}$, and an asymptotic one-particle mode.*

Proof. An isolated simple pole singles out a stable branch of the resolvent and therefore a stable asymptotic excitation. The derivative of the inverse branch fixes the residue normalization, which determines the one-particle wavefunction weight. □

Example 15.6 (Nonrelativistic pole expansion)**.** If the dominant branch admits the low-momentum expansion

$$\eta_*(q) = m_{\mathfrak{F}} + \frac{|q|^2}{2M_{\mathfrak{F}}} + \cdots,$$

then the one-pole propagator becomes

$$G_{\mathfrak{F}}(\eta, q) \approx \frac{Z_{\mathfrak{F}}}{\eta - m_{\mathfrak{F}} - |q|^2/(2M_{\mathfrak{F}}) + i0},$$

which is the minimal observer-level particle pole formula.

15.3 Effective Quantization Axioms and Fock-Like Sectors

Definition 15.7 (Mode decomposition)**.** Let $\Phi_{\mathfrak{F}}(x)$ be the continuum field image of a field sector in a homogeneous observer background. Its mode decomposition is the observer-level expansion over momentum modes and the associated creation-annihilation operators.

Definition 15.8 (Effective quantization axiom)**.** For bosonic sectors one imposes effective commutation relations, and for fermionic sectors one imposes effective anticommutation relations, on the mode operators. At the level of the present book these relations are to be treated as low-energy quantization axioms rather than already derived theorems.

Definition 15.9 (Fock-like space)**.** For each field sector $\mathfrak{F}$, define the associated Fock-like space by

$$\mathcal{F}_{\mathfrak{F}} := \bigoplus_{n \geq 0} \mathcal{H}_{\mathfrak{F}}^{(n)},$$

where the n-particle component records n-fold excitations of the same asymptotic kernel branch.

Remark 15.10. The word "Fock-like" is intentional. The construction is observer-level and low-energy. Its exact microscopic derivation from structural histories is deferred and should not be hidden.

Definition 15.11 (Sector factorization inside a many-sector observer window)**.** When several field sectors are simultaneously resolved by the observer, the effective Hilbert space is written as

$$\mathcal{H}_{\mathrm{tot}}^{\mathrm{eff}} = \bigotimes_{\mathfrak{F} \in \mathcal{W}_{\mathcal{O}}} \mathcal{F}_{\mathfrak{F}},$$

where $\mathcal{W}_{\mathcal{O}}$ is the observer-selected window of stable sectors and $\mathcal{F}_{\mathfrak{F}}$ is the Fock-like space of sector $\mathfrak{F}$. Later reduced-dynamics constructions are always understood relative to such an effective factorization rather than to a microscopic ontological split.

15.4 Interaction Kernels and Dyson-Type Expansion

Definition 15.12 (Interaction kernel)**.** Let the full effective inverse propagator decompose as

$$\mathcal{D} = \mathcal{D}_0 - \mathcal{V},$$

where $\mathcal{D}_0$ is the free quadratic part and $\mathcal{V}$ is the interaction kernel.

Definition 15.13 (Full propagator)**.** The full propagator of the interacting theory is

$$G = (\mathcal{D}_0 - \mathcal{V})^{-1}.$$

Proposition 15.14 (Dyson-type expansion)**.** *If the interaction kernel is sufficiently small in the relevant observer window, then the full propagator expands as*

$$G = G_0 + G_0\mathcal{V}G_0 + G_0\mathcal{V}G_0\mathcal{V}G_0 + \cdots, \qquad G_0 := \mathcal{D}_0^{-1}.$$

Proof. This is the geometric series expansion of the inverse operator around the free kernel block. □

Definition 15.15 (Local interaction density)**.** After observer decategorification and continuum expansion, the interaction kernel yields a local density

$$S_{\text{int}} = \int d^d x\, dt\, \mathcal{L}_{\text{int}}(\phi_a, \psi_b, A_\mu, \partial\phi, \partial\psi, \dots).$$

Typical low-energy forms include cubic bosonic couplings, Yukawa-type couplings, and current-gauge couplings.

Definition 15.16 (Vertex factor)**.** The observer-level vertex factor associated with an elementary interaction channel is the appropriate functional derivative of S_{int} with respect to the participating fields.

15.5 Generating Functionals and Correlators

Definition 15.17 (Generating functional)**.** For field multiplet Φ_A with sources J_A, the generating functional is

$$\mathcal{Z}[J] := \int \mathcal{D}\Phi\, e^{iS[\Phi] + i\int J_A \Phi_A}.$$

The Euclidean continuation is written as

$$\mathcal{Z}_E[J] := \int \mathcal{D}\Phi\, e^{-S_E[\Phi] + \int J_A \Phi_A}.$$

Definition 15.18 (Correlators)**.** Time-ordered correlators are obtained by differentiating the generating functional with respect to the external sources and then setting the sources to zero.

Proposition 15.19 (Two-point function as propagator)**.** *For a free field sector, the time-ordered two-point correlator is the position-space propagator of the corresponding kernel block.*

Definition 15.20 (Connected generating functional)**.** Define

$$W[J] := -i \log \mathcal{Z}[J].$$

Its derivatives generate the connected correlators.

15.6 Asymptotic Sectors and Scattering Theory

Definition 15.21 (In and out asymptotic sectors)**.** Let $\mathfrak{F}_1, \dots, \mathfrak{F}_n$ be isolated one-particle sectors. Their asymptotic in and out states are built as Fock-like excitations of the corresponding free poles.

Definition 15.22 (S-matrix and T-matrix)**.** The scattering matrix is defined by

$$S_{\beta\alpha} := \langle \beta, \text{out} \mid \alpha, \text{in} \rangle,$$

and the T-matrix by

$$S = I + iT.$$

Proposition 15.23 (Born approximation)**.** *At leading order in the interaction kernel,*

$$T^{(1)}_{\beta\alpha} = \langle \beta \mid \mathcal{V} \mid \alpha \rangle.$$

Theorem 15.24 (Target theorem: LSZ-like reduction)**.** *If asymptotic one-particle poles are stable and isolated, then scattering amplitudes are extracted from connected time-ordered correlators by amputating external propagators and multiplying by the appropriate residue factors. In schematic form,*

$$\mathcal{A}_{\beta\alpha} \sim \prod_{\text{ext}} Z_{\mathfrak{F}}^{-1/2} \times \textit{amputated connected correlator on shell.}$$

Remark 15.25. This statement is a target theorem because it depends on the precise relation between asymptotic sector isolation and the observer-level correlator calculus.

15.7 Bound States, Resonances, and Composite Channels

Definition 15.26 (Bound-state pole)**.** A composite sector $\mathfrak{B}$ is a bound state if the composite propagator has a pole below continuum threshold:

$$G_{\mathrm{comp}}(E, q) \sim \frac{Z_{\mathfrak{B}}}{E - E_{\mathfrak{B}}(q)}$$

with rest energy below the threshold energy.

Definition 15.27 (Binding energy)**.** The binding energy of the composite sector is

$$E_{\mathrm{bind}} := E_{\mathrm{thresh}} - E_{\mathfrak{B}}(0).$$

Definition 15.28 (Resonance pole)**.** If the pole moves off the real axis to

$$E_* = M_* - \frac{i}{2}\Gamma_*,$$

then the composite sector is a resonance with width Γ_*.

Proposition 15.29 (Breit–Wigner-type form)**.** *Near a narrow resonance one has the effective propagator form*

$$G(E) \approx \frac{Z_*}{E - M_* + i\Gamma_*/2}.$$

Theorem 15.30 (Target theorem: Emergence of composite poles)**.** *If an assembly-level attractive channel and the relevant residue compatibility conditions hold, then the corresponding EFU or assembly sector survives as a bound-state or resonance pole of the full observer-level resolvent.*

15.8 Gauge-Like Sectors, Effective Action, and Euclidean Bridge

Definition 15.31 (Gauge-like massless sector)**.** A gauge-like sector is a massless bosonic field sector whose coupling to matter is governed by a transfer-stable internal observable, so that its low-energy interaction takes current form

$$\mathcal{L}_{\mathrm{int}} = J_Q^\mu A_\mu.$$

Theorem 15.32 (Target theorem: Ward-like consistency)**.** *If the corresponding charge is transfer-stable and the interaction kernel is compatible with the observer-local symmetry structure, then the scattering amplitudes satisfy Ward-like relations.*

Definition 15.33 (One-particle irreducible effective action)**.** Let Φ_c denote the classical field. The one-particle irreducible effective action is defined by the Legendre transform of $W[J]$ with respect to the sources.

Definition 15.34 (Self-energy and pole shift)**.** For a two-point function write

$$G^{-1}(p) = G_0^{-1}(p) - \Sigma(p).$$

The physical pole mass is then determined by the pole of the dressed inverse propagator.

Definition 15.35 (Euclidean continuation)**.** Under suitable analytic continuation $t \mapsto -i\tau$, the quantum generating functional becomes a Euclidean functional and thus links field-theoretic amplitudes to statistical partition functionals.

Proposition 15.36 (Quantum-statistical bridge)**.** *Under Euclidean continuation, the field-theoretic vacuum functional becomes the observer-level statistical functional corresponding to the effective Hamiltonian or Euclidean action of the sector.*

15.9 Chapter Audit Summary

- Definitions introduced: on-shell conditions, residues, asymptotic multiparticle sectors, generating functionals, scattering amplitudes, bound states, and resonance poles.
- Standing assumptions: stable isolated one-particle poles, admissible asymptotic sectors, and interaction kernels compatible with the observer-level symmetry structure.
- Internally established results: the pole-residue organization of interacting field sectors, the effective scattering language, and the Euclidean bridge back to statistical functionals.
- Imported identifications used: standard QFT language for propagators, correlators, LSZ-type extraction, and Ward-type relations at observer level.
- Open targets deferred: full LSZ-like reduction, complete proof of composite-pole emergence, and the Ward-like consistency theorem beyond the current target status.
- Downstream dependence: the reduced-dynamics and Standard Model chapters use this field-theoretic pole-residue layer as their base.

15.10 Lean Orientation for the Next Chapter

For later formalization, the scattering chapter should be split into five files.

1. A pole-residue file defining on-shell conditions, residues, and one-pole propagators.
2. A quantization-and-Fock file recording the effective quantization axioms and the construction of asymptotic multiparticle sectors.
3. An interaction-and-correlator file defining Dyson expansion, local vertex factors, generating functionals, and connected correlators.
4. A scattering-and-composites file separating Born approximation, LSZ-like reduction targets, bound states, and resonance poles.
5. A Euclidean-bridge file connecting effective action, self-energy, and Euclidean continuation back to the statistical mechanics of histories.

This decomposition keeps the logic of the chapter explicit: all observer-level QFT objects are secondary expressions of pole-residue dynamics, not the primitive starting point of the theory.

15.11 Conclusion of the Chapter

This chapter formulated observer-level quantum field theory as the effective language of poles, residues, interaction kernels, correlators, asymptotic sectors, scattering amplitudes, bound states, and resonances. The Euclidean bridge back to statistical mechanics was also fixed, so the field-theoretic and thermodynamic parts of the book remain two views of the same kernel-based framework.

The next chapter returns to information, but now on the effective sector factorization

$$\mathcal{H}_{\mathrm{tot}}^{\mathrm{eff}} = \bigotimes_{\mathfrak{F}\in\mathcal{W}_{\mathcal{O}}} \mathcal{F}_{\mathfrak{F}},$$

so reduced dynamics, entanglement structure, measurement channels, and information flow are all formulated directly in the language of interacting field sectors.

Chapter 16

Quantum Information, the Observer, and Reduced Dynamics

16.1 Why the Informational Layer Must Return at the Level of Interacting Field Sectors

The early informational chapters introduced density operators, reduced states, record entropy, mutual information, and the observer–measurement–decoherence block in a minimal setting. The previous chapter then moved to interacting field sectors, asymptotic poles, scattering amplitudes, and composite channels. That shift changes the informational problem. The question is no longer how a toy detector reads out a toy sector, but how information is stored, transferred, degraded, and recovered when the subsystems themselves are interacting field sectors of the observer-level theory.

This chapter therefore does not repeat the earlier measurement material. It lifts it to the interacting field level. Reduced states must now be defined on sector decompositions, entanglement must be measured across field-theoretic cuts, measurement channels must be stated as completely positive maps acting on these sectors, and decoherence must be understood as the suppression of off-diagonal sector coherences inside an open interacting theory. The observer becomes a distinguished information-processing subsystem of the field dynamics rather than an external narrator of it.

The chapter has five tasks. First, it fixes the observer-dependent subsystem split and the algebra of accessible observables. Second, it defines entanglement, mutual information, relative entropy, and distinguishability for interacting sectors. Third, it builds measurement instruments and reduced dynamical maps. Fourth, it formulates decoherence rates, classicalization error, and information flow between sectors. Fifth, it states the observer-memory interpretation and the thermodynamic cost of erasure.

16.2 Observer-Accessible Decomposition and Reduced States

Definition 16.1 (Observer-accessible sector decomposition)**.** At the interacting field level, the observer works with a decomposition

$$\mathcal{H}_{\text{tot}}^{\text{eff}} = \mathcal{H}_S \otimes \mathcal{H}_D \otimes \mathcal{H}_E,$$

where S is the measured sector, D the detector or pointer sector, and E the inaccessible environment sector. This decomposition is observer-dependent and reflects the coarse partition of the effective theory rather than a fundamental split of the ontology.

Definition 16.2 (Total and reduced states)**.** The total state is a density operator satisfying

$$\rho_{\text{tot}} \geq 0, \qquad \text{Tr}(\rho_{\text{tot}}) = 1.$$

The reduced states are defined by partial trace:

$$\rho_S := \text{Tr}_{D,E}(\rho_{\text{tot}}), \qquad \rho_D := \text{Tr}_{S,E}(\rho_{\text{tot}}), \qquad \rho_E := \text{Tr}_{S,D}(\rho_{\text{tot}}).$$

Pairwise reduced states ρ_{SD}, ρ_{SE}, and ρ_{DE} are defined analogously.

Definition 16.3 (Observer algebra of accessible observables)**.** The observer has access only to a subalgebra

$$\mathcal{A}_{\mathcal{O}} \subset \text{End}(\mathcal{H}_{\text{tot}}^{\text{eff}})$$

compatible with the observer coarse-graining. This algebra determines which reduced observables and which information measures are operationally meaningful.

16.3 Entanglement Structure and Information Measures

Definition 16.4 (Product and entangled states)**.** A bipartite state is factorized across a cut $A|B$ if

$$\rho_{AB} = \rho_A \otimes \rho_B.$$

A pure state is entangled if it cannot be written as a simple tensor product across the same cut.

Definition 16.5 (Schmidt form)**.** Every pure bipartite state admits a decomposition

$$|\Psi\rangle = \sum_{i=1}^{r} \sqrt{\lambda_i}\, |i_A\rangle \otimes |i_B\rangle, \qquad \lambda_i \geq 0, \qquad \sum_i \lambda_i = 1.$$

Proposition 16.6 (Equal subsystem entropies for pure bipartite states)**.** *If* $\rho_{AB} = |\Psi\rangle\langle\Psi|$ *is pure, then*

$$S(\rho_A) = S(\rho_B).$$

Proof. The nonzero spectra of ρ_A and ρ_B coincide in Schmidt form, so the corresponding von Neumann entropies agree. □

Definition 16.7 (Entanglement entropy, mutual information, and conditional entropy)**.** For a bipartite cut $A|B$, define

$$S_{\mathrm{ent}}(A|B) := S(\rho_A)$$

for pure global states, together with

$$I(A:B) := S(\rho_A) + S(\rho_B) - S(\rho_{AB})$$

and

$$S(A|B) := S(\rho_{AB}) - S(\rho_B).$$

Definition 16.8 (Relative entropy and trace distinguishability)**.** For two admissible states define

$$S(\rho\|\sigma) := \mathrm{Tr}(\rho(\log\rho - \log\sigma))$$

and

$$\mathcal{D}_{\mathrm{tr}}(\rho,\sigma) := \frac{1}{2}\|\rho - \sigma\|_1.$$

Proposition 16.9 (Positivity of relative entropy)**.** *For all admissible states one has*

$$S(\rho\|\sigma) \geq 0,$$

with equality if and only if $\rho = \sigma$*.*

16.4 Measurement Instruments and Reduced Dynamical Maps

Definition 16.10 (Measurement instrument)**.** A measurement is described by a family of completely positive maps

$$\mathfrak{M} = \{\mathcal{M}_m\}_{m=1}^{M}$$

such that $\sum_m \mathcal{M}_m$ is trace preserving. The outcome probabilities and conditioned states are

$$p_m = \mathrm{Tr}(\mathcal{M}_m(\rho_S)), \qquad \rho_{S|m} = \frac{\mathcal{M}_m(\rho_S)}{p_m}.$$

Definition 16.11 (Pointer projectors and projective case)**.** Let

$$P_m = |r_m\rangle\langle r_m|, \qquad \sum_m P_m = I_D,$$

be the stable pointer projectors of the detector sector. The projective special case is

$$\mathcal{M}_m(\rho) = P_m \rho P_m.$$

Definition 16.12 (Reduced dynamical map)**.** If the total field-sector evolution is generated by a unitary operator $U(t)$, then the reduced system dynamics is

$$\rho_S(t) = \Phi_t^{S\leftarrow DE}(\rho_S(0)),$$

with

$$\Phi_t^{S\leftarrow DE}(\rho_S) := \mathrm{Tr}_{D,E}\Big(U(t)[\rho_S \otimes \rho_{DE}(0)]U(t)^\dagger\Big).$$

Proposition 16.13 (Complete positivity of reduced dynamics)**.** *If the initial state factorizes as*

$$\rho_{\mathrm{tot}}(0) = \rho_S(0) \otimes \rho_{DE}(0),$$

then the reduced map $\Phi_t^{S\leftarrow DE}$ is completely positive and trace preserving.

Proof. This is the standard consequence of unitary evolution on the enlarged space followed by partial trace over a fixed ancillary state. □

Definition 16.14 (Kraus form and effective Markovian generator)**.** The reduced map admits a Kraus representation

$$\Phi_t^{S\leftarrow DE}(\rho) = \sum_\alpha K_\alpha(t)\rho K_\alpha(t)^\dagger, \qquad \sum_\alpha K_\alpha(t)^\dagger K_\alpha(t) = I.$$

When the dynamics forms a semigroup, one may further describe it by an effective Lindblad-type generator. At the level of this book, that generator is an effective open-system axiom unless derived for the specific sector under study.

16.5 Decoherence, Classicalization, and Information Flow

Definition 16.15 (Decoherence basis and rate)**.** Let $\{|r_m\rangle\}$ be the pointer basis of the detector sector. The off-diagonal coherences are

$$c_{mn}(t) := \langle r_m|\rho_D(t)|r_n\rangle, \qquad m \neq n.$$

If these decay as

$$|c_{mn}(t)| \sim e^{-\Gamma_{mn}t},$$

then Γ_{mn} is the decoherence rate between channels m and n.

Definition 16.16 (Classicalization error)**.** Let the classicalized detector state be

$$\rho_D^{\mathrm{cl}} := \sum_m p_m |r_m\rangle\langle r_m|.$$

The classicalization error may be measured by

$$\epsilon_{\mathrm{cl}}(t) := \|\rho_D(t) - \rho_D^{\mathrm{cl}}(t)\|_1$$

or by relative entropy distance to the same classicalized state.

Theorem 16.17 (Target theorem: Exponential classicalization)**.** *Under stable pointer basis, sufficient amplification, and environmental dephasing, one expects*

$$\epsilon_{\mathrm{cl}}(t) \leq Ce^{-\Gamma_{\mathrm{dec}}t},$$

and hence

$$\rho_D(t) \to \rho_D^{\mathrm{cl}}(t).$$

Definition 16.18 (Information current between sectors)**.** For two interacting sectors A and B, define the information current by

$$\mathcal{I}_{A\to B}(t) := \frac{d}{dt} I(A:B)(t).$$

Proposition 16.19 (Interaction-generated correlation)**.** *If initially uncorrelated sectors interact through a genuinely entangling channel, then the interaction can generate positive mutual information between them.*

16.6 Observer Memory, Information Gain, and Erasure Cost

Definition 16.20 (Observer memory sector)**.** The observer memory is a distinguished subspace

$$\mathcal{H}_{\mathrm{mem}} \subset \mathcal{H}_D$$

whose stable basis states encode persistent records.

Definition 16.21 (Observer information gain)**.** The information gain of a successful

observation is

$$\Delta I_{\text{obs}} := I(S : \text{mem})_{\text{after}} - I(S : \text{mem})_{\text{before}}.$$

Proposition 16.22 (Positive information gain in successful measurement)**.** *If the measurement channel successfully correlates the measured sector with pointer memory, then*

$$\Delta I_{\text{obs}} > 0.$$

Definition 16.23 (Landauer-type erasure cost)**.** If observer memory is reset to a standard blank state, the coarse thermodynamic erasure cost satisfies a Landauer-type lower bound

$$\Delta Q_{\text{erase}} \geq T_{\text{eff}}\, \Delta S_{\text{mem}}.$$

16.7 Selective and Unselective Reduced Evolution

Definition 16.24 (Selective trajectory)**.** Conditioned on a record history $m(t)$, the reduced system follows the selective evolution

$$\rho_S(t) \mapsto \rho_{S|m(t)}.$$

Definition 16.25 (Unselective evolution)**.** If record outcomes are ignored, the reduced state is the average

$$\rho_S(t) = \sum_m p_m(t) \rho_{S|m}(t).$$

Proposition 16.26 (Selective versus unselective information)**.** *Selective dynamics retains more information than the corresponding unselective averaged evolution. Averaging over records increases effective entropy and hides branch-level distinctions.*

16.8 Lean Orientation for the Next Chapter

For later formalization, the reduced-dynamics chapter should be split into five files.

1. A reduced-state file defining observer-accessible decompositions, reduced states, and the algebra of accessible observables.

2. An information-measures file defining entanglement entropy, mutual information, conditional entropy, relative entropy, and distinguishability.

3. A measurement-instruments file defining completely positive instruments, pointer projectors, and conditioned post-measurement states.

4. A reduced-dynamics file defining Kraus maps, complete positivity, effective Lindblad generators, and decoherence rates.

5. An observer-memory file defining information gain, erasure cost, and selective versus unselective reduced evolution.

This split makes the dependency chain explicit: the observer-information layer over interacting field sectors is built on top of the earlier density-operator language, the measurement block, and the field-theoretic sector decomposition.

16.9 Conclusion of the Chapter

This chapter completed the observer-information layer for interacting field sectors. Reduced states, entanglement measures, measurement instruments, completely positive reduced maps, decoherence rates, classicalization error, information currents, observer memory, and erasure cost were all organized in a common language.

The next chapter can therefore move from interacting sectors to many-body organization: phases, symmetry breaking, collective modes, quasi-particles, and topological sectors.

Chapter 17

Many-Body Phases, Symmetry Breaking, and Topological Sectors

17.1 Why Many-Body Theory Must Be Introduced After Reduced Dynamics

The previous chapter completed the observer-information layer for interacting field sectors. Reduced states, effective sector factorizations, completely positive maps, decoherence rates, and information currents are now available. This is the natural point within the present framework at which one passes from few-sector interaction language to the organization of large collections of coupled sectors. Earlier than that, one could write down formal many-body symbols, but one would not yet know how macroscopic order, phase stability, or topological protection are read by an observer through coarse-grained access to the effective theory.

The present chapter moves from interacting sectors to stable macrophases. Its guiding rule is the same as in the earlier parts of the book: phases are not added as a new ontology over the kernel-residue theory. They are the stable large-scale regimes of many-body sector dynamics, distinguished by order parameters, collective propagator poles, gap structure, defect sectors, and topological invariants. Materials and bodies begin to appear only when these many-body regimes become sufficiently stable under coarse-graining.

This chapter therefore does eight things. First, it defines many-body sectors, effective many-body state spaces, and many-body effective actions. Second, it introduces thermodynamic effective potentials and order parameters as observer-accessible diagnostics of phase structure. Third, it formulates first-order and continuous phase transitions together with the minimal Landau-type expansion. Fourth, it builds correlation functions, correlation length, and collective modes around phase minima. Fifth, it introduces quasi-particle branches, spectral gaps, and paired phases. Sixth, it formalizes domain walls, vortices, and solitons as defect sectors. Seventh, it defines topological sectors and topologically

ordered phases. Eighth, it links entanglement markers back to phase classification.

17.2 Many-Body Sectors and Effective Potentials

Definition 17.1 (Many-body sector)**.** A many-body sector $\mathfrak{M}$ is an observer-resolved subsystem built from a large family of compatible elementary or composite sectors,

$$\mathfrak{M} = \{\mathfrak{F}_1, \dots, \mathfrak{F}_N\}, \qquad N \gg 1,$$

where each $\mathfrak{F}_i$ is an elementary field sector, a bound-state sector, a quasi-particle branch, or another transfer-stable composite excitation.

Definition 17.2 (Effective many-body state space)**.** The effective state space of a many-body sector is

$$\mathcal{H}^{\text{eff}}_{\mathfrak{M}} := \bigotimes_{i=1}^{N} \mathcal{H}^{\text{eff}}_{\mathfrak{F}_i},$$

or, for indistinguishable excitations, the corresponding symmetrized or antisymmetrized Fock-like reduction. This effective space is a coarse many-sector window inside $\mathcal{H}^{\text{eff}}_{\text{tot}}$.

Definition 17.3 (Many-body effective action)**.** On coarse-grained order fields $\Phi_A(x,t)$ one defines the many-body effective action

$$\Gamma_{\mathfrak{M}}[\Phi;\lambda] = \int d^d x\, dt\, \mathcal{L}_{\text{eff}}(\Phi_A, \partial\Phi_A; \lambda),$$

where λ collects control parameters such as temperature-like variables, density-like variables, couplings, external sources, and geometric data.

Definition 17.4 (Thermodynamic effective potential)**.** For homogeneous configurations, define the effective potential density by

$$V_{\text{eff}}(\Phi;\lambda) := -\frac{1}{\text{Vol}\cdot T}\log Z[\Phi;\lambda].$$

Stable phases correspond to robust minima of V_{eff} in the observer window in which the thermodynamic description is valid.

17.3 Order Parameters and Phase Criteria

Definition 17.5 (Order parameter)**.** An order parameter is an observer-accessible coarse-grained observable or field expectation value

$$\mathcal{O}_\alpha := \langle \widehat{\mathcal{O}}_\alpha \rangle$$

such that distinct macrophases correspond to distinct stable values of $\mathcal{O}_\alpha$.

Example 17.6 (Typical order parameters)**.** Typical order parameters include a scalar condensate $\langle\phi\rangle$, a density contrast $\delta\rho$, a magnetization-like observable M, a pairing amplitude $\Delta = \langle\psi\psi\rangle$, and a topological marker Q_{top}.

Definition 17.7 (Symmetric and broken phases)**.** Let G be the relevant observer-level symmetry group. A phase is symmetric if the order parameter vanishes or remains invariant under G. It is symmetry-broken if the order parameter is nonzero and is preserved only by a proper subgroup $H \subsetneq G$.

Proposition 17.8 (Phase criterion by effective-potential minima)**.** *If $V_{\text{eff}}(\Phi;\lambda)$ has isolated stable minima $\Phi_*^{(1)},\dots,\Phi_*^{(k)}$, then each stable phase corresponds to one connected family of such minima together with the associated stable fluctuation spectrum around it.*

Proof. The effective potential records the coarse free-energy cost of homogeneous configurations. Stable observer-level regimes are therefore precisely those minima that survive coarse perturbations and support a stable expansion of fluctuations. □

Definition 17.9 (Susceptibility and criticality)**.** If h is a source coupled to an order parameter $\mathcal{O}$, then the susceptibility is

$$\chi_{\mathcal{O}} := \frac{\partial \mathcal{O}}{\partial h}.$$

Critical behavior is signaled by divergence or nonanalyticity of $\chi_{\mathcal{O}}$, or equivalently by nonanalyticity of derivatives of $\log Z$.

17.4 Phase Transitions and Landau-Type Expansion

Definition 17.10 (Phase boundary)**.** A phase boundary is a locus in control-parameter space at which the effective free-energy density becomes nonanalytic, competing minima exchange dominance, or the correlation length diverges.

Definition 17.11 (First-order transition)**.** A first-order transition is a phase change at which the order parameter jumps discontinuously and metastable local minima coexist on the transition line.

Definition 17.12 (Continuous transition)**.** A continuous transition is a phase change at which the order parameter varies continuously while susceptibility or correlation length diverges.

Proposition 17.13 (Landau-type expansion)**.** *Near a continuous transition with scalar order parameter ϕ, and assuming the relevant symmetry forbids odd terms, one has the expansion*

$$V_{\text{eff}}(\phi) = a(\lambda)\phi^2 + b(\lambda)\phi^4 + O(\phi^6), \qquad b(\lambda) > 0.$$

If $a(\lambda) > 0$, the stable minimum is $\phi_* = 0$. If $a(\lambda) < 0$, the broken minima are

$$\phi_* = \pm\sqrt{\frac{-a(\lambda)}{2b(\lambda)}}.$$

Corollary 17.14 (Mean-field critical law)**.** *Within the mean-field regime, the broken-phase order parameter satisfies*

$$\phi_* \sim (\lambda_c - \lambda)^{1/2}$$

as one approaches the critical point from the ordered side.

17.5 Correlation Functions and Collective Modes

Definition 17.15 (Connected two-point correlation function)**.** For a local observable $O(x)$, define the connected correlator by

$$C_O(x - y) := \langle O(x)O(y)\rangle - \langle O(x)\rangle\langle O(y)\rangle.$$

Definition 17.16 (Correlation length)**.** If the connected correlator decays as

$$C_O(r) \sim e^{-r/\xi},$$

then ξ is the correlation length of the phase in that observer window.

Proposition 17.17 (Diverging correlation length at continuous transition)**.** *At a continuous transition one expects*

$$\xi(\lambda) \to \infty \qquad as \qquad \lambda \to \lambda_c.$$

Definition 17.18 (Collective mode)**.** A collective mode is a stable pole or narrow branch in the fluctuation propagator of an order field,

$$G_{\delta\Phi}(\omega, q) \sim \frac{Z_{\text{coll}}}{\omega^2 - \omega_{\text{coll}}(q)^2 + i0}.$$

Proposition 17.19 (Amplitude and phase modes)**.** *If a complex order parameter is written as*

$$\Phi = (v + \sigma)e^{i\theta},$$

then fluctuations split into an amplitude mode σ and a phase mode θ. In a broken continuous-symmetry phase, the amplitude mode is typically gapped whereas the phase mode may be gapless.

Theorem 17.20 (Target theorem: Goldstone-type mode)**.** *If a continuous global symmetry*

G is spontaneously broken to H, then each broken generator yields a gapless collective branch provided no gauge-like locking removes that degree of freedom.

Theorem 17.21 (Target theorem: Higgs or amplitude mode)**.** *Near a broken phase with scalar condensate $v \neq 0$, the radial fluctuation branch has an effective mass determined by*

$$m_\sigma^2 = \frac{\partial^2 V_{\text{eff}}}{\partial \phi^2}\Big|_{\phi=\phi_*}.$$

17.6 Quasi-Particles, Gap Structure, and Paired Phases

Definition 17.22 (Quasi-particle branch)**.** A quasi-particle is a long-lived pole or narrow resonance in a many-body propagator,

$$G(\omega, q) \sim \frac{Z_{\text{qp}}}{\omega - \omega_{\text{qp}}(q) + i\Gamma(q)/2}.$$

If $\Gamma(q) \ll \omega_{\text{qp}}(q)$, the excitation is long-lived on the observer timescale.

Definition 17.23 (Spectral gap)**.** The many-body spectral gap is

$$\Delta_{\text{gap}} := \inf_q \omega_{\text{qp}}(q).$$

The phase is gapless if $\Delta_{\text{gap}} = 0$ and gapped if $\Delta_{\text{gap}} > 0$.

Proposition 17.24 (Phase distinction by gap structure)**.** *Two phases can differ even when they share the same local symmetry pattern, provided their gap structure, long-distance response, or topological invariants are different.*

Definition 17.25 (Pairing field)**.** For fermionic sectors, define the pairing order parameter by

$$\Delta(x) := \langle \psi_\downarrow(x)\psi_\uparrow(x)\rangle.$$

Definition 17.26 (Paired phase)**.** A paired phase is a phase for which $\Delta \neq 0$ in the observer window of interest.

Proposition 17.27 (Bogoliubov-type spectrum)**.** *In a homogeneous paired phase the elementary excitation spectrum takes the form*

$$E(q) = \sqrt{\xi(q)^2 + |\Delta|^2}.$$

Therefore the pairing field opens a gap whenever $\Delta \neq 0$.

17.7 Defect Sectors: Domain Walls, Vortices, and Solitons

Definition 17.28 (Domain wall)**.** If V_{eff} has disconnected stable minima, then a configuration interpolating between two distinct minima defines a domain wall sector.

Definition 17.29 (Domain-wall tension)**.** For a one-dimensional interpolation field $\phi(x)$, the domain-wall tension is

$$T_{\text{DW}} = \int dx \left[\frac{1}{2}(\partial_x \phi)^2 + V_{\text{eff}}(\phi) - V_{\text{eff}}(\phi_*)\right].$$

Definition 17.30 (Vortex)**.** For a complex order parameter with phase field θ, a vortex defect is characterized by a nontrivial winding number

$$\nu = \frac{1}{2\pi} \oint \nabla\theta \cdot dl \in \mathbb{Z}.$$

Definition 17.31 (Soliton)**.** A soliton is a localized nonlinear stable configuration with finite total energy or action and with topological or dynamical protection against decay.

Proposition 17.32 (Defect sectors as composite branches)**.** *If a defect configuration remains stable under coarse-grained evolution, then it appears as a stable composite sector, a long-lived resonance branch, or a protected topological excitation in the many-body resolvent.*

17.8 Topological Sectors and Topologically Ordered Phases

Definition 17.33 (Topological sector label)**.** A topological sector is a superselection-like class labeled by an invariant

$$Q_{\text{top}} \in \mathcal{T}$$

that cannot change under admissible local deformations preserving the relevant gap and boundary conditions.

Definition 17.34 (Topological current and charge)**.** If a conserved topological current exists,

$$\partial_\mu j^\mu_{\text{top}} = 0,$$

then the corresponding charge is

$$Q_{\text{top}} = \int_\Sigma j^0_{\text{top}}\, d^d x.$$

Proposition 17.35 (Topological stability)**.** *If Q_{top} is integer-valued and locally conserved, then configurations with different topological labels cannot be continuously deformed into one another without singular behavior or gap closure.*

Definition 17.36 (Topologically ordered phase)**.** A phase is topologically ordered if it is not fully characterized by local order parameters alone, if it carries stable topological sector labels, and if its boundary or defect excitations depend on the topological class.

Theorem 17.37 (Target theorem: Gap-protected topological stability)**.** *If a phase is gapped and carries a nontrivial topological invariant, then local perturbations that preserve the gap cannot change its topological sector.*

17.9 Gauge Locking and Effective Phase Diagrams

Definition 17.38 (Gauge-like locking)**.** If a broken continuous order parameter couples to a massless gauge-like sector, the would-be phase mode may be absorbed into the gauge sector, leaving a massive vector branch rather than an observable Goldstone branch.

Proposition 17.39 (Gauge-locking mass scale)**.** *In the simplest effective description one obtains a vector mass scale of the form*

$$m_A^2 \propto g^2 v^2,$$

where g is the effective coupling and v is the broken-phase amplitude.

Definition 17.40 (Phase diagram)**.** A phase diagram is the partition of control-parameter space

$$\lambda = (\beta, \mu, g, h, \rho, \dots)$$

into stable phases, with boundaries determined by nonanalyticity, order-parameter changes, gap closure, or changes of topological sector.

Definition 17.41 (Topological transition)**.** A topological transition is a phase transition in which the local symmetry pattern may remain unchanged while the gap structure or the invariant Q_{top} changes.

Proposition 17.42 (Gap-closing criterion for many topological transitions)**.** *Many topological phase transitions occur when*

$$\Delta_{\mathrm{gap}} \to 0$$

at the transition point.

17.10 Entanglement Markers of Phase Structure

Definition 17.43 (Entanglement Hamiltonian and entanglement spectrum)**.** For a bipartition $A|B$, define the entanglement Hamiltonian H_E by

$$\rho_A = \frac{e^{-H_E}}{\mathrm{Tr}(e^{-H_E})}.$$

Its eigenvalues define the entanglement spectrum.

Proposition 17.44 (Phase information in the entanglement spectrum)**.** *Distinct many-body phases may be distinguished not only by local order parameters but also by qualitative properties of the entanglement spectrum.*

Definition 17.45 (Topological entanglement marker)**.** In a topological phase, the subsystem entropy may admit the expansion

$$S(A) = \alpha|\partial A| - \gamma_{\mathrm{top}} + \cdots,$$

where γ_{top} is a topological entanglement marker.

Theorem 17.46 (Target theorem: Entanglement-topology link)**.** *If a phase is genuinely topological, then $\gamma_{\mathrm{top}} \neq 0$ and remains stable under local perturbations that preserve the gap.*

17.11 Programmatic Synthesis and Failure Modes

Proposition 17.47 (Programmatic synthesis for many-body phase structure)**.** *Assume the kernel-residue theory admits: a stable many-body observer window, an effective potential V_{eff}, local or quasi-local order fields, stable collective-mode propagators, and admissible topological charges. Then stable phases correspond to minima of V_{eff}, spontaneous symmetry breaking is detected by nonzero order parameters, collective modes arise as poles of fluctuation propagators, paired phases open spectral gaps, and topological sectors are classified by invariants stable under allowed local deformations.*

Remark 17.48 (Failure modes)**.** The many-body description can fail in several ways: coarse-grained order fields may not stabilize; the observer window may be too far from the thermodynamic regime; strong nonlocality may invalidate local order-parameter language; collective poles may be too broad to define good quasi-particles; or candidate topological labels may fail to be protected under allowed perturbations.

17.12 Chapter Audit Summary

- Definitions introduced: many-body sectors, effective potentials, order parameters, phase transitions, collective modes, quasi-particle branches, defect sectors, and topological sectors.
- Standing assumptions: a stable many-body observer window, effective thermodynamic potentials, and admissible local or quasi-local order fields.
- Internally established results: the observer-level phase language, gap and quasi-particle criteria, defect-sector organization, and the entanglement-marker interface.
- Imported identifications used: standard many-body and condensed-matter terminology for phases, symmetry breaking, and topological order.
- Open targets deferred: complete Goldstone-, amplitude-mode-, and topological-protection theorems beyond the current target-theorem status.
- Downstream dependence: the chemistry and materials chapters rely on this many-body stability layer as their immediate background.

17.13 Lean Orientation for the Next Chapter

For later formalization, the many-body chapter should be split into six files.

1. A many-body-sector file defining observer windows, effective many-body spaces, and effective actions.
2. An order-parameter file defining phase criteria, susceptibilities, and Landau-type expansions.
3. A correlator-and-collective-mode file defining correlation length, collective propagators, and Goldstone-type targets.
4. A quasiparticle-and-pairing file defining gaps, paired phases, and Bogoliubov-type spectra.
5. A defect-and-topology file defining domain walls, vortices, solitons, topological charges, and topological stability.
6. An entanglement-and-phase-markers file connecting entanglement spectra and topological entanglement markers back to phase classification.

This split keeps the dependency chain explicit. The chapter depends on the earlier thermodynamic, field-theoretic, and reduced-dynamics blocks, and it prepares the ground for chemistry by showing how stable bodies and materials first appear as many-body phases of interacting sectors.

17.14 Conclusion of the Chapter

This chapter introduced the many-body layer of the theory. Many-body sectors, effective potentials, order parameters, phase transitions, collective modes, quasi-particles, paired phases, defects, and topological sectors were all organized in one observer-level language. The chapter also connected phase classification to entanglement markers, so the informational and many-body parts of the theory remain joined.

The next chapter can therefore pass from general macrophases to chemistry proper. The key transition is now explicit: atoms and molecules will be read not as exceptions to many-body theory, but as stable or metastable composite sectors selected by the same effective-potential, gap, and resolvent logic that already classifies phases, defects, and collective branches.

Chapter 18

Chemistry, Atomic and Molecular Binding, and Emergent Material Classes

18.1 Why Chemistry Must Follow Many-Body Theory

The previous chapter introduced the many-body language of phases, collective modes, gap structure, defect sectors, and topological protection. This is the natural point within the present framework at which chemistry is introduced. Earlier than that, atoms and molecules would appear as isolated names with no place in the hierarchy of the theory. After the many-body chapter, however, the manuscript adopts the following rule: chemistry is the theory of stable and metastable composite sectors built from charged and correlated many-body branches.

This rule matters methodologically. Atoms, molecules, chemical bonds, reactions, and materials are not a separate layer added on top of physics. They are observer-resolved composite pole sectors of the same kernel-residue theory, but now in regimes where shell structure, bonding channels, slow-fast separation, and metastable reaction graphs become the dominant coarse description. The language of chemistry is therefore a specialization of the composite-sector language rather than an independent ontology.

This chapter does ten things. First, it defines atomic constituent sectors, atomic assembly sectors, molecular sectors, and their reduced resolvents. Second, it formulates atomic binding, shell sectors, and ionization thresholds. Third, it formulates molecular binding, reaction coordinates, potential-energy surfaces, and vibrational or rotational branches. Fourth, it distinguishes ionic, covalent, metallic, and hydrogen-bond-like sectors. Fifth, it describes chemistry as a graph of metastable sectors connected by barrier-controlled reactions. Sixth, it defines catalyst sectors. Seventh, it classifies crystalline, liquid, glassy, polymeric, and electronic material families. Eighth, it records atomic and molecular spectra and selection-rule logic. Ninth, it states the Born–Oppenheimer-type slow-fast limit. Tenth, it introduces informational chemical structures and the chemical memory

criterion.

18.2 Atomic and Molecular Composite Sectors

Definition 18.1 (Atomic constituent sectors)**.** Assume the theory contains stable nucleus-like composite sectors $\mathfrak{N}_A$ and stable electron-like sectors $\mathfrak{e}$. Their observer-level interactions are mediated by long-range charge channels together with short-range correlation, exclusion, and structural-constraint channels.

Definition 18.2 (Atomic assembly sector)**.** For a nucleus-like core $\mathfrak{N}_A$ and charge number Z, define the atomic assembly sector by

$$\mathfrak{A}_{A,Z} := \mathfrak{N}_A \otimes \mathfrak{e}^{\otimes Z} \otimes \mathfrak{R}_{\mathrm{corr}},$$

where $\mathfrak{R}_{\mathrm{corr}}$ is the correlation sector encoding screening, exclusion, pairing, and internal rearrangement.

Definition 18.3 (Molecular assembly sector)**.** For atomic sectors $\mathfrak{A}_1, \ldots, \mathfrak{A}_n$, define the molecular sector

$$\mathfrak{M}_{\mathrm{mol}} := \mathfrak{A}_1 \otimes \cdots \otimes \mathfrak{A}_n \otimes \mathfrak{B}_{\mathrm{bond}},$$

where $\mathfrak{B}_{\mathrm{bond}}$ is the bond sector encoding shared density, charge transfer, exchange correlations, and geometric stabilization.

Definition 18.4 (Composite resolvent)**.** For any composite sector X, define its observer-level reduced resolvent by

$$G_X(E, q) := \mathcal{D}_X(E, q)^{-1}.$$

An atomic or molecular sector is physically admissible when G_X has a stable isolated pole or a sufficiently narrow resonance branch.

18.3 Atomic Binding, Shell Structure, and Ionization

Definition 18.5 (Atomic bound-state pole)**.** An atomic state exists when the composite resolvent has an isolated pole

$$G_{\mathfrak{A}}(E, q) \sim \frac{Z_{\mathfrak{A}}}{E - E_{\mathfrak{A}}(q)},$$

with rest energy $E_{\mathfrak{A}}(0) = M_{\mathfrak{A}}$.

Definition 18.6 (Atomic binding energy)**.** For a nucleus-like core $\mathfrak{N}_A$ and Z separated

electron-like excitations, define

$$E_{\mathrm{bind}}^{\mathrm{atom}} := E_{\mathfrak{N}_A} + ZE_{\mathfrak{e}} - E_{\mathfrak{A}}.$$

Atomic stability requires $E_{\mathrm{bind}}^{\mathrm{atom}} > 0$.

Proposition 18.7 (Atomic stability criterion)**.** *The atomic sector $\mathfrak{A}_{A,Z}$ is stable if there exists a simple dominant composite pole below the ionization threshold, if the residue $Z_{\mathfrak{A}}$ is finite and nonzero, and if the corresponding branch survives observer coarse-graining and admissible transfer.*

Definition 18.8 (Shell sectors)**.** Electronic organization inside $\mathfrak{A}_{A,Z}$ is encoded by a finite family of shell sectors

$$\mathcal{S}_{\mathfrak{A}} = \{S_1, \dots, S_k\},$$

with occupation numbers $n_a \in \mathbb{N}_0$ satisfying $\sum_{a=1}^{k} n_a = Z$.

Proposition 18.9 (Shell energy decomposition)**.** *In the first coarse approximation one has*

$$E_{\mathfrak{A}} \approx E_{\mathfrak{N}_A} + \sum_{a=1}^{k} n_a \varepsilon_a + E_{\mathrm{corr}}(n_1, \dots, n_k),$$

where ε_a are shell energies and E_{corr} records exclusion, screening, exchange, and other many-body corrections.

Definition 18.10 (Ionization threshold)**.** The first ionization threshold is

$$I_1 := E(\mathfrak{A}_{A,Z-1}) + E_{\mathfrak{e}} - E(\mathfrak{A}_{A,Z}),$$

and more generally

$$I_k := E(\mathfrak{A}_{A,Z-k}) + kE_{\mathfrak{e}} - E(\mathfrak{A}_{A,Z}).$$

18.4 Molecular Binding and Potential-Energy Surfaces

Definition 18.11 (Molecular binding energy)**.** For a molecular sector assembled from atoms $\mathfrak{A}_1, \dots, \mathfrak{A}_n$, define

$$E_{\mathrm{bind}}^{\mathrm{mol}} := \sum_{i=1}^{n} E(\mathfrak{A}_i) - E(\mathfrak{M}_{\mathrm{mol}}).$$

Molecular stability requires $E_{\mathrm{bind}}^{\mathrm{mol}} > 0$.

Definition 18.12 (Reaction coordinate)**.** A reaction coordinate is an observer-level coarse variable Q_{rxn} parametrizing interpolation between different molecular sectors, for

example an internuclear distance, transferred charge, shared pairing amplitude, or shell-rearrangement index.

Definition 18.13 (Chemical effective potential surface)**.** For a fixed slow nucleus-like configuration R and an optimized fast electronic sector, define

$$V_{\mathrm{eff}}^{\mathrm{chem}}(R) := E_{\min}(R).$$

A stable molecule corresponds to a local minimum satisfying

$$\frac{\partial V_{\mathrm{eff}}^{\mathrm{chem}}}{\partial R} = 0, \qquad \frac{\partial^2 V_{\mathrm{eff}}^{\mathrm{chem}}}{\partial R^2} > 0.$$

Proposition 18.14 (Harmonic vibrational limit)**.** *Near a stable minimum R_*, one has the approximation*

$$V_{\mathrm{eff}}^{\mathrm{chem}}(R) \approx V_* + \frac{1}{2}K(R - R_*)^2,$$

which yields the vibrational frequency

$$\omega_{\mathrm{vib}} = \sqrt{K/\mu_{\mathrm{red}}},$$

where μ_{red} is the reduced inertia of the bonded sectors.

Definition 18.15 (Rotational branch)**.** For an approximately rigid molecule, the rotational levels satisfy

$$E_J = \frac{J(J+1)}{2I_{\mathrm{mol}}}, \qquad J = 0, 1, 2, \ldots,$$

where I_{mol} is the effective moment of inertia.

18.5 Bond Types as Composite-Sector Regimes

Definition 18.16 (Ionic bond sector)**.** An ionic bond is a stable molecular sector in which the coarse-grained charge transfer satisfies $\Delta Q \neq 0$ between atomic subclusters.

Definition 18.17 (Covalent bond sector)**.** A covalent bond is a stable molecular sector with shared electron-like assembly and negligible net transfer, characterized schematically by $\Delta Q \approx 0$ together with nonzero intersite coherence.

Example 18.18 (Minimal two-site covalent model)**.** In the minimal two-site description,

$$H_{\mathrm{cov}} = \begin{pmatrix} \varepsilon_A & -t \\ -t & \varepsilon_B \end{pmatrix},$$

with eigenvalues

$$E_{\pm} = \frac{\varepsilon_A + \varepsilon_B}{2} \pm \sqrt{\left(\frac{\varepsilon_A - \varepsilon_B}{2}\right)^2 + t^2}.$$

Bond formation occurs when the lower branch lies below the separated-atom reference.

Definition 18.19 (Metallic bond sector)**.** A metallic bond corresponds to a delocalized many-body electron-like sector with partially filled extended bands and with approximately gapless transport channels in the observer window.

Definition 18.20 (Hydrogen-bond-like sector)**.** A hydrogen-bond-like sector is a weak directional bond in which donor polarization, acceptor polarization, and geometric selectivity lower the effective energy without full charge transfer or full covalent sharing.

Proposition 18.21 (Unified bond criterion)**.** *Ionic, covalent, metallic, hydrogen-bond-like, and mixed bonds may all be read as distinct stable minima or narrow resonances of one composite effective functional built from local energies, transfer energy, exchange terms, polarization terms, and repulsion terms.*

18.6 Chemistry as a Graph of Metastable Sectors

Definition 18.22 (Chemical state graph)**.** Define the chemical graph

$$\mathcal{G}_{\mathrm{chem}} = (\mathcal{V}_{\mathrm{chem}}, \mathcal{E}_{\mathrm{chem}}),$$

whose vertices are metastable atomic or molecular sectors and whose edges are reaction channels between them.

Definition 18.23 (Reaction barrier)**.** For sectors X and Y, define the barrier

$$\Delta^{\ddagger}(X \to Y) := E_{\mathrm{TS}} - E_X,$$

where E_{TS} is the transition-state or saddle energy along an optimal reaction coordinate.

Proposition 18.24 (Arrhenius-type rate law)**.** *In a thermally activated regime one expects*

$$k_{X \to Y} \approx A e^{-\beta \Delta^{\ddagger}(X \to Y)},$$

which is the reaction-graph form of the earlier history-thermodynamic kinetics.

Definition 18.25 (Catalyst sector)**.** A catalyst is a sector C such that the barrier in its presence satisfies

$$\Delta^{\ddagger}_{X \to Y}(C) < \Delta^{\ddagger}_{X \to Y}(\varnothing),$$

while the overall free-energy difference of the reaction is left approximately unchanged.

18.7 Material Classes as Many-Body Chemical Regimes

Definition 18.26 (Material class)**.** A material class is a stable many-body phase of atomic or molecular sectors characterized by its composition pattern, bond structure, symmetry or topology class, gap structure, and coarse mechanical, electrical, or thermal response.

Definition 18.27 (Crystalline class)**.** A crystalline material has long-range positional order, equivalently detected by periodic long-range density structure or by nonzero Bragg-type amplitudes.

Definition 18.28 (Liquid class)**.** A liquid has no rigid long-range positional order, but retains finite density and strong short-range coordination.

Definition 18.29 (Glass class)**.** A glass is an amorphous metastable material class with no long-range crystalline order, slow relaxation, and a rugged landscape of metastable minima. A simple criterion is $\tau_{\text{relax}} \gg \tau_{\text{obs}}$.

Definition 18.30 (Polymeric class)**.** A polymeric material is one in which the stable chemical sectors are long chain-like assembly structures whose chain statistics control elasticity, transport, and coarse morphology.

Definition 18.31 (Electronic material classes)**.** Using gap structure, one distinguishes conductor-like, insulator-like, semiconductor-like, and paired superconducting-like regimes by the effective spectral gap and transport response of the many-body sector.

Proposition 18.32 (Material classification by three invariants)**.** *Much of material behavior is classified by the triple*

$$(\textit{order-parameter pattern},\ \textit{gap structure},\ \textit{topological class}).$$

This directly links chemistry and materials back to the previous many-body chapter.

18.8 Atomic and Molecular Spectra

Definition 18.33 (Spectral line)**.** A spectral line corresponds to a transition between two observer-level branches with frequency

$$\omega_{if} = E_f - E_i,$$

or $\hbar\omega_{if} = E_f - E_i$ if one restores explicit $\hbar$.

Proposition 18.34 (Selection-rule principle)**.** *A transition is allowed only if the matrix element of the relevant interaction operator is nonzero. Selection rules therefore arise from residue observables, symmetry classes, and the observer-decategorified coupling channel.*

Definition 18.35 (Molecular spectroscopy decomposition)**.** When adiabatic separation is valid, a molecular level decomposes approximately as

$$E \approx E_{\mathrm{elec}} + E_{\mathrm{vib}} + E_{\mathrm{rot}} + E_{\mathrm{spin}} + \cdots .$$

18.9 Born–Oppenheimer-Type Slow-Fast Separation

Definition 18.36 (Slow-fast decomposition)**.** Let R denote slow nucleus-like coordinates and r denote fast electron-like coordinates. If the slow sector is much heavier than the fast one, then the total state approximately factorizes as

$$\Psi(R, r) \approx \chi(R)\, \psi_r(r; R).$$

Proposition 18.37 (Born–Oppenheimer-type reduction)**.** *The fast sector solves an effective equation*

$$H_{\mathrm{el}}(R)\psi_r(r; R) = E_{\mathrm{el}}(R)\psi_r(r; R),$$

and the slow sector then moves in the effective potential

$$V_{\mathrm{BO}}(R) = E_{\mathrm{el}}(R) + V_{\mathrm{nuc}}(R).$$

This is the chemistry-level consequence of separating heavy and light transport sectors.

18.10 Information-Bearing Chemical Structures

Definition 18.38 (Informational chemical structure)**.** A chemical structure is informational if it admits several metastable states, if transitions between them are selective rather than purely random, and if those states can encode reproducible observer-distinguishable records.

Example 18.39 (Informational chemical motifs)**.** Abstract examples include conformational bits, switching networks, templating chains, and catalytic cycles with stable readout states.

Proposition 18.40 (Chemical memory criterion)**.** *A molecular system can function as memory if there exist two metastable sectors X_0 and X_1 such that*

$$\tau_{\mathrm{flip}} \gg \tau_{\mathrm{read}}$$

and

$$\mathcal{D}_{\mathrm{tr}}(\rho_{X_0}, \rho_{X_1}) \approx 1.$$

This links chemistry back to the observer-information layer.

18.11 Programmatic Synthesis and Failure Modes

Proposition 18.41 (Programmatic synthesis for chemistry and material classes)**.** *Assume the structural-physical theory admits EFU or assembly decomposition, composite resolvent language, interaction channels that produce bound-state poles, many-body phase structure, and observer-level material observables. Then atoms arise as stable nucleus-electron composite sectors, molecules arise as stable or metastable bonded sectors, bond types correspond to different minima of a common composite functional, reactions become transitions on a graph of metastable sectors with barrier-controlled rates, material classes arise as many-body phases of atomic or molecular sectors, and some chemical structures serve as stable information-bearing physical subsystems.*

Remark 18.42 (Failure modes)**.** Chemistry can fail at several levels: atomic poles may be absent; molecular sectors may be too broad or too short-lived; slow-fast separation may fail; material classes may lack stable coarse observables; or reaction barriers may be too ill-defined to support a metastable reaction graph.

18.12 Lean Orientation for the Next Chapter

For later formalization, the chemistry chapter should be split into six files.

1. An atomic-sectors file defining atomic constituent sectors, shell sectors, and ionization thresholds.
2. A molecular-binding file defining molecular sectors, potential-energy surfaces, vibrational branches, and rotational branches.
3. A bond-classes file separating ionic, covalent, metallic, and hydrogen-bond-like regimes.
4. A reaction-graph file defining metastable nodes, barriers, Arrhenius-type rates, and catalyst sectors.
5. A material-classes file defining crystalline, liquid, glassy, polymeric, and electronic material regimes.
6. An informational-chemistry file defining informational chemical structures and the chemical-memory criterion.

This decomposition keeps the dependency chain visible. The chemistry layer depends on the many-body, field-theoretic, thermodynamic, and observer-information layers, and it prepares the ground for biology by showing how memory-bearing and selectively transformable chemical sectors already arise within ordinary physical dynamics.

18.13 Conclusion of the Chapter

This chapter recast chemistry as the theory of stable and metastable composite sectors. Atomic binding, shell structure, molecular potentials, bond classes, reaction graphs, catalysts, material families, spectroscopy, slow-fast separation, and informational chemical structures were all placed in one kernel-residue language.

The next chapter can therefore move from chemistry to biological organization. The transition point is now explicit: once chemical sectors are simultaneously metastable, selectively transformable, and capable of supporting durable records, one can ask when they cease to be merely stable chemistry and become self-maintaining, memory-bearing, and reproducing systems.

Chapter 19

Biological Organization, Adaptive Systems, and Self-Replication

19.1 Why Biology Must Be Read as a Special Regime of Chemistry

The previous chapter established stable chemistry, metastable reaction graphs, catalyst sectors, material classes, and informational chemical structures. This is the natural point within the present framework at which biological organization is introduced. Earlier than that, one could speak about molecules and reactions, but not yet about systems that maintain themselves, preserve usable internal information, repair damage, and reproduce. Biology is therefore treated here not as a new nonphysical layer, but as a special regime of chemistry in which information-bearing composite sectors actively stabilize their own continued existence.

This methodological point is essential. A living system is not a primitive exception to physics and chemistry. It is a chemical-information-maintaining composite system whose compartment, catalytic network, hereditary sector, repair channels, and resource exchanges remain jointly viable over time. The observer-level language of life is therefore built from the already developed notions of composite sectors, reduced information, thermodynamic budgets, and metastable transformations.

This chapter does nine things. First, it defines proto-biological systems, internal state variables, viability regions, and viability functionals. Second, it formulates homeostatic attractors, maintenance cost, and the minimal resource inequality. Third, it defines catalytic networks and catalytic closure. Fourth, it introduces compartments, boundary permeability, and selective boundary criteria. Fifth, it defines hereditary sectors, replication channels, fidelity matrices, and error thresholds. Sixth, it introduces repair channels and post-repair fidelity. Seventh, it defines adaptation, predictive information, and sensing-action loops. Eighth, it formulates self-replication and persistence criteria at both individual

and population levels. Ninth, it sketches morphogenesis as a reaction-diffusion-type extension of the same viability logic.

19.2 Proto-Biological Systems and Viability

Definition 19.1 (Proto-biological system)**.** A proto-biological system is a composite object

$$\mathfrak{B} = (\mathfrak{G}, \mathfrak{M}, \mathfrak{C}, \mathfrak{K}, \mathfrak{R}, \mathcal{E}),$$

where $\mathfrak{G}$ is the hereditary template sector, $\mathfrak{M}$ the metabolic or catalytic sector, $\mathfrak{C}$ the compartment or boundary sector, $\mathfrak{K}$ the control sector, $\mathfrak{R}$ the replication and repair channel family, and $\mathcal{E}$ the effective environment.

Definition 19.2 (Internal state)**.** At the coarse-grained level, the system state is written as

$$X(t) = \big(g(t), m(t), c(t), k(t)\big),$$

where g records hereditary configuration, m metabolic or catalytic state, c compartment organization, and k regulatory state.

Definition 19.3 (Viability region)**.** There exists a subset

$$\mathcal{V}_{\text{life}} \subset \mathcal{X}$$

such that $X(t) \in \mathcal{V}_{\text{life}}$ means the system remains operationally viable: it preserves a boundary, supports catalytic activity, and does not lose hereditary records too quickly.

Definition 19.4 (Viability functional)**.** Define the viability functional by

$$\mathcal{W}[X] = \alpha_1 B[X] + \alpha_2 C_{\text{cat}}[X] + \alpha_3 I_{\text{mem}}[X] - \alpha_4 L[X],$$

where $B[X]$ measures boundary integrity, $C_{\text{cat}}[X]$ catalytic closure, $I_{\text{mem}}[X]$ retained hereditary information, and $L[X]$ leakage, disorder, or damage cost.

19.3 Homeostasis and Self-Maintenance

Definition 19.5 (Homeostatic state)**.** A homeostatic macrostate X_* satisfies

$$\dot{X} = F(X, \mathcal{E}) = 0$$

and is locally stable in the sense that the eigenvalues of the Jacobian satisfy

$$\operatorname{Re} \lambda_i\big(DF(X_*)\big) < 0.$$

Proposition 19.6 (Homeostasis criterion)**.** *If the system receives a positive resource inflow, if repair and closure channels are sufficiently fast, and if leakage and damage remain sufficiently small, then one expects a homeostatic attractor inside* $\mathcal{V}_{\text{life}}$*.*

Definition 19.7 (Maintenance cost)**.** For a system history Γ over a time interval $[0, T]$, define the maintenance cost

$$\mathcal{S}_{\text{maint}}(\Gamma) := \mathcal{S}_{\text{tot}}(\Gamma)\big|_{\text{repair+transport+regulation}},$$

and the mean maintenance power by

$$P_{\text{maint}} := \frac{\langle \mathcal{S}_{\text{maint}} \rangle}{T}.$$

Proposition 19.8 (Minimal viability inequality)**.** *For a long-lived system one requires an effective resource inequality of the form*

$$J_{\text{in}} > P_{\text{maint}} + P_{\text{leak}} + P_{\text{repl}}^{\min},$$

where J_{in} *is resource inflow,* P_{leak} *dissipative loss, and* $P_{\text{repl}}^{\min}$ *the minimal cost of hereditary upkeep or reproduction.*

19.4 Catalytic Closure

Definition 19.9 (Catalytic network)**.** Let the internal chemistry be described by a non-negative matrix A_{ij}, where A_{ij} records the contribution of component j to the production or activation of component i.

Definition 19.10 (Autocatalytic closure)**.** A catalytic network is autocatalytically closed if every essential component lies on a directed cycle of positive catalytic links.

Proposition 19.11 (Perron–Frobenius viability criterion)**.** *If the effective catalytic matrix is irreducible and its dominant eigenvalue satisfies*

$$\lambda_{\max}(A) > \delta,$$

where δ *is the effective decay or leakage rate, then the catalytic sector can amplify itself faster than it decays.*

Corollary 19.12 (Minimal autocatalytic growth law)**.** *For total catalytic mass* $M(t)$*, one obtains the coarse law*

$$\dot{M} \approx \big(\lambda_{\max}(A) - \delta\big) M.$$

Growth occurs when $\lambda_{\max}(A) > \delta$*, and collapse occurs when* $\lambda_{\max}(A) < \delta$*.*

19.5 Compartments and Selective Boundaries

Definition 19.13 (Compartment sector)**.** A compartment sector $\mathfrak{C}$ is a boundary-forming sector that separates interior from exterior, permits selective exchange, and stabilizes internal concentrations.

Definition 19.14 (Boundary permeability)**.** For each observable or particle class a, define a permeability coefficient $\kappa_a \geq 0$. The boundary flux is then written as

$$J_a^{\mathrm{bdry}} = \kappa_a \big(c_a^{\mathrm{out}} - c_a^{\mathrm{in}}\big).$$

Proposition 19.15 (Selective-boundary criterion)**.** *A compartment is biologically useful only if nutrient and waste permeabilities are sufficiently large while hereditary and catalytic-core permeabilities remain sufficiently small. The boundary must therefore pass resource flow without destroying the sectors that store memory and maintain organization.*

Definition 19.16 (Boundary maintenance energy)**.** Define the boundary-maintenance cost by

$$E_{\mathrm{bdry}} = \sigma_{\mathrm{bdry}} A_{\mathrm{surf}} + E_{\mathrm{repair}},$$

where σ_{bdry} is an effective cohesion parameter and A_{surf} is surface area.

19.6 Heredity, Replication, and Error Thresholds

Definition 19.17 (Hereditary sector)**.** A hereditary sector $\mathfrak{G}$ is an internal structure whose state space contains distinguished templates

$$g \in \mathcal{G} = \{g_1, \ldots, g_N\},$$

and whose configurations can be copied with nonzero fidelity.

Definition 19.18 (Replication channel)**.** Replication is a map of the form

$$\mathcal{R} : (g, m, c, k) + \text{resources} \longrightarrow (g, m, c, k) \oplus (g', m', c', k').$$

In the ideal case $g' = g$; more generally $g' = \mu(g)$ for a mutation or noise channel μ.

Definition 19.19 (Fidelity matrix)**.** Define

$$Q_{ij} = P(g_j \to g_i \text{ after replication}), \qquad \sum_i Q_{ij} = 1.$$

The diagonal elements Q_{ii} are replication fidelities.

Proposition 19.20 (Replicator-mutator dynamics)**.** *For population fractions $x_i(t)$ of hereditary types one has the effective dynamics*

$$\dot{x}_i = \sum_j Q_{ij} w_j x_j - \Phi x_i, \qquad \Phi = \sum_k w_k x_k,$$

where w_j are type-dependent growth rates.

Definition 19.21 (Error threshold)**.** If the master template g_0 has fitness w_0, then an error threshold occurs when fidelity becomes too small to maintain concentration near g_0. A simple threshold criterion is

$$Q_{00} w_0 \leq \max_{i \neq 0} w_i.$$

Corollary 19.22 (Template stability criterion)**.** *Long hereditary memory requires*

$$Q_{00} w_0 > \max_{i \neq 0} w_i.$$

19.7 Repair, Adaptation, and Informational Closure

Definition 19.23 (Repair channel)**.** A repair channel is a correction map

$$\mathcal{C} : g_{\text{damaged}} \mapsto g_{\text{restored}},$$

driven by catalytic or template-guided sectors.

Definition 19.24 (Effective post-repair fidelity)**.** If the raw fidelity is q and repair suppresses error by a factor $r \in [0, 1]$, define the effective fidelity by

$$q_{\text{eff}} = 1 - r(1 - q).$$

Proposition 19.25 (Repair-assisted heredity)**.** *Replacing Q_{00} by an effective repaired fidelity Q_{00}^{eff} can restore the template-stability inequality even when the unrepaired replication process would fail it.*

Definition 19.26 (Environment-dependent fitness)**.** Let the growth or viability weight of hereditary type i depend on environment,

$$w_i = w_i(\mathcal{E}).$$

Adaptation then means that the system changes its state distribution or regulatory sector so as to improve viability under a changing environment.

Definition 19.27 (Adaptive control functional)**.** Define

$$\mathcal{A}[k] = \beta_1 \mathcal{W}[X_k] + \beta_2 I(\text{mem} : \mathcal{E}) - \beta_3 C_{\text{ctrl}}[k],$$

where $\mathcal{W}$ is viability, $I(\text{mem} : \mathcal{E})$ useful environmental information, and C_{ctrl} control cost.

Proposition 19.28 (Adaptive update law)**.** *A natural mesoscopic adaptive law is the gradient-type equation*

$$\dot{k} = \eta \nabla_k \mathcal{A}[k].$$

Definition 19.29 (Homeostatic adaptation)**.** A system is adaptive if, after an environmental perturbation $\mathcal{E} \to \mathcal{E}'$, it can return to the viability region at a later time.

Definition 19.30 (Informational closure)**.** A system has informational closure if its internal dynamics depends not only on the raw environment but on an internal encoded model,

$$\dot{X} = F\big(X, \hat{\mathcal{E}}(X)\big),$$

where $\hat{\mathcal{E}}(X)$ is an internal representation of relevant conditions.

Definition 19.31 (Predictive information)**.** For memory M and future environment E^+, define the predictive information

$$I_{\text{pred}} := I(M : E^+).$$

Proposition 19.32 (Useful-information criterion)**.** *Memory is biologically useful only if increasing predictive information increases viability, at least in the effective observer description.*

19.8 Self-Replication and Population Persistence

Definition 19.33 (Sensing-action loop)**.** A sensing-action loop is a closed feedback structure

$$\mathcal{E} \to \text{sensor} \to \text{memory/control} \to \text{action} \to \mathcal{E}.$$

Definition 19.34 (Self-replicating system)**.** A system is self-replicating if there exists a channel

$$\mathcal{R}_{\text{self}} : X + \text{resources} \to X + X' + \text{waste},$$

such that X' remains structurally close to X, both copies remain viable for nonzero time, and the process is repeatable.

Definition 19.35 (Effective reproductive number)**.** Define the effective reproductive number by

$$\mathcal{R}_0 := \text{expected viable offspring count per parent system.}$$

Proposition 19.36 (Persistence criterion)**.** *A lineage persists only if*

$$\mathcal{R}_0 > 1.$$

If $\mathcal{R}_0 < 1$, the lineage dies out.

Theorem 19.37 (Target theorem: Self-replication criterion)**.** *A proto-biological system becomes a genuine self-replicating system if catalytic closure holds, the compartment remains selectively stable, heredity stays above the error threshold, repair maintains sufficiently high effective fidelity, and the reproductive number satisfies $\mathcal{R}_0 > 1$.*

Definition 19.38 (Population dynamics)**.** For populations $N_i(t)$ of several types, define

$$\dot{N}_i = (b_i - d_i)N_i + \sum_j M_{ij}N_j,$$

where b_i are birth or replication rates, d_i decay rates, and M_{ij} conversion or mutation couplings.

Definition 19.39 (Resource competition)**.** With shared resource $R(t)$ one may write

$$\dot{R} = J_{\text{in}} - \sum_i \alpha_i N_i R - J_{\text{out}},$$

so growth rates become resource dependent.

Proposition 19.40 (Ecological fixed point)**.** *An ecological steady state (N_i^*, R^*) satisfies $\dot{N}_i = 0$ and $\dot{R} = 0$, and its stability is determined by the Jacobian of the coupled population-resource system.*

19.9 Morphogenesis and Pattern Formation

Definition 19.41 (Morphogenetic field)**.** A developing structured system may be described by coarse fields $u_a(x, t)$ obeying reaction-diffusion-type equations

$$\partial_t u_a = D_a \Delta u_a + F_a(u_1, \ldots, u_n).$$

Proposition 19.42 (Pattern-formation criterion)**.** *A homogeneous state becomes morphogenetically unstable when it is stable to homogeneous perturbations but unstable to some inhomogeneous mode with nonzero wave number. Spatial organization then emerges as a self-organized pattern sector.*

19.10 Programmatic Synthesis and Failure Modes

Proposition 19.43 (Programmatic synthesis for biological organization)**.** *Assume the theory admits stable chemical sectors, a catalytic network, a selective compartment, hereditary templates with fidelity matrix Q, repair channels, an adaptive control sector, and*

resource flux from the environment. Then self-maintaining systems are characterized by positive viability, catalytic closure requires $\lambda_{\max}(A) > \delta$, stable heredity requires survival above the error threshold, adaptation is the restoration of viability under environmental perturbation, and true self-replication requires simultaneous homeostasis, heredity above threshold, and reproductive excess $\mathcal{R}_0 > 1$.

Remark 19.44 (Failure modes)**.** Biological organization can fail if catalytic closure is absent, if the boundary is not selective, if fidelity falls below error threshold, if repair costs exceed the action budget, or if $\mathcal{R}_0 \leq 1$ so that no lineage persists.

19.11 Lean Orientation for the Next Chapter

For later formalization, the biology chapter should be split into six files.

1. A viability-and-homeostasis file defining proto-biological systems, viability regions, and viability functionals.
2. A catalytic-closure file defining catalytic matrices, closure criteria, and growth laws.
3. A compartment-and-boundary file defining permeability, selective-boundary criteria, and boundary-maintenance cost.
4. A heredity-and-replication file defining template sectors, fidelity matrices, error thresholds, and repair channels.
5. An adaptation-and-information file defining adaptive control, informational closure, predictive information, and sensing-action loops.
6. A self-replication-and-population file defining reproductive number, persistence criteria, ecological fixed points, and morphogenetic extensions.

This split keeps the logic explicit: biology is introduced as a special class of chemical systems that preserve and use information to maintain themselves. It depends on the chemistry, many-body, thermodynamic, and observer-information chapters, and it prepares the ground for cognition by making internal memory, control, and predictive use of information into central dynamical objects.

19.12 Conclusion of the Chapter

This chapter introduced biology as a chemical-information-maintaining regime of the theory. Viability, homeostasis, catalytic closure, selective boundaries, heredity, repair, adaptation,

predictive information, self-replication, population persistence, and morphogenesis were all stated in one composite-sector language.

The next chapter can therefore move to cognition, agency, language, and collective symbolic systems. The transition point is now explicit: once internal memory and control stop serving only viability maintenance and begin to support prediction, policy choice, reusable symbols, and inter-agent coordination, the theory passes from adaptive life-like organization to cognition proper.

Chapter 20

Cognition, Agency, Language, and Collective Symbolic Systems

20.1 Why Cognition Must Follow Adaptive Biological Organization

The previous chapter introduced viability, homeostasis, catalytic closure, heredity, repair, adaptation, predictive information, and self-replication. This is the natural point within the present framework at which cognition is introduced. Earlier than that, one could describe memory-bearing adaptive systems, but not yet systems that build internal models, select actions through those models, exchange reusable symbols, and stabilize collective codes. Cognition is therefore treated here not as a foreign addition to biology, but as a further specialization of adaptive information-maintaining systems.

This point is structurally important for the whole book. The project began with symbolic structures, semantic maps, admissible transformations, and quotient levels. Once adaptive living systems begin to use internal states as persistent, action-guiding surrogates for the world, the book has reached the point where symbolic organization reappears inside biology itself. Cognition is therefore the controlled use of internal symbolic states for prediction and action, and language is the multi-agent stabilization of that same symbolic organization.

This chapter does ten things. First, it defines cognitive systems, internal symbolic states, and perception maps. Second, it defines representation, predictive models, and a minimal cognition criterion. Third, it defines action sets, policies, utility functionals, and the agency criterion. Fourth, it introduces learning dynamics and model-complexity tradeoffs. Fifth, it defines symbol sets, semantics, and compositionality. Sixth, it defines communication channels, messages, and shared codes. Seventh, it introduces collective lexica, semantic alignment, collective memory, norms, and institutions. Eighth, it defines cognitive hierarchies and predictive-control decompositions. Ninth, it defines self-models

and reflexive cognition. Tenth, it states the chapter's programmatic synthesis for cognition and collective symbolic systems.

20.2 Cognitive Systems and Internal Symbolic States

Definition 20.1 (Cognitive system)**.** A cognitive system is a composite object

$$\mathfrak{C} = (X, M, S, A, \Pi, \mathcal{U}, \mathcal{L}),$$

where X is the external or environmental sector, M the internal memory or model sector, S the sensory encoding sector, A the action or control sector, Π the perception map, $\mathcal{U}$ the internal update dynamics, and $\mathcal{L}$ the learning dynamics.

Definition 20.2 (Internal symbolic state)**.** An internal symbolic state is an element $m \in M$ that is observer-distinguishable, is not reducible to instantaneous sensory input alone, and influences future action or inference.

Definition 20.3 (Perception map)**.** If $x \in X$ is an external state, perception is the map

$$\Pi : X \to S,$$

and memory is updated by a rule of the form

$$m_{t+1} = \mathcal{U}(m_t, \Pi(x_t), a_t).$$

20.3 Internal Models and Prediction

Definition 20.4 (Representational adequacy)**.** An internal state m represents an external feature $f(x)$ if there exists a decoding map

$$D : M \to \mathcal{F}$$

such that $D(m) \approx f(x)$ in the operationally relevant regime.

Definition 20.5 (Predictive model)**.** An internal model is predictive if, from m_t and candidate action a_t, it can estimate future-relevant observables in the form

$$\widehat{x}_{t+\Delta t} = \mathcal{P}(m_t, a_t).$$

Proposition 20.6 (Minimal cognition criterion)**.** *A system is minimally cognitive if it possesses nontrivial memory, a decode map, and a predictive map such that expected viability under predictive control exceeds expected viability under purely reactive action.*

20.4 Agency, Policy, and Value

Definition 20.7 (Action set)**.** Let the system have an action repertoire

$$\mathcal{A} = \{a_1, \dots, a_n\}.$$

An action changes the environment, the system state, or both.

Definition 20.8 (Policy)**.** A policy is a map

$$\pi : M \to \Delta(\mathcal{A}),$$

where $\Delta(\mathcal{A})$ is the probability simplex over actions.

Definition 20.9 (Goal or utility functional)**.** Define the effective objective

$$\mathcal{J} = \mathbb{E}\Big[\sum_{t \geq 0} \gamma^t U(x_t, m_t, a_t)\Big], \qquad 0 < \gamma \leq 1.$$

In survival-centered regimes one may take $U \sim \mathcal{W}$, where $\mathcal{W}$ is the viability functional of the previous chapter.

Proposition 20.10 (Agency criterion)**.** *A system is agent-like if its action distribution depends irreducibly on internal state, so that*

$$P(a_t \mid x_t, m_t) \neq P(a_t \mid x_t).$$

Definition 20.11 (Effective value function)**.** For a policy π, define

$$V^\pi(m) = \mathbb{E}_\pi\Big[\sum_{t \geq 0} \gamma^t U_t \mid m_0 = m\Big].$$

Proposition 20.12 (Bellman-type recursion)**.** *The effective value function satisfies the recursion*

$$V^\pi(m) = \sum_a \pi(a \mid m)\Big[U(m, a) + \gamma\, \mathbb{E}\big[V^\pi(m') \mid m, a\big]\Big].$$

20.5 Learning Dynamics

Definition 20.13 (Learning update)**.** Learning is a slow dynamics on parameters θ of the perception map, predictive model, or policy,

$$\theta_{t+1} = \theta_t + \eta\, \Delta_\theta.$$

Definition 20.14 (Prediction error)**.** Define the prediction error by

$$\varepsilon_t = d(\widehat{x}_{t+1}, x_{t+1}),$$

where d is a relevant distance on coarse-grained observables.

Proposition 20.15 (Error-driven learning)**.** *A natural learning rule is*

$$\Delta_\theta = -\nabla_\theta \mathbb{E}[\varepsilon_t],$$

or more generally a gradient on the effective objective $\mathcal{J}$.

Definition 20.16 (Model-class complexity)**.** If θ parametrizes a model class, define its complexity cost by $C_{\text{model}}(\theta)$.

Proposition 20.17 (Bias-complexity tradeoff)**.** *Effective cognitive performance balances prediction quality and model cost through a functional of the form*

$$\mathcal{J}_{\text{cog}} = -\mathbb{E}[\varepsilon] - \lambda C_{\text{model}}.$$

20.6 Symbols, Semantics, and Compositionality

Definition 20.18 (Symbol set)**.** A symbol system is a finite or countable set

$$\Sigma = \{\sigma_1, \dots, \sigma_k\}$$

together with admissible composition rules.

Definition 20.19 (Internal symbolic encoding)**.** A system uses symbols if there exists an encoding map

$$E_{\text{sym}} : M \to \Sigma^*$$

and a decoding map

$$D_{\text{sym}} : \Sigma^* \to M'$$

such that symbolic strings preserve or transmit task-relevant structure.

Definition 20.20 (Task-relative semantics)**.** A symbol string $w \in \Sigma^*$ has semantics relative to a task if its interpreted content stably changes prediction, action, or coordination in the relevant observer regime.

Definition 20.21 (Compositionality)**.** A symbolic system is compositional if there exists a composition operation $\circ$ such that

$$[\![w_1 \circ w_2]\!] = \mathcal{C}\big([\![w_1]\!], [\![w_2]\!]\big)$$

for an appropriate semantic composition rule $\mathcal{C}$.

Proposition 20.22 (Minimal language criterion)**.** *A system has language-like organization if it has a reusable symbol set, symbols can be composed, composite strings systematically alter interpretation or action, and the same symbol tokens recur across contexts.*

20.7 Communication and Shared Codes

Definition 20.23 (Communication channel)**.** For agents A and B, a communication channel is a map

$$\mathcal{T}_{A\to B} : \Sigma_A^* \to \Sigma_B^*,$$

possibly noisy.

Definition 20.24 (Message)**.** A message is a symbolic object $w \in \Sigma^*$ whose transmission changes the receiver's posterior internal state.

Definition 20.25 (Communication fidelity)**.** Communication fidelity is the probability that the receiver's decoded content approximately matches the sender's intended decoded content.

Proposition 20.26 (Useful communication criterion)**.** *Communication is useful only if it improves the receiver's expected objective relative to the no-message baseline.*

Definition 20.27 (Shared code)**.** A shared code exists if two or more agents use approximately compatible encoding and decoding maps on task-relevant states.

20.8 Collective Symbolic Systems

Definition 20.28 (Symbolic population)**.** A symbolic population is a set of agents

$$\mathcal{P} = \{A_1, \ldots, A_N\}$$

with communication graph $\mathcal{G}_{\text{comm}} = (V, E)$.

Definition 20.29 (Collective lexicon)**.** A collective lexicon is the shared statistical distribution of symbol use across the population.

Definition 20.30 (Semantic alignment)**.** Semantic alignment is measured by the average decode agreement across communication links. It records how consistently the same symbolic strings are interpreted inside the population.

Proposition 20.31 (Alignment dynamics)**.** *If communication is repeated and feedback exists, semantic alignment evolves dynamically and may approach a stable attractor above a critical threshold needed for a persistent language-like regime.*

Definition 20.32 (Collective memory)**.** Collective memory is information stored not in any one agent alone but in persistent distributed structures such as shared symbolic records, institutions, external archives, or stable network specializations.

Definition 20.33 (Norm)**.** A norm is a symbolic rule that modifies policy selection.

Definition 20.34 (Institution)**.** An institution is a stable collectively enforced symbolic rule system consisting of symbolic prescriptions, role-update rules, and enforcement channels.

Proposition 20.35 (Institutional stability criterion)**.** *An institution is stable if its coordination, prediction, and trust gains exceed the costs of enforcement, communication, and conflict.*

20.9 Cognitive Hierarchies and Reflexive Self-Models

Definition 20.36 (Cognitive hierarchy)**.** A system is multiscale cognitive if its internal model decomposes into levels

$$M = M^{(0)} \oplus M^{(1)} \oplus \cdots \oplus M^{(L)},$$

with different temporal or spatial scopes.

Definition 20.37 (Predictive-control decomposition)**.** At each scale ℓ, one has a model state $m^{(\ell)}$, a prediction $\widehat{x}^{(\ell)}$, and a control policy $\pi^{(\ell)}$, and the levels evolve by coupled update laws.

Proposition 20.38 (Agency amplification by hierarchy)**.** *Hierarchical cognition improves control if long-horizon prediction gains exceed the maintenance cost of the hierarchy.*

Definition 20.39 (Self-model)**.** A self-model is an internal representation $m_{\text{self}} \in M$ encoding coarse-grained information about the system's own state, capabilities, or expected trajectories.

Definition 20.40 (Reflexive cognition)**.** A system is reflexive if its updates depend on such self-models in addition to external perception and action.

Proposition 20.41 (Reflexive stability criterion)**.** *A self-model is useful only if the information it carries about the system improves control or viability.*

20.10 Programmatic Synthesis and Failure Modes

Proposition 20.42 (Programmatic synthesis for cognition and collective symbolic systems)**.** *Assume an adaptive self-maintaining system with an internal memory sector, a*

predictive model, a policy, learning dynamics, symbolic encoding and decoding maps, and communication channels between agents. Then cognition appears when internal symbolic states improve future prediction and action; agency appears when action selection depends irreducibly on internal states; language-like structure appears when reusable compositional symbols stably alter prediction and coordination; collective symbolic systems appear when populations stabilize shared codes, collective memory, and institutional rules; and reflexive cognition appears when the system constructs and uses self-models in control.

Remark 20.43 (Failure modes)**.** Cognition can fail if internal memory is not stable, if internal states do not affect policy, if symbols are not reusable or compositional, if communication fidelity is too low for a shared code, or if the costs of collective symbolic maintenance exceed the gains.

20.11 Chapter Audit Summary

- Definitions introduced: cognitive systems, internal symbolic states, predictive models, policies, symbol systems, communication channels, institutions, and self-models.
- Standing assumptions: adaptive self-maintaining systems with stable internal memory, reusable symbols, and communication channels of sufficient fidelity.
- Internally established results: the organization of cognition as predictive symbolic control, the agency criterion, and the collective-symbolic-system layer linking language, norms, and institutions.
- Imported identifications used: standard decision-theoretic, communication-theoretic, and symbolic-system terminology at observer level.
- Open targets deferred: rigorous quantitative semantics and a complete formal closure of collective symbolic dynamics.
- Downstream dependence: the mathematics-and-logic chapter uses this stabilized symbolic layer as its immediate precursor.

20.12 Lean Orientation for the Next Chapter

For later formalization, the cognition chapter should be split into six files.

1. A cognitive-system file defining memory sectors, perception maps, and internal symbolic states.
2. A prediction-and-agency file defining predictive models, policies, utility functionals, and agency criteria.

3. A learning file defining update rules, prediction error, and model-complexity tradeoffs.
4. A symbols-and-language file defining symbol sets, semantics, compositionality, and minimal language criteria.
5. A communication-and-collective-systems file defining shared codes, semantic alignment, collective memory, norms, and institutions.
6. A hierarchy-and-reflexivity file defining multiscale cognitive hierarchies, self-models, and reflexive control.

This split keeps the dependence chain explicit. The cognition chapter depends on adaptive biology, observer-information theory, and the original symbolic-structure layer, and it prepares the ground for mathematics and logic by making collective symbolic stabilization a precise dynamical object rather than a metaphor.

20.13 Conclusion of the Chapter

This chapter introduced cognition as the controlled use of internal symbolic states for prediction and action. It then extended that same logic to agency, learning, language, communication, collective symbolic systems, institutions, cognitive hierarchies, and self-models.

The next chapter can therefore move to mathematics, logic, and formal knowledge as symbolic fixed points. The transition point is now explicit: once collective symbolic systems stabilize shared codes, archives, norms of acceptance, and reusable inferential practices, they no longer merely coordinate action but begin to stabilize proof, computation, abstraction, and verification as persistent symbolic regimes.

Chapter 21

Mathematics, Logic, and Formal Knowledge as Symbolic Fixed Points

21.1 Why Mathematics Must Be Read as a Stable Regime of Collective Symbolic Dynamics

The previous chapter introduced cognition, agency, language, communication, collective symbolic systems, institutional rules, and reflexive self-models. This is the natural point within the present framework at which mathematics and logic are introduced. Earlier than that, one could speak about symbols and shared codes, but not yet about stabilized proof objects, admissible inference, trusted verification, or collective archives of formal knowledge. Mathematics is therefore treated here not as an external metadiscipline imposed from outside the theory, but as a special stable regime of collective symbolic dynamics.

This point returns the book to its own starting layer. The project began from symbolic structures, semantic maps, admissible transformations, and towers of equivalence. Mathematics now reappears as the special case in which symbolic transformations are controlled so strictly that proof, computation, abstraction, and verification become durable collective fixed points. Logic is the control layer on admissible symbolic transformations; formal knowledge is the fixed point of proof, checking, revision, and archival stabilization.

This chapter does ten things. First, it defines formal symbolic systems, proof objects, provability, and semantic truth. Second, it defines logic as the control of admissible symbolic transformations through inference operators and theory closure. Third, it introduces consistency, soundness, and completeness. Fourth, it formulates computation as symbolic dynamics with structural cost. Fifth, it defines mathematics as invariant extraction across levels of equivalence. Sixth, it defines verified knowledge and verification closure. Seventh, it introduces formal communities, shared proof repositories, and collective theorem acceptance. Eighth, it defines self-correcting formal cultures and revision dynamics. Ninth, it formulates machine verification and trusted kernels. Tenth, it states self-reference

and incompleteness boundaries.

21.2 Formal Symbolic Systems, Proof, and Truth

Definition 21.1 (Formal symbolic system)**.** A formal symbolic system is an object

$$\mathfrak{F} = (X, [\![\cdot]\!], R, \mathcal{A}, \mathcal{J}),$$

where X is the class of well-formed symbolic objects, $[\![\cdot]\!]$ the semantics or interpretation map, R the admissible transformations, $\mathcal{A} \subset X$ the accepted axioms or initial objects, and $\mathcal{J}$ the judgment structure distinguishing formulas, sequents, proofs, programs, or other admissible judgments.

Definition 21.2 (Proof object)**.** A proof object π is a causal-compatible history of admissible symbolic transformations,

$$\pi : \quad x_0 \xrightarrow{r_1} x_1 \xrightarrow{r_2} \cdots \xrightarrow{r_n} x_n, \qquad r_k \in R,$$

where x_0 starts from axioms or hypotheses and x_n is the final theorem or judgment.

Definition 21.3 (Provability relation)**.** The notation

$$\mathcal{A} \vdash_{\mathfrak{F}} \varphi$$

means that there exists a proof object leading from the accepted basis $\mathcal{A}$ to the judgment φ.

Definition 21.4 (Semantic truth)**.** The notation

$$\mathcal{A} \models \varphi$$

means that the judgment φ is true in all admissible interpretations compatible with $\mathcal{A}$.

21.3 Logic as Control of Admissible Transformation

Definition 21.5 (Logic kernel)**.** The logic kernel is the distinguished subset of admissible rules

$$R_{\log} \subset R$$

that acts on symbolic judgments and derivations rather than on the broader physical or cognitive state space.

Definition 21.6 (Inference operator)**.** For a theory fragment $K \subset X$, define the inference operator

$$T(K) := K \cup \mathrm{Infer}(K),$$

where $\mathrm{Infer}(K)$ is the set of all judgments obtainable from K in one admissible inference step.

Definition 21.7 (Theory closure)**.** The closure of the theory is

$$\mathrm{Cl}(K) := \bigcup_{n \geq 0} T^n(K).$$

Equivalently, one may write the resulting theory as the least fixed point

$$\mathrm{Th}(K) = \mathrm{lfp}(T_K).$$

Proposition 21.8 (Monotonicity of closure)**.** *If $K_1 \subseteq K_2$, then*

$$\mathrm{Cl}(K_1) \subseteq \mathrm{Cl}(K_2).$$

Proposition 21.9 (Idempotence of closure)**.** *One has*

$$\mathrm{Cl}(\mathrm{Cl}(K)) = \mathrm{Cl}(K).$$

21.4 Consistency, Soundness, and Completeness

Definition 21.10 (Consistency)**.** A theory K is consistent if there is no judgment φ such that both $K \vdash \varphi$ and $K \vdash \neg\varphi$.

Definition 21.11 (Soundness)**.** A formal system is sound relative to its semantics if

$$K \vdash \varphi \Rightarrow K \models \varphi.$$

Definition 21.12 (Completeness)**.** A formal system is complete relative to its semantics if

$$K \models \varphi \Rightarrow K \vdash \varphi.$$

Proposition 21.13 (Proof as semantic-preserving history)**.** *If each rule in the logic kernel preserves semantic validity, then an entire proof history preserves semantic validity from premises to conclusion.*

21.5 Computation as Symbolic Dynamics with Structural Cost

Definition 21.14 (Program or algorithm)**.** An algorithm is a symbolic object $p \in X$ together with an evaluation dynamics of the form

$$(p, \text{input}) \xrightarrow{R^*_{\text{comp}}} \text{output}.$$

Definition 21.15 (Computation history)**.** A computation history is a finite sequence of symbolic states

$$\Gamma_{\text{comp}} : \quad y_0 \to y_1 \to \cdots \to y_n.$$

Definition 21.16 (Computed function)**.** The function computed by program p is the map $f_p(u) = v$ whenever a computation history starting from (p, u) terminates at output v.

Definition 21.17 (Computational structural cost)**.** Using the structural action already introduced in the earlier physical chapters, define the computational cost by

$$C_{\text{comp}}(p, u) := \min_{\Gamma_{\text{comp}}:(p,u)\to v} \mathcal{S}(\Gamma_{\text{comp}}).$$

Proposition 21.18 (Proofs as computations)**.** *Every proof may be treated as a computation over judgments that produces a proof witness from accepted premises and target statement.*

Proposition 21.19 (Programs as proofs and proofs as programs)**.** *In appropriately typed systems one expects a Curry–Howard-like correspondence relating propositions to types and proofs to programs. In the present book this is best treated as a target structural bridge rather than as an already derived theorem of the general framework.*

21.6 Mathematics as Invariant Extraction Across Equivalence Levels

Definition 21.20 (Mathematical theory)**.** A mathematical theory is a formal symbolic system whose closure under inference is stable, whose semantics is sufficiently sharp, whose proofs are externally checkable, and whose abstractions are reusable across many concrete realizations.

Definition 21.21 (Abstract object)**.** An abstract mathematical object is an equivalence class

$$[x]_{\sim}$$

under an admissible structural equivalence, such as normalization equivalence, reduction equivalence, h-equivalence, semantic equivalence, or categorical isomorphism.

Definition 21.22 (Mathematical invariant)**.** A property $I(x)$ is a mathematical invariant if

$$x \sim y \Rightarrow I(x) = I(y).$$

Proposition 21.23 (Mathematics as invariant extraction)**.** *In the language of the present project, mathematics is the disciplined extraction and study of invariants of symbolic structures under admissible transformations and equivalence relations.*

Definition 21.24 (Symbolic fixed point of knowledge)**.** Let

$$\mathcal{V}(K) = \text{validated closure of } K$$

denote proof, checking, normalization, and consistency filtering applied to K. Then a formal knowledge state K_* is a symbolic fixed point if

$$\mathcal{V}(K_*) = K_*.$$

21.7 Verified Knowledge and Verification Closure

Definition 21.25 (Knowledge base)**.** A knowledge base is a pair

$$\mathfrak{K} = (K, \mathcal{P}_K),$$

where K is the set of accepted statements and $\mathcal{P}_K$ stores proof certificates, derivation witnesses, or verification metadata.

Definition 21.26 (Verification map)**.** The verification map is

$$\mathrm{Ver} : (\varphi, \pi) \mapsto \{0, 1\},$$

with value 1 meaning that π is an admissible proof certificate for φ.

Definition 21.27 (Verified knowledge)**.** A statement φ belongs to verified knowledge if there exists a certificate π such that

$$\mathrm{Ver}(\varphi, \pi) = 1.$$

Proposition 21.28 (Verification closure)**.** *Define the verified extension operator by*

$$\mathcal{V}(K) := K \cup \{\varphi : \exists \pi,\ \mathrm{Ver}(\varphi, \pi) = 1\}.$$

Then verified knowledge is the least fixed point of repeated verified extension,

$$K_* = \mathrm{lfp}(\mathcal{V}).$$

21.8 Formal Communities and Collective Stabilization

Definition 21.29 (Formal community)**.** A formal community is a population of agents sharing a symbol code, verification rules, admissible proof formats, and an archive of accepted statements.

Definition 21.30 (Shared proof repository)**.** A shared proof repository is the structured archive of tuples $(\varphi, \pi, \text{meta})$ storing accepted statements together with proof certificates and metadata.

Definition 21.31 (Collective theorem acceptance)**.** A theorem is collectively accepted if a certificate exists in the shared repository and every agent in the formal community verifies it according to the agreed proof standards.

Proposition 21.32 (Institutional stabilization of mathematics)**.** *Formal mathematics becomes socially stable only when the gains of reuse, prediction, and coordination exceed the costs of verification, communication, and disagreement.*

21.9 Revision, Machine Verification, and Trusted Kernels

Definition 21.33 (Error-correcting formal culture)**.** A symbolic community is self-correcting if it contains channels for proof checking, contradiction detection, revision of flawed derivations, and preservation of correct derivations.

Definition 21.34 (Revision dynamics)**.** If K_t is the currently accepted body of knowledge, a revision step has the form

$$K_{t+1} = \mathcal{R}_{\text{rev}}(K_t, \Delta_t),$$

where Δ_t is new proof, refutation, countermodel, or consistency conflict.

Proposition 21.35 (Stable formal-culture criterion)**.** *A formal culture is stable if revision and repair reduce inconsistency faster than novelty and error create it.*

Definition 21.36 (Machine-checkable proof)**.** A proof certificate π is machine-checkable if

$$\text{Ver}_{\text{machine}}(\varphi, \pi) = 1$$

in a fixed trusted kernel.

Definition 21.37 (Trusted kernel)**.** A trusted kernel is the minimal rule set

$$\mathcal{K}_{\text{trust}}$$

used by a proof assistant or verifier to decide certificate validity.

Proposition 21.38 (Formal compression of trust)**.** *Machine verification compresses social trust from the whole informal derivation into trust in the kernel and in the encoding of the formal object.*

21.10 Self-Reference and Incompleteness Boundaries

Definition 21.39 (Self-reference)**.** A formal system admits self-reference if some symbolic objects can encode statements about their own derivability, truth, size, or verification.

Proposition 21.40 (Fixed-point or diagonal principle)**.** *In sufficiently expressive symbolic systems one expects a fixed-point construction producing a sentence ψ satisfying a relation of the form*

$$\psi \leftrightarrow F(\ulcorner \psi \urcorner).$$

Definition 21.41 (Incompleteness boundary)**.** A theory is incompleteness-limited if there exist semantically meaningful statements whose truth is not fully captured by derivability inside the theory, or if consistency is preserved only by leaving some such statements undecided. Here "semantically meaningful" is always understood relative to the semantic layer adopted in this chapter.

21.11 Programmatic Synthesis and Failure Modes

Proposition 21.42 (Programmatic synthesis for mathematics, logic, and formal knowledge)**.** *Assume a collective symbolic system with symbolic objects, semantics, admissible transformations, an inference operator, a verification map, a collective archive, and a trusted proof kernel. Then logic appears as control of admissible symbolic transformations, formal theories appear as least fixed points of inference closure, verified knowledge appears as a fixed point of proof plus checking plus revision dynamics, mathematics appears as the study of invariants stable under admissible transformations and equivalences, machine formalization creates especially robust collective symbolic fixed points by compressing trust into a small kernel, and sufficiently expressive systems acquire self-reference and corresponding internal limits.*

Remark 21.43. This proposition is a structured synthesis of the chapter rather than a single narrow theorem with homogeneous proof status.

Remark 21.44 (Failure modes)**.** Formal knowledge can fail if no stable syntax or encoding layer persists, if admissible transformations do not preserve intended semantics, if verification is too costly or socially unstable, if the trusted kernel is ambiguous or unstable, or if the symbolic system is either too weak for abstraction or too uncontrolled to remain consistent.

21.12 Chapter Audit Summary

- Definitions introduced: formal symbolic systems, provability, semantic truth, inference closure, verified knowledge, trusted kernels, and incompleteness boundaries.
- Standing assumptions: stable symbolic syntax, admissible inference operators, verification rules, and a collectively maintained proof archive.
- Internally established results: the fixed-point language for theories and verified knowledge, the computation-as-symbolic-dynamics viewpoint, and the role of machine-checkable kernels.
- Imported identifications used: standard logical, proof-theoretic, and computation-theoretic terminology.
- Open targets deferred: a complete formal Curry–Howard-type bridge for the whole framework and a full machine-checked formalization of the theory.
- Downstream dependence: the final audit chapter uses this chapter to frame formalization policy and theorem-status discipline.

21.13 Lean Orientation for the Next Chapter

For later formalization, the mathematics-and-logic chapter should be split into six files.

1. A formal-system file defining symbolic objects, judgments, proof objects, and semantic truth.
2. A logic-and-closure file defining inference operators, closure, fixed points, consistency, soundness, and completeness.
3. A computation-and-cost file defining computation histories, structural cost, and proof-computation correspondences.
4. An invariants-and-abstraction file defining abstract objects, invariants, and mathematical theory.
5. A verification-and-community file defining verification maps, proof repositories, collective theorem acceptance, revision, and trusted kernels.
6. A self-reference-and-limits file defining diagonal principles, incompleteness boundaries, and machine-checkable permanence targets.

This split keeps the return to the symbolic layer explicit. The chapter depends on the original structural core and on the later cognition-and-language chapters, and it closes the long arc of the book by showing how mathematics and formal knowledge emerge as especially stable symbolic regimes inside the same overall ontology.

21.14 Conclusion of the Chapter

This chapter recast mathematics, logic, proof, computation, and verification as stable symbolic fixed points. Formal systems, closure operators, soundness, structural computation cost, abstraction by invariants, verified knowledge, formal communities, trusted kernels, and incompleteness boundaries were all organized inside one symbolic-dynamical language.

The next step in the manuscript is therefore no longer another macro-domain above mathematics. It is the return to specialized microphysics. The transition point is now explicit: once the book has recovered symbolic invariants, admissible transformations, and stable verification layers, it can re-enter particle physics at a more disciplined level and reinterpret charges, sectors, and kernels as internal residue data rather than as externally attached labels.

Chapter 22

Residue Calculus as Preparation for the Standard Model

22.1 Why the Standard Model Block Must Begin with Residue Calculus

The previous chapter returned the manuscript to symbolic structures, invariants, proof objects, and fixed points of verified knowledge. That return is not an interruption of physics but the final preparation for re-entering specialized microphysics. The Standard Model block cannot begin by postulating gauge groups, charge tables, and particle names. It must begin by identifying the internal residue data from which those observer-level labels later emerge.

This methodological choice is essential for the whole book. The earlier chapters developed structural histories, transfer kernels, marked observables, dominant spectral branches, observer decategorification, and field-theoretic poles. The Standard Model can therefore be introduced only as a later specialization of that machinery. Charges are not primitive decorations attached to particles. They are numerical shadows of residue observables carried by stable sectors of the kernel-residue theory.

This chapter does six things. First, it defines the full residue space relevant for the Standard Model block. Second, it separates electroweak, color, and family residues. Third, it defines the internal observables T^3, Y, Q, and C^a. Fourth, it relates residue data to kernel branches and dominant poles. Fifth, it formulates observer decategorification as the map from residue data to ordinary physical labels. Sixth, it states the target statement that charge data are to be reconstructed as internal observables rather than imposed externally.

22.2 The Full Residue Space

Definition 22.1 (Full residue space)**.** The full residue space of the Standard Model block is

$$\mathcal{R}_{\mathrm{SM}} := \mathcal{R}_{\mathrm{EW}} \times \mathcal{R}_c \times \mathcal{R}_F,$$

where $\mathcal{R}_{\mathrm{EW}}$ is the electroweak residue space, $\mathcal{R}_c$ the color residue space, and $\mathcal{R}_F$ the family residue space.

Definition 22.2 (Residue map)**.** The residue map is a projection

$$\rho : W \to \mathcal{R}_{\mathrm{SM}},$$

from the structural state space W to the observer-relevant residue data carried by the corresponding sector.

Remark 22.3. The residue map is coarser than the full structural description. It is not intended to retain assembly history, normalization data, or the full hierarchy of equivalence classes. It isolates only the internal observables that survive into the later particle-physics image.

22.3 Electroweak, Color, and Family Residues

Definition 22.4 (Electroweak residue space)**.** The electroweak residue space $\mathcal{R}_{\mathrm{EW}}$ records the internal data from which weak isospin, hypercharge, chirality, and weak-ladder transitions are later read by the observer.

Definition 22.5 (Color residue space)**.** The color residue space is

$$\mathcal{R}_c = \{0, r, g, b\},$$

or equivalently its linearized observer image in the fundamental color triplet together with the neutral singlet value.

Definition 22.6 (Family residue space)**.** The family residue space $\mathcal{R}_F$ records the internal family geometry responsible for generation structure, family operators, and later flavour mixing.

Definition 22.7 (Residue decomposition of a sector)**.** For a stable sector $\mathfrak{F}$, the residue data decomposes as

$$\rho(\mathfrak{F}) = \big(\rho_{\mathrm{EW}}(\mathfrak{F}), \rho_c(\mathfrak{F}), \rho_F(\mathfrak{F})\big).$$

Example 22.8 (Three residue layers of one observer sector)**.** An observer-level matter sector may simultaneously carry electroweak residue data, nontrivial family residue data,

and either singlet or nonsinglet color residue data. These are not three different particles but three internal projections of one and the same structural sector.

22.4 Internal Residue Observables

Definition 22.9 (Electroweak residue observables)**.** On $\mathcal{R}_{\mathrm{EW}}$ define the internal observables

$$T^3, \qquad Y, \qquad Q := T^3 + Y,$$

together with the ladder operators $T^{\pm}$ when the relevant weak transitions are defined.

Definition 22.10 (Color observables)**.** On the color residue space define the internal color observables

$$C^a, \qquad a = 1, \dots, 8,$$

which form the observer-level numerical shadow of the adjoint color structure.

Definition 22.11 (Family operators)**.** On the family residue space define the family operators that later control generation structure, nearest-neighbor family coupling, and flavour geometry.

Proposition 22.12 (Charges as residue combinations)**.** *The electric charge observable is not primitive but is built from electroweak residue observables through*

$$Q = T^3 + Y.$$

Remark 22.13. This is the first place in the Standard Model block where the book must be rigid about language. The symbols T^3, Y, Q, and C^a are internal residue observables before they are read as ordinary quantum numbers of observer-level particles.

22.5 Residue Data and Kernel Branches

Definition 22.14 (Kernel-residue branch)**.** For each flavour or matter sector f, let

$$\lambda_i^{(f)}(\eta, q)$$

be a dominant spectral branch of the corresponding effective kernel. The associated on-shell mass is determined by

$$\lambda_i^{(f)}(m_i^{(f)}, 0) = 1.$$

Definition 22.15 (Residue factor of a spectral branch)**.** The residue factor of the branch is

$$Z_i^{(f)} := \Big[\partial_\eta \big(1 - \lambda_i^{(f)}(\eta, 0)\big)\Big]^{-1}_{\eta = m_i^{(f)}}.$$

Proposition 22.16 (Residue observables are transfer-stable)**.** *If an internal observable is defined on the residue layer and is compatible with admissible transfer, then its observer-level numerical shadow is stable under re-encoding of the same sector.*

Proposition 22.17 (Pole language from residue branches)**.** *Once a dominant branch and its residue factor are fixed, the observer-level particle interpretation arises from the same pole-residue calculus already developed in the earlier field-theory chapters. The Standard Model block therefore imports no new spectral ontology here.*

22.6 Observer Decategorification and Numerical Shadows

Definition 22.18 (Observer decategorification of residue data)**.** Observer decategorification is the map that sends internal residue data to ordinary observer-level labels, numerical charges, and effective field-sector identifiers.

Definition 22.19 (Numerical shadow)**.** The numerical shadow of a residue observable is its value in the observer-level effective description after coarse projection, transfer stabilization, and spectral selection.

Proposition 22.20 (Residue-to-particle interpretation)**.** *After observer decategorification, a stable sector is described by its numerical residue shadows together with the corresponding kernel branch, pole location, and residue factor. This is the bridge from the general structural-physical theory to the language of later Standard Model sectors.*

Example 22.21 (Charges as observer shadows)**.** What ordinary particle physics calls electric charge, weak charge assignment, or color representation is interpreted in the present framework as the observer-level shadow of an internal residue configuration selected by the relevant stable kernel sector.

22.7 Programmatic Bridge and Failure Modes

Proposition 22.22 (Target statement: Charges as internal residue observables)**.** *The Standard Model charge data should admit reconstruction as internal observables of the residue calculus, not as externally attached labels. In particular, electroweak and color charges are intended to arise from internal residue observables whose numerical shadows survive observer decategorification.*

Remark 22.23 (Failure modes)**.** The residue-calculus program can fail in several ways: the residue map may be too coarse to separate physically relevant sectors; internal observables may fail to be transfer-stable; observer decategorification may lose the distinction between

internal charges and arbitrary labels; or kernel branches may fail to support stable numerical shadows for the sectors of interest.

22.8 Chapter Audit Summary

- Definitions introduced: the full residue space, the residue map, residue observables, dominant kernel branches, residue factors, observer decategorification, and numerical shadows.
- Standing assumptions: transfer-stable residue extraction, admissible dominant branches, and an observer projection that preserves the residue data needed for later charge interpretation.
- Internally established results: charge combination $Q = T^3 + Y$ as a residue observable, transfer-stability of residue observables, and the pole-residue bridge from internal branches to observer-level sector language.
- Imported identifications used: ordinary electroweak and color charge notation as the observer-level language for the later effective field-theoretic image.
- Open targets deferred: full reconstruction of audited Standard Model charge data from internal observables without any stronger external charge input.
- Downstream dependence: the electroweak matter chapter uses this residue layer as its charge and sector data source.

22.9 Lean Orientation for the Next Chapter

For later formalization, the residue-calculus chapter should be split into five files.

1. A residue-space file defining $\mathcal{R}_{\mathrm{SM}}$, its electroweak, color, and family components, and the residue map.
2. A residue-observables file defining T^3, Y, Q, $T^{\pm}$, and C^a as internal observables.
3. A spectral-residue file defining dominant branches, on-shell conditions, and residue factors.
4. An observer-shadow file defining observer decategorification and numerical shadows of residue data.
5. A bridge-to-Standard-Model file stating the target theorem that charge data must arise from internal observables rather than external labels.

This split keeps the Standard Model block disciplined from the start. The present chapter depends on the structural core, transfer kernels, pole-residue calculus, and observer projection already developed earlier, and it prepares the next chapter in which the electroweak sector and matter content are written explicitly.

22.10 Conclusion of the Chapter

This chapter introduced the residue calculus that prepares the Standard Model block. It defined the full residue space, separated electroweak, color, and family residues, introduced the internal observables T^3, Y, Q, and C^a, connected them to kernel branches and residue factors, and interpreted physical charge data as observer-level shadows of internal residue structure.

The next chapter can therefore write the electroweak sector and matter content in explicit form. The transition point is now explicit: residue observables have been identified, so the next task is to show how Higgs residue data, vacuum neutrality, weak generators, and Yukawa-compatible kernels convert those internal observables into the broken-phase electroweak block and the standard charge table.

Chapter 23

The Electroweak Sector and Matter Content

23.1 Why the Standard Model Block Must First Specialize to the Broken Electroweak Sector

The previous chapter identified the full residue space, the electroweak and color observables, and the observer-level numerical shadows of residue data. This is the natural point within the present framework at which the Standard Model block first specializes to explicit particle-physics structure. The first specialization is the broken electroweak sector, because that is where internal observables first become recognizable as familiar charges, currents, bosonic modes, and matter assignments.

This chapter therefore does not yet try to solve flavour or confinement. It isolates the electroweak core. Its task is to show how Higgs residue data, vacuum neutrality, weak generators, bosonic and fermionic kernels, and Yukawa bridges together produce the ordinary broken-phase electroweak image. The chapter must also make clear that matter content is not inserted as a ready-made charge table. It is reconstructed from residue compatibility conditions.

This chapter does seven things. First, it defines the Higgs residue and the neutral vacuum configuration. Second, it identifies the unbroken electromagnetic generator and the broken-phase $(A, Z, W^{\pm})$ basis. Third, it defines the unified bosonic kernel. Fourth, it defines the unified fermionic electroweak kernel. Fifth, it formulates the Yukawa bridges. Sixth, it derives hypercharge reduction and the standard charge table. Seventh, it states the matter-content theorem as the target theorem of the section.

23.2 Higgs Residue and Vacuum Neutrality

Definition 23.1 (Higgs residue)**.** The Higgs residue is the electroweak residue sector H carrying the weak-doublet structure needed to mediate Yukawa bridges and to define the broken electroweak vacuum.

Definition 23.2 (Neutral Higgs vacuum)**.** The broken-phase vacuum is the observer-level sector H_{vac} selected by the Higgs residue such that its electric charge vanishes:

$$Q(H_{\mathrm{vac}}) = 0.$$

Proposition 23.3 (Unbroken electromagnetic generator)**.** *If the Higgs vacuum is neutral, then the unbroken generator in the electroweak sector is*

$$Q = T^3 + Y.$$

Proof. Vacuum neutrality means the broken-phase vacuum is annihilated by the unique linear combination of electroweak residue observables that survives as an unbroken observer-level symmetry. By construction that combination is the electric charge observable $Q = T^3 + Y$. □

Example 23.4 (Vacuum neutrality and the unbroken generator)**.** Suppose the Higgs residue has weak-doublet components with opposite values of T^3 and a common hypercharge assignment. Requiring the vacuum component to satisfy $Q(H_{\mathrm{vac}}) = 0$ forces the surviving generator to be the charge combination $T^3 + Y$, so the later photon-like branch is tied to vacuum neutrality rather than postulated independently.

23.3 Broken-Phase Bosonic Basis

Definition 23.5 ($(A, Z, W^{\pm})$ basis)**.** The broken-phase bosonic basis is the observer-level basis of electroweak modes

$$A, \qquad Z, \qquad W^{\pm},$$

obtained after the Higgs residue selects the neutral vacuum and the electroweak sector is written in the basis adapted to the surviving charge observable Q.

Remark 23.6. The role of this basis is purely organizational at the present stage. It is the broken-phase basis in which later masses, currents, and charge assignments are written transparently.

23.4 Unified Bosonic Kernel

Definition 23.7 (Unified bosonic kernel)**.** The broken-phase bosonic kernel is the effective inverse kernel

$$\mathcal{D}_B(A, Z, W^{\pm}, h),$$

whose dominant branches describe the observer-level bosonic modes of the electroweak sector together with the Higgs fluctuation h around the broken vacuum.

Proposition 23.8 (Broken electroweak bosonic spectrum)**.** *In the broken-phase observer window, the unified bosonic kernel admits a spectrum consisting of a massless photon-like branch A, massive weak branches $W^{\pm}$ and Z, and a massive Higgs-like fluctuation h.*

Remark 23.9. At the level of the present chapter, the exact low-energy normalization of these masses is part of the later electroweak dictionary. The current chapter fixes the structural origin of the broken-phase basis and the corresponding kernel blocks.

Example 23.10 (Minimal low-energy electroweak bosonic block)**.** In the simplest low-energy observer image, the bosonic block is organized so that the A branch remains gapless, the Z and $W^{\pm}$ branches acquire a Higgs-induced gap, and the scalar fluctuation h records the local response of the Higgs residue around the broken vacuum. This is the minimal structure needed before one writes the effective mass formulas in the later Standard Model dictionary.

23.5 Unified Fermionic Electroweak Kernel

Definition 23.11 (Unified fermionic kernel)**.** The unified fermionic electroweak kernel is the effective inverse kernel

$$\mathcal{D}_F^{\mathrm{EW}},$$

whose diagonal blocks are controlled by the electroweak residue observables Q and T^3, and whose off-diagonal weak blocks are controlled by the ladder operators $T^{\pm}$.

Remark 23.12. The flavour-dependent refinements of the fermionic kernel are deferred to the next chapter. In the present chapter, one isolates only the electroweak matter-content architecture.

Definition 23.13 (Left and right matter sectors)**.** The electroweak fermionic kernel distinguishes left and right matter sectors, with weak-doublet action carried on the left sector and singlet behavior carried on the appropriate right sector. This chirality split is part of the residue assignment rather than an externally imposed bookkeeping device.

23.6 Yukawa Bridges and Hypercharge Reduction

Definition 23.14 (Yukawa bridges)**.** The admissible Yukawa bridges are the electroweak residue channels

$$Q_L + \widetilde{H} \to u_R, \qquad Q_L + H \to d_R, \qquad L_L + H \to e_R, \qquad L_L + \widetilde{H} \to \nu_R,$$

where Q_L and L_L are the left quark and lepton sectors and u_R, d_R, e_R, and ν_R are the corresponding right sectors.

Proposition 23.15 (Hypercharge reduction)**.** *If the Yukawa bridges exist and are compatible with residue conservation, then all right-handed hypercharges are determined by the two left-handed parameters*

$$Y(Q_L) = y_Q, \qquad Y(L_L) = y_L.$$

Proposition 23.16 (Neutral neutrino branch)**.** *If the neutrino branch is selected so that*

$$Q(\nu_L) = 0,$$

then one obtains the standard left-lepton hypercharge assignment

$$Y(L_L) = -\frac{1}{2}.$$

Proposition 23.17 (Standard charge table under electroweak reduction)**.** *If one combines Higgs vacuum neutrality, Yukawa-bridge compatibility, and the neutral-neutrino branch, then the electroweak charge assignments reduce to the standard table*

$$\begin{aligned} Y(Q_L) = \frac{1}{6}, \quad Y(L_L) = -\frac{1}{2}, \quad Y(u_R) = \frac{2}{3}, \\ Y(d_R) = -\frac{1}{3}, \quad Y(e_R) = -1, \quad Y(\nu_R) = 0. \end{aligned}$$

Example 23.18 (How Higgs neutrality fixes the electromagnetic generator)**.** Take the Higgs vacuum to be neutral and demand that all Yukawa bridges preserve the electroweak residue balance. The same charge combination $Q = T^3 + Y$ that survives as the unbroken generator then constrains the hypercharge assignments of all right sectors, because each bridge must close in the residue calculus. The standard charge table is therefore read as a closure condition, not as an external input.

23.7 Matter-Content Theorem

Theorem 23.19 (Target theorem: Electroweak matter-content theorem)**.** *The broken-phase electroweak matter content should admit reconstruction from the residue calculus by combining Higgs vacuum neutrality, weak generators, Yukawa-compatible bridges, and the neutral-neutrino branch. In that reconstruction, the standard electroweak charge table appears as an internal consequence of residue compatibility rather than as a prior table of labels.*

Remark 23.20 (Failure modes)**.** The electroweak reconstruction can fail if Higgs vacuum neutrality does not isolate a unique unbroken generator, if the Yukawa bridges are not residue-compatible, if the weak-doublet versus singlet distinction is not stable under observer decategorification, or if hypercharge reduction leaves multiple incompatible low-energy branches.

23.8 Chapter Audit Summary

- Definitions introduced: the Higgs residue, neutral vacuum, broken-phase bosonic basis, bosonic and fermionic electroweak kernels, and Yukawa bridges.
- Standing assumptions: Higgs vacuum neutrality, residue-compatible Yukawa bridges, and a stable chirality split under observer decategorification.
- Internally established results: the unbroken charge generator, hypercharge reduction to two left-handed parameters, the neutral-neutrino branch selection, and the resulting one-generation electroweak charge table.
- Imported identifications used: the ordinary broken-phase electroweak bosonic basis notation and the continuum reading of the electroweak charge table.
- Open targets deferred: complete reconstruction of all electroweak matter content beyond the already audited one-generation charge-compatibility statement.
- Downstream dependence: the flavour chapter treats generation structure only after this electroweak matter architecture is fixed.

23.9 Lean Orientation for the Next Chapter

For later formalization, the electroweak chapter should be split into six files.

1. A Higgs-and-vacuum file defining the Higgs residue, the broken vacuum, and the neutrality condition.

2. A weak-generators file defining Q, T^3, $T^{\pm}$, and the broken-phase $(A, Z, W^{\pm})$ basis.

3. A bosonic-kernel file defining the unified bosonic electroweak block.

4. A fermionic-kernel file defining the chirally split fermionic electroweak block.

5. A Yukawa-bridges file defining the four bridge types and the hypercharge-reduction chain.

6. A matter-content file stating the charge table and the target theorem that it arises from residue compatibility.

This split keeps the theorem chain explicit. The present chapter depends on the residue calculus developed previously and prepares the flavour chapter by fixing the electroweak matter architecture before any generational structure is added.

23.10 Conclusion of the Chapter

This chapter specialized the residue calculus to the broken electroweak block. It introduced the Higgs residue, vacuum neutrality, the unbroken charge generator, the broken-phase $(A, Z, W^{\pm})$ basis, the unified bosonic and fermionic electroweak kernels, the Yukawa bridges, and the standard charge table as a residue-compatibility consequence.

The next chapter can therefore treat flavour as a genuine refinement of an already fixed electroweak matter sector. The transition point is now explicit: charge assignments, weak generators, and Yukawa bridge types are fixed, so flavour can be introduced solely as the internal family geometry of masses, residues, and mixing matrices without re-opening the charge-assignment problem.

Chapter 24

Unified Flavour Module

24.1 Why Flavour Must Be Introduced Only After the Electroweak Matter Sector Is Fixed

The previous chapter fixed the electroweak matter architecture: Higgs residue data, vacuum neutrality, broken-phase bosonic modes, chirally split fermionic kernels, Yukawa bridge types, and the standard electroweak charge table. This is the natural point within the present framework at which flavour is introduced. Earlier than that, one would risk confusing generation structure with charge assignment. The flavour chapter therefore treats generations, masses, and mixing as refinements of an already fixed electroweak matter content.

This methodological separation is necessary. Electroweak charges determine what kinds of sectors can couple. Flavour determines how the different branches of those sectors are organized, how their pole data is related, and how mass eigenstates differ from interaction eigenstates. The present chapter therefore studies the family geometry of the kernel-residue theory rather than re-deriving the electroweak block.

This chapter does seven things. First, it defines the family core and the family operators. Second, it defines kernel-residue poles and the flavour residue law. Third, it states the low-energy closure needed for geometric-mean-type flavour relations. Fourth, it formulates the charged-lepton and neutrino flavour blocks, including a seesaw variant. Fifth, it formulates the up and down quark flavour blocks. Sixth, it derives the emergence of PMNS and CKM matrices. Seventh, it records the leading flavour relations and the target theorem that small CKM versus large PMNS mixing should arise from one family geometry.

24.2 Family Core and Family Operators

Definition 24.1 (Family core)**.** All flavour sectors are governed by one family core consisting of the family operators

$$L_F, \qquad E_{12}, \qquad E_{23},$$

which encode family level, nearest-neighbor family coupling, and family geometry.

Definition 24.2 (Family edge data)**.** The effective family couplings are controlled by edge data

$$G_{12}, \qquad G_{23},$$

which appear in the residue law for off-diagonal family mixing.

Remark 24.3. The flavour chapter is the first place where the Standard Model block uses family data as a genuine internal geometry. Family indices are not arbitrary copies of one matter generation; they are structured branches of the same kernel-residue architecture.

24.3 Kernel-Residue Poles and the Flavour Residue Law

Definition 24.4 (Kernel-residue poles)**.** For each flavour sector f and family mode i, let

$$\lambda_i^{(f)}(\eta, q)$$

be the dominant spectral branch. The on-shell mass satisfies

$$\lambda_i^{(f)}\big(m_i^{(f)}, 0\big) = 1,$$

and the corresponding residue factor is

$$Z_i^{(f)} := \Big[\partial_\eta\big(1 - \lambda_i^{(f)}(\eta, 0)\big)\Big]^{-1}_{\eta = m_i^{(f)}}.$$

Definition 24.5 (Flavour residue law)**.** The off-diagonal family couplings are governed by the residue law

$$c_{ij}^{(f)} = \chi_f G_{ij} \sqrt{Z_i^{(f)} Z_j^{(f)}},$$

where χ_f is a flavour-sector coefficient and G_{ij} is the appropriate family-edge parameter.

Proposition 24.6 (Geometric-mean law under low-energy closure)**.** *If the one-pole low-energy closure holds in the form*

$$Z_i^{(f)} \propto m_i^{(f)},$$

then the off-diagonal flavour couplings satisfy the geometric-mean-type relation

$$c_{ij}^{(f)} \propto \sqrt{m_i^{(f)} m_j^{(f)}}.$$

Remark 24.7. This proposition must be interpreted carefully. The low-energy closure is a derived or effective relation in the observer regime, not the deepest statement of the theory. The underlying residue law is more fundamental than the geometric-mean approximation that it yields after low-energy closure.

24.4 Lepton Flavour Blocks

Definition 24.8 (Charged-lepton flavour block)**.** The charged-lepton flavour block is the family-resolved mass or kernel block

$$M_\ell,$$

whose pole data determine the charged-lepton masses and the unitary transformation that diagonalizes the charged-lepton sector.

Definition 24.9 (Neutrino flavour block)**.** The neutrino flavour sector is described by a Dirac block

$$M_D^{(\nu)}$$

and, when the heavy-right-sector variant is used, by a Majorana block

$$M_R^{(\nu)}.$$

Proposition 24.10 (Seesaw variant)**.** *When a heavy Majorana sector is present, the light-neutrino block is approximated by*

$$M_\nu^{\text{light}} \approx -M_D^{(\nu)} \big(M_R^{(\nu)}\big)^{-1} \big(M_D^{(\nu)}\big)^T.$$

Example 24.11 (Why lepton mixing need not be small)**.** If the charged-lepton Dirac block is close to the family basis selected by the electroweak bridges while the effective light-neutrino block is strongly shaped by the seesaw geometry, then the relative diagonalization can produce large PMNS mixing even when the underlying family core is the same. This is the basic structural explanation pursued in the present framework.

24.5 Quark Flavour Blocks

Definition 24.12 (Up and down quark flavour blocks)**.** The quark flavour sector is organized by two family-resolved blocks

$$M_u, \qquad M_d,$$

which determine the up-type and down-type pole data and their relative diagonalization.

Remark 24.13. The flavour chapter does not yet require color confinement. It treats quark flavour at the level of family geometry and mass textures inside the electroweak matter architecture. The color block later refines observability and asymptotic-state questions.

24.6 Emergence of PMNS and CKM

Definition 24.14 (CKM matrix)**.** If U_u and U_d diagonalize the up-type and down-type quark flavour blocks, then the quark mixing matrix is

$$V_{\mathrm{CKM}} = U_u^{\dagger} U_d.$$

Definition 24.15 (PMNS matrix)**.** If the effective light-neutrino block is diagonalized relative to the charged-lepton block, then the lepton mixing matrix is the unitary matrix U_{PMNS} satisfying

$$U_{\mathrm{PMNS}}^{T} M_{\nu}^{\mathrm{light}} U_{\mathrm{PMNS}} = \mathrm{diag}(m_{\nu_i}).$$

Proposition 24.16 (Mixing as relative diagonalization)**.** *Both PMNS and CKM matrices arise as relative diagonalizations of flavour blocks rather than as independent external inputs. In this framework, mixing is therefore an observable shadow of family geometry.*

24.7 Leading Flavour Relations and the Unified Flavour Theorem

Proposition 24.17 (Leading flavour relations)**.** *In the low-energy closure regime, one expects leading CKM entries to admit geometric-mean-type relations controlled by the family-edge data. Schematic examples are*

$$V_{us} \sim G_{12}\Big(\sqrt{\frac{m_d}{m_s}} + \sqrt{\frac{m_u}{m_c}}\Big),$$

$$V_{cb} \sim G_{23}\Big(\sqrt{\frac{m_s}{m_b}} + \sqrt{\frac{m_c}{m_t}}\Big),$$

with higher-order suppression for V_{ub}.

Remark 24.18 (Observable low-energy closure versus fundamental residue law)**.** The leading flavour relations are observer-level consequences of the low-energy closure regime. They must not be confused with the fundamental residue law itself. The manuscript must keep these two levels separate and state clearly which numerical features remain unresolved beyond the leading closure approximation.

Theorem 24.19 (Target theorem: Unified flavour theorem)**.** *The flavour sector should admit reconstruction from one family geometry such that PMNS and CKM both arise from the same family core, with small CKM mixing explained by near-alignment of Dirac quark sectors and larger PMNS mixing explained by the seesaw-amplified misalignment of the neutrino sector.*

Example 24.20 (Minimal flavour diagonalization example)**.** Take two nearly aligned quark flavour blocks and a neutrino block whose effective light-sector geometry differs substantially from the charged-lepton one because of a heavy-right-sector correction. Then the relative diagonalization of the quark sector is small while the relative diagonalization of the lepton sector can be large. This is the minimal concrete example the reader should keep in mind throughout the chapter.

24.8 Chapter Audit Summary

- Definitions introduced: the family core, family-edge data, flavour residue law, low-energy closure regime, flavour blocks, and the CKM and PMNS matrices as relative diagonalizations.
- Standing assumptions: one-pole low-energy closure, stable family operators, and the electroweak matter architecture fixed in the previous chapter.
- Internally established results: the residue-law control of off-diagonal couplings, the geometric-mean flavour approximation in the closure regime, the seesaw variant, and the interpretation of mixing matrices as observer-level shadows of family geometry.
- Imported identifications used: standard CKM and PMNS notation and the continuum language of mass textures and flavour blocks.
- Open targets deferred: a complete universal flavour-edge law beyond the audited closure regime and a full numerical derivation of flavour data.
- Downstream dependence: the color and anomaly chapters assume flavour geometry is fixed and no longer reopen the charge-assignment problem.

24.9 Lean Orientation for the Next Chapter

For later formalization, the flavour chapter should be split into six files.

1. A family-core file defining L_F, E_{12}, E_{23}, and the family-edge parameters.
2. A residue-law file defining kernel-residue poles, residue factors, and the off-diagonal flavour law.
3. A low-energy-closure file defining the closure approximation and the geometric-mean-type consequences.
4. A lepton-flavour file defining charged-lepton blocks, neutrino Dirac and Majorana blocks, and the seesaw variant.
5. A quark-flavour file defining the up and down quark flavour blocks and the CKM matrix.
6. A unified-mixing file defining PMNS, CKM, and the theorem chain relating both to one family geometry.

This split keeps the flavour logic explicit. The chapter depends on the electroweak matter-content chapter and prepares the color and anomaly chapters by separating generational geometry from both charge assignment and confinement questions.

24.10 Conclusion of the Chapter

This chapter introduced the unified flavour module of the kernel-residue theory. It defined the family core, the residue law, the low-energy closure approximation, the charged-lepton and neutrino flavour blocks, the quark flavour blocks, and the emergence of PMNS and CKM matrices as relative diagonalizations of one family geometry.

The next chapter can therefore turn to the color and QCD module. The transition point is now explicit: flavour geometry has fixed generation structure and mixing, so the remaining Standard Model specialization concerns color observables, physical observability, and confinement rather than family mixing.

Chapter 25

The Color/QCD Module, Observability, and Confinement

25.1 Why the Color Chapter Must Separate Internal Color Structure from Physical Observability

The previous chapter fixed the flavour architecture of the Standard Model block: family core, residue law, low-energy closure, and the emergence of PMNS and CKM matrices. This is the natural point within the present framework at which the color chapter is written. The color sector is not introduced merely as another charge table. It must distinguish two levels that are often conflated in loose presentations: internal color structure and physical asymptotic observability.

This distinction is essential. The kernel-residue theory can carry nontrivial color data at the internal level while still allowing only color-singlet asymptotic sectors to survive in the observer-level physical resolvent. The chapter must therefore treat color not merely as an internal label on quark branches, but as a structural degree of freedom whose observability is constrained by the projected resolvent of the theory.

This chapter does seven things. First, it defines the color residue space and the color observables. Second, it defines the gluon kernel and the quark color-covariant kernel. Third, it introduces the three-resolvent architecture needed to separate microscopic, sector-resolved, and physical observer-level resolvents. Fourth, it defines the notion of a physical pole and gives a criterion of observability. Fifth, it formulates the color-singlet rule as an observer-level admissibility condition. Sixth, it gives necessary and sufficient criteria for a physical asymptotic state. Seventh, it states the confinement theorem as a target theorem of the program.

25.2 Color Residue Space and Color Observables

Definition 25.1 (Color residue space)**.** The color residue space is

$$\mathcal{R}_c = \{0, r, g, b\},$$

or equivalently its observer-level linearized image in the color triplet together with the singlet neutral value.

Definition 25.2 (Color observables)**.** On the color residue space define the eight internal color observables

$$C^a, \qquad a = 1, \dots, 8,$$

which act as the observer-level numerical shadows of the adjoint color structure.

Remark 25.3. Color observables are internal structural quantities before they are interpreted as QCD-like charges. Their physical status depends on the observability criteria introduced later in the chapter.

25.3 Gluon Kernel and Color-Covariant Quark Kernel

Definition 25.4 (Gluon kernel)**.** The gluon sector is described by an adjoint color bosonic kernel

$$\mathcal{D}_g^{ab}(\omega, q),$$

whose branches determine the observer-level propagation of color-adjoint bosonic modes.

Definition 25.5 (Quark color-covariant kernel)**.** The quark sector is described by a color-covariant fermionic kernel in which the flavour-resolved quark blocks are lifted to color-valued form and coupled to the gluon kernel through the internal color observables.

Proposition 25.6 (Local QCD dictionary at the kernel level)**.** *After observer decategorification, the color-covariant quark kernel and the gluon kernel yield the local QCD-like interaction pattern in which quark branches couple to adjoint gluon modes through the color observables.*

25.4 Three-Resolvent Architecture

Definition 25.7 (Microscopic resolvent)**.** For the full Standard Model kernel define the microscopic resolvent

$$\mathcal{R}_{\mathrm{micro}}(\eta, q) := \big(\mathcal{D}_{\mathrm{SM}}(\eta, q)\big)^{-1}.$$

This object still contains unprojected color and gauge structure.

Definition 25.8 (Sector-resolved resolvent)**.** If P_σ projects onto a chosen residue sector σ, define the sector-resolved resolvent by

$$\mathcal{R}_\sigma(\eta, q) := P_\sigma \, \mathcal{R}_{\text{micro}}(\eta, q) \, P_\sigma.$$

Definition 25.9 (Physical observer resolvent)**.** Let P_{phys} denote the projector onto physically admissible observer sectors. The physical observer resolvent is

$$\mathcal{R}_{\text{phys}}(\eta, q) := P_{\text{phys}} \, \mathcal{R}_{\text{micro}}(\eta, q) \, P_{\text{phys}}.$$

Remark 25.10 (Canonical versus merely chosen physical resolvent)**.** When the later selection conditions isolate a unique admissible observer projector, $\mathcal{R}_{\text{phys}}$ is canonical. In the weaker pre-classification situation treated in this chapter, it should be read as a chosen admissible projected resolvent rather than as an already proved unique object.

Remark 25.11. The point of the three-resolvent architecture is to separate mere internal existence of a branch from physical asymptotic observability. The color chapter must never identify poles of the microscopic resolvent directly with observable particles.

25.5 Physical Poles and the Criterion of Observability

Definition 25.12 (Physical pole)**.** A sector σ has a physically observable asymptotic branch if the projected physical resolvent

$$P_\sigma \, \mathcal{R}_{\text{phys}}(\eta, q) \, P_\sigma$$

has an isolated simple pole at

$$\eta = E_\sigma(q)$$

with nonzero residue factor.

Definition 25.13 (Criterion of observability)**.** A sector is physically observable only if its observer-level projected resolvent contains such a physical pole or another admissible asymptotic branch in the observer sense. Mere internal representation-theoretic existence is not enough.

Proposition 25.14 (Necessary condition for physical observability)**.** *If a sector contains no color-singlet component, then its projected physical resolvent cannot contain a physical asymptotic pole.*

Proposition 25.15 (Sufficient criterion for a physical asymptotic state)**.** *If a sector contains a color-singlet component and the corresponding projected effective kernel has*

a simple real on-shell branch with nonzero residue, then the projected physical resolvent contains a physical asymptotic pole for that sector.

25.6 The Color-Singlet Rule

Definition 25.16 (Color-singlet observability rule)**.** A sector is kinematically admissible as a physical asymptotic observer state only if its color representation contains the singlet component

$$\mathbf{1} \subset \mathrm{Rep}_c(\sigma).$$

Remark 25.17. This rule is not yet the full confinement statement. It is the observer-level admissibility criterion that restricts which sectors may even qualify as physically observable asymptotic states.

Example 25.18 (Single quark, meson, baryon, and gluonic composite)**.** A single quark branch is color nonsinglet and so fails the color-singlet observability rule. By contrast, meson-like $q\bar{q}$ sectors, baryon-like qqq sectors, and some gluonic composite sectors can contain the singlet representation and are therefore kinematically admissible as candidate physical sectors.

25.7 Confinement as a Target Statement

Theorem 25.19 (Target theorem: Confinement criterion)**.** *Colored one-particle branches should fail to survive as isolated poles of the physical observer resolvent, whereas suitable color-singlet composite sectors should survive as physical asymptotic branches. This is the precise confinement target of the kernel-residue program.*

Remark 25.20 (Failure modes)**.** The color program can fail if the singlet projector is not stable under observer decategorification, if the physical resolvent does not separate kinematical singlet admissibility from genuine dynamical survival, if colored poles remain in the projected observer resolvent, or if singlet composite sectors fail to produce stable asymptotic poles.

25.8 Chapter Audit Summary

- Definitions introduced: the color residue space, color observables, gluon and quark color-covariant kernels, and the three-resolvent architecture.
- Standing assumptions: observer-stable singlet projection, admissible projected resolvents, and a meaningful distinction between internal color structure and physical asymptotic observability.

- Internally established results: the separation between microscopic and projected resolvents, necessary and sufficient observability criteria, and the kinematical color-singlet admissibility rule.
- Imported identifications used: the local QCD reading of the color-covariant kernel after observer decategorification.
- Open targets deferred: the full nonperturbative confinement theorem and the complete physical color spectrum.
- Downstream dependence: the anomaly chapter assumes this observability split and tests consistency only on the admissible electroweak-color matter content.

25.9 Lean Orientation for the Next Chapter

For later formalization, the color chapter should be split into six files.

1. A color-residue file defining $\mathcal{R}_c$ and the color observables C^a.
2. A gluon-kernel file defining the adjoint color bosonic kernel.
3. A quark-color-kernel file defining the color-covariant quark kernel.
4. A resolvent-architecture file defining microscopic, sector-resolved, and physical observer resolvents.
5. An observability file defining physical poles, the singlet rule, and necessary versus sufficient observability criteria.
6. A confinement-target file stating the confinement theorem and the examples of quark, meson-like, baryon-like, and gluonic composite sectors.

This split keeps the logic strict: internal color structure is separated from physical observability, and confinement is kept as a precise target theorem rather than blurred together with the singlet rule.

25.10 Conclusion of the Chapter

This chapter introduced the color and QCD module of the kernel-residue theory. It defined the color residue space and color observables, introduced the gluon and color-covariant quark kernels, separated microscopic from physical resolvents, formulated the criterion of observability, stated the color-singlet rule, and fixed confinement as a precise target theorem.

The next chapter can therefore turn to anomalies and branch selection. At that stage the main logical task is no longer to decide which color sectors are observable, but to test whether the electroweak-color matter content is internally consistent and to distinguish rigorously between the anomaly constraints that already prove admissibility of the standard one-generation branch and the stronger uniqueness statement that still remains a target theorem.

Chapter 26

Anomalies, Consistency, and Selection of the Standard Branch

26.1 Why the Anomaly Chapter Must Separate Proven Consistency from Target Uniqueness

The previous chapter fixed the color module at the level of internal color structure, physical observability, the singlet rule, and the confinement target. That still leaves one decisive question open inside the Standard Model block: whether the combined electroweak-color matter content is internally consistent as a chiral theory.

This chapter is therefore not an optional technical appendix. It is the place where the kernel-residue program must perform the analogue of the standard anomaly test in its own internal language. If the anomaly constraints fail, then the electroweak-color branch is not admissible no matter how natural its charge tables or flavour relations may look. If the anomaly constraints hold, then the branch is internally consistent at the level required for the effective Standard Model image.

The chapter must also be methodologically strict. Some statements here are genuine derived results once Higgs bridges, vacuum neutrality, and the residue content have been fixed. Other statements remain stronger selection principles that the theory aims to justify but has not yet reduced to the same level of closure. The book must not blur these levels.

This chapter does five things. First, it defines the anomaly observables in the residue language. Second, it proves the Yukawa-compatible reduction of the mixed color and mixed gravitational anomalies. Third, it shows that the remaining local anomaly conditions reduce to one hypercharge consistency relation. Fourth, it derives the standard anomaly-free one-generation branch and checks the global weak consistency condition. Fifth, it states uniqueness of the anomaly-free electroweak-color branch as a target theorem rather than as a completed proof.

26.2 Anomaly Observables in the Residue Language

Let $\mathfrak{R}$ be the residue content of one chiral matter generation. For each residue sector $r \in \mathfrak{R}$, let $\chi(r) \in \{+1, -1\}$ denote chirality weight, let $Y(r)$ be its hypercharge, let $d_c(r)$ and $d_w(r)$ be its color and weak multiplicities, and let $I_c(r)$ and $I_w(r)$ denote the standard color and weak index factors appropriate to the representation carried by r.

Definition 26.1 (Anomaly observables)**.** Define the anomaly observables of the generation $\mathfrak{R}$ by

$$\mathcal{A}_{3^2Y}(\mathfrak{R}) := \sum_{r\in\mathfrak{R}} \chi(r)\, I_c(r)\, d_w(r)\, Y(r),$$

$$\mathcal{A}_{2^2Y}(\mathfrak{R}) := \sum_{r\in\mathfrak{R}} \chi(r)\, I_w(r)\, d_c(r)\, Y(r),$$

$$\mathcal{A}_{Y^3}(\mathfrak{R}) := \sum_{r\in\mathfrak{R}} \chi(r)\, d_c(r)\, d_w(r)\, Y(r)^3,$$

$$\mathcal{A}_{\mathrm{grav}^2Y}(\mathfrak{R}) := \sum_{r\in\mathfrak{R}} \chi(r)\, d_c(r)\, d_w(r)\, Y(r),$$

and define the global weak consistency observable by

$$\mathcal{W}_2(\mathfrak{R}) := \sum_{\substack{r\in\mathfrak{R} \\ m(r)=2,\ \chi(r)=L}} d_c(r) \pmod 2.$$

Remark 26.2. The local observables $\mathcal{A}_{3^2Y}$, $\mathcal{A}_{2^2Y}$, $\mathcal{A}_{Y^3}$, and $\mathcal{A}_{\mathrm{grav}^2Y}$ are the residue-language shadows of the standard local gauge and mixed gauge-gravitational anomaly checks, while $\mathcal{W}_2$ is the observer-level image of the global $SU(2)$ consistency constraint.

Definition 26.3 (Anomaly-compatible matter content)**.** The chiral residue content $\mathfrak{R}$ is called anomaly-compatible if

$$\mathcal{A}_{3^2Y}(\mathfrak{R}) = \mathcal{A}_{2^2Y}(\mathfrak{R}) = \mathcal{A}_{Y^3}(\mathfrak{R}) = \mathcal{A}_{\mathrm{grav}^2Y}(\mathfrak{R}) = 0, \qquad \mathcal{W}_2(\mathfrak{R}) = 0.$$

26.3 Yukawa-Compatible Reduction of the Anomaly System

Let the unknown one-generation hypercharges be written as

$$Y(Q_L) = y_Q, \quad Y(L_L) = y_L, \quad Y(u_R) = y_u,$$
$$Y(d_R) = y_d, \quad Y(e_R) = y_e, \quad Y(\nu_R) = y_\nu.$$

From the previous electroweak chapter we already have the Higgs residue with

$$Y(H) = +\tfrac{1}{2}, \qquad Y(\widetilde{H}) = -\tfrac{1}{2},$$

and the existence of the Yukawa bridges

$$Q_L + \widetilde{H} \to u_R, \qquad Q_L + H \to d_R, \qquad L_L + H \to e_R, \qquad L_L + \widetilde{H} \to \nu_R.$$

Proposition 26.4 (Yukawa relations)**.** *The Yukawa bridges force*

$$y_u = y_Q + \tfrac{1}{2}, \qquad y_d = y_Q - \tfrac{1}{2}, \qquad y_e = y_L - \tfrac{1}{2}, \qquad y_\nu = y_L + \tfrac{1}{2}.$$

Hence all one-generation hypercharges reduce to the two parameters y_Q and y_L.

Proposition 26.5 (Yukawa-compatible anomaly reduction)**.** *If the Yukawa bridges exist, then the mixed color anomaly and the mixed gravitational anomaly vanish automatically:*

$$\mathcal{A}_{3^2Y}(\mathfrak{R}) = 0, \qquad \mathcal{A}_{\mathrm{grav}^2Y}(\mathfrak{R}) = 0.$$

Proof. Using the Yukawa relations,

$$\mathcal{A}_{3^2Y} \propto 2y_Q - y_u - y_d = 2y_Q - (y_Q + \tfrac{1}{2}) - (y_Q - \tfrac{1}{2}) = 0.$$

Similarly,

$$\mathcal{A}_{\mathrm{grav}^2Y} = 6y_Q - 3y_u - 3y_d + 2y_L - y_e - y_\nu = 0$$

after substitution of the same four relations. □

Remark 26.6. This is the first place where the chapter must be explicit about what is genuinely proved. The vanishing of $\mathcal{A}_{3^2Y}$ and $\mathcal{A}_{\mathrm{grav}^2Y}$ is not a conjectural selection principle here; it is a derived consequence of the already established Higgs and Yukawa structure.

26.4 Hypercharge Consistency

Proposition 26.7 ($(\mathrm{SU}(2)^2\mathrm{U}(1))$ anomaly condition)**.** *For one generation the weak mixed anomaly reduces to*

$$\mathcal{A}_{2^2Y}(\mathfrak{R}) \propto 3y_Q + y_L.$$

Therefore anomaly compatibility requires

$$3y_Q + y_L = 0.$$

Proposition 26.8 ($(\mathrm{U}(1)^3$ anomaly condition)**.** *After imposing the Yukawa relations, the cubic hypercharge anomaly reduces to*

$$\mathcal{A}_{Y^3}(\mathfrak{R}) = -\tfrac{3}{2}\,(y_L + 3y_Q).$$

Hence

$$\mathcal{A}_{Y^3}(\mathfrak{R}) = 0 \iff y_L = -3y_Q.$$

Theorem 26.9 (Hypercharge consistency)**.** *Assume Higgs bridges and Yukawa compatibility. Then cubic hypercharge anomaly cancellation and weak mixed anomaly cancellation reduce to the same linear relation*

$$y_L = -3y_Q.$$

Remark 26.10. This theorem is the rigorous version of the familiar statement that anomaly cancellation almost fixes the Standard Model hypercharges once one simultaneously requires mass-generating Yukawa couplings. At this stage the chapter has proved a consistency relation, not yet the full numerical standard branch.

26.5 Selection of the Standard Anomaly-Free Generation

Proposition 26.11 (Neutral neutrino branch selection)**.** *If the physical one-generation lepton branch contains a neutral neutrino with*

$$Q(\nu_L) = 0,$$

then since $T^3(\nu_L) = +\frac{1}{2}$ and $Q = T^3 + Y$, one obtains

$$y_L = -\tfrac{1}{2}.$$

Consequently,

$$y_Q = \tfrac{1}{6}, \qquad y_u = \tfrac{2}{3}, \qquad y_d = -\tfrac{1}{3}, \qquad y_e = -1, \qquad y_\nu = 0.$$

Theorem 26.12 (Standard anomaly-free generation)**.** *The standard one-generation residue content*

$$Y(Q_L) = \tfrac{1}{6}, \quad Y(L_L) = -\tfrac{1}{2}, \quad Y(u_R) = \tfrac{2}{3},$$
$$Y(d_R) = -\tfrac{1}{3}, \quad Y(e_R) = -1, \quad Y(\nu_R) = 0$$

is anomaly-compatible. Explicitly,

$$\mathcal{A}_{3^2Y} = \mathcal{A}_{2^2Y} = \mathcal{A}_{Y^3} = \mathcal{A}_{\mathrm{grav}^2Y} = 0, \qquad \mathcal{W}_2 = 0.$$

Proof. The four local anomaly observables vanish by direct substitution of the standard values. For the global weak condition, one generation contains three left-handed quark doublets, counted by color, and one left-handed lepton doublet. Hence the total number of left-handed weak doublets is

$$3 + 1 = 4,$$

which is even, so $\mathcal{W}_2 = 0$. □

Example 26.13 (Explicit one-generation check)**.** The standard branch satisfies

$$2 \cdot \tfrac{1}{6} - \tfrac{2}{3} - (-\tfrac{1}{3}) = 0, \qquad 3 \cdot \tfrac{1}{6} + (-\tfrac{1}{2}) = 0,$$

$$6 \cdot \tfrac{1}{6} - 3 \cdot \tfrac{2}{3} - 3 \cdot (-\tfrac{1}{3}) + 2 \cdot (-\tfrac{1}{2}) - (-1) - 0 = 0,$$

and

$$6\left(\tfrac{1}{6}\right)^3 - 3\left(\tfrac{2}{3}\right)^3 - 3\left(-\tfrac{1}{3}\right)^3 + 2\left(-\tfrac{1}{2}\right)^3 - (-1)^3 - 0 = 0.$$

This is exactly the observer-level anomaly cancellation check for one Standard Model generation.

26.6 What Is Proved and What Remains a Target Theorem

Remark 26.14 (What is proved)**.** Given the previously established Higgs residue, neutral vacuum, electroweak charge relation $Q = T^3 + Y$, and Yukawa bridges, this chapter proves four claims inside the residue calculus: the reduction of the mixed color and mixed gravitational anomalies, the hypercharge consistency relation $y_L = -3y_Q$, the standard numerical one-generation hypercharge assignment after neutral neutrino branch selection, and anomaly compatibility of that standard one-generation branch.

Theorem 26.15 (Target theorem: Uniqueness of the anomaly-free electroweak-color branch)**.** *Among admissible chiral residue contents compatible with Higgs bridges and the neutral vacuum, the standard one-generation branch should be uniquely selected up to vectorlike extensions and family replication.*

Remark 26.16 (Why uniqueness is still a target)**.** The chapter has shown that the standard branch is consistent. It has not yet shown that no other genuinely distinct anomaly-compatible chiral branch can survive the same structural constraints. That stronger exclusion statement requires a sharper classification of admissible residue contents than has been built so far, so it must still be treated as a target theorem rather than as completed closure.

26.7 Chapter Audit Summary

- Definitions introduced: the anomaly observables, anomaly-compatible matter content, and the global weak consistency observable.
- Standing assumptions: Higgs vacuum neutrality, Yukawa-compatible bridges, and the one-generation chiral residue content audited in this chapter.
- Internally established results: automatic vanishing of the mixed color and mixed gravitational anomalies, the hypercharge consistency relation, the standard one-generation branch after neutral-neutrino selection, and anomaly compatibility of that one-generation branch.
- Imported identifications used: the standard gauge-anomaly interpretation of the residue observables and the usual one-generation weak-doublet counting for the global consistency check.
- Open targets deferred: uniqueness of the anomaly-free electroweak-color branch beyond the already proved one-generation compatibility statement.
- Downstream dependence: the dictionary chapter assumes an admissible one-generation branch and does not strengthen this chapter's uniqueness status.

26.8 Lean Orientation for the Next Chapter

For later formalization, the anomaly chapter should be split into five files.

1. An anomaly-observables file defining $\mathcal{A}_{3^2Y}$, $\mathcal{A}_{2^2Y}$, $\mathcal{A}_{Y^3}$, $\mathcal{A}_{\mathrm{grav}^2Y}$, and $\mathcal{W}_2$.
2. A Yukawa-reduction file proving the reduction from six raw hypercharges to y_Q and y_L and showing automatic vanishing of the mixed color and mixed gravitational anomalies.
3. A hypercharge-consistency file proving equivalence of the cubic and weak mixed anomaly constraints.
4. A standard-branch file proving anomaly compatibility of the standard one-generation branch and checking the global weak condition.
5. A branch-classification file stating uniqueness of the anomaly-free electroweak-color branch as a target theorem and listing the missing classification lemmas needed to prove it.

This split keeps the chapter logically honest: proven consistency statements are isolated from the stronger uniqueness claim.

26.9 Conclusion of the Chapter

This chapter closed the internal consistency check of the Standard Model block. It defined the anomaly observables, derived the Yukawa-compatible reduction of the anomaly system, proved the hypercharge consistency relation, reconstructed the standard one-generation hypercharge assignment from neutral branch selection, and verified anomaly compatibility of the standard one-generation matter content.

The chapter also fixed the sharp boundary between what the present theory already proves and what remains a target theorem. The existence of an anomaly-compatible standard branch is proved; uniqueness of that branch is not yet proved and remains a classification target.

The next chapter can therefore turn from consistency constraints to the observer-level dictionary into the familiar Standard Model Lagrangian. At that point the main task is no longer to test admissibility of the branch, but to show that the bosonic, fermionic, weak-current, Yukawa, flavour, and QCD sectors of the standard field-theoretic language arise as the observer-decategorified effective image of the already constructed kernel-residue theory rather than as a second independent postulate.

Chapter 27

Dictionary into a Standard-Model-Type Lagrangian

27.1 Why the Dictionary Chapter Must Not Postulate the Standard Model Again

The previous chapter closed the internal consistency question of the Standard Model block at the level of anomaly compatibility. Once that has been done, the next task is not to add new physical content by hand. The task is to show how the familiar field-theoretic language of elementary-particle physics appears as the observer-level image of the structures already built in the kernel-residue theory.

This distinction is essential. If a Standard-Model-type Lagrangian were now simply inserted as an external postulate, then the preceding chapters would lose their mathematical force. The role of the dictionary chapter is precisely to prevent that collapse. The bosonic kernels, fermionic kernels, electroweak charge observables, Yukawa bridges, flavour module, and color module have all already been constructed internally. What remains is to exhibit the map under which these internal structures become the usual gauge, Higgs, fermion, Yukawa, flavour, and QCD terms of the effective low-energy Standard-Model-type description.

This chapter therefore does seven things. First, it defines the observer-decategorification map from kernel data to local field-theoretic data. Second, it gives the bosonic dictionary. Third, it gives the fermionic and weak-current dictionaries. Fourth, it gives the Yukawa dictionary. Fifth, it gives the flavour dictionary. Sixth, it gives the QCD dictionary together with the separation between local QCD structure and singlet-state observability. Seventh, it assembles the audited effective image of the Standard Model.

27.2 Observer-Decategorification as the Dictionary Map

Definition 27.1 (Observer-decategorification map)**.** Let $\mathcal{C}_{\mathrm{int}}^{\mathrm{SM}}$ denote the class of admissible internal Standard-Model-sector configurations carrying bosonic kernels, fermionic kernels, residue observables, Higgs bridges, flavour data, and projected sector labels. Let $\mathcal{O}_{\mathrm{eff}}^{\mathrm{SM}}$ denote the class of observer-level low-energy effective field descriptions modulo the reparametrizations that preserve the audited sector content, pole data, and current structure. Then the observer-decategorification map is the observer-level translation

$$\chi_{\mathrm{obs}} : \mathcal{C}_{\mathrm{int}}^{\mathrm{SM}} \to \mathcal{O}_{\mathrm{eff}}^{\mathrm{SM}}$$

from kernels, residue observables, bridge data, and projected sector information to local field operators and effective actions.

Remark 27.2. The dictionary chapter uses χ_{obs} as a translation map, not as a new dynamical assumption. The dynamical content lies in the previously constructed kernel-residue objects; the observer map identifies their low-energy continuum image.

Remark 27.3 (Preserved and forgotten structure)**.** In the present manuscript, observer decategorification is used as an audited effective translation step. It is intended to preserve residue labels, sector projectors, and the pole data needed for low-energy continuum interpretation, while forgetting assembly history, higher categorical structure, and the internal multiplicity of equivalent structural realizations.

Proposition 27.4 (Well-posedness on audited observer classes)**.** *Under the standing assumptions of the Standard Model block, the map χ_{obs} is well defined on classes of internal configurations that agree on residue labels, projected sector data, and the pole information used in the low-energy dictionary. On that audited domain it preserves the charge labels, current assignments, and pole/mass data required by the bosonic, fermionic, Yukawa, flavour, and local QCD dictionary statements.*

Remark 27.5 (Canonical versus chosen representative)**.** The observer-decategorified image is canonical only when the observer-projection and branch-selection conditions isolate a unique admissible low-energy representative. In the weaker situation treated here, the dictionary should be read as using a chosen admissible representative of the audited observer class rather than as asserting global uniqueness of the translation.

Remark 27.6 (Failure cases for the dictionary step)**.** The translation fails if observer projection does not stabilize the residue labels needed for low-energy reading, if distinct projected branches collapse to inequivalent local terms, or if the continuum expansion loses the sector distinctions required by the audited dictionary.

Let the unified kernel-side action be written schematically as

$$S_{\mathrm{ker}} = \tfrac{1}{2} X^{\dagger} \mathcal{D}_B X + \bar{\Psi}\, \mathcal{D}_F\, \Psi,$$

where X collects the bosonic modes and Ψ collects the matter multiplet. The dictionary claim of this chapter is that after observer decategorification and the appropriate low-energy continuum expansion, S_{ker} becomes an audited effective Standard-Model-type action.

27.3 Bosonic Dictionary

Proposition 27.7 (Bosonic dictionary)**.** *The bosonic kernel maps to the gauge and Higgs quadratic sector:*

$$\mathcal{D}_B \longmapsto \mathcal{L}_{\mathrm{gauge}} + \mathcal{L}_{H,\mathrm{quad}}.$$

Remark 27.8. At low momentum, quadratic kernel factors of the form $\omega^2 - v^2\hat{q}^2 - m^2$ are read as local wave operators of the form $\Box + m^2$. Under this translation the photon branch remains massless, the broken electroweak W and Z branches acquire masses, the Higgs fluctuation becomes a scalar mode, and the gluon sector remains massless at bare level.

Proposition 27.9 (Higgs-potential dictionary)**.** *The Higgs residue and its background order parameter map to the standard scalar sector*

$$\mathcal{L}_H = (D_\mu \Phi)^{\dagger}(D^\mu \Phi) - \mu_H^2 \Phi^{\dagger}\Phi - \lambda_H (\Phi^{\dagger}\Phi)^2.$$

In particular, after choosing the neutral vacuum one obtains the broken-phase mass pattern

$$m_W = \tfrac{gv}{2}, \qquad m_Z = \tfrac{v}{2}\sqrt{g^2 + g'^2}, \qquad m_A = 0.$$

27.4 Fermionic and Weak-Current Dictionary

Proposition 27.10 (Fermionic dictionary)**.** *The diagonal fermionic kernel maps to the standard kinetic and covariant-derivative structure:*

$$\mathcal{D}_F^{\mathrm{diag}} \longmapsto \bar{\psi} i\gamma^\mu D_\mu \psi.$$

Remark 27.11. The residue observables Q, T^3, Y, and the color generators determine the gauge couplings inside D_μ. In this way the familiar field-theoretic charge table is not inserted from outside; it is the observer-level numerical shadow of the residue observables constructed in the previous chapters.

Proposition 27.12 (Weak-current dictionary)**.** *The off-diagonal electroweak blocks map to the standard current structure: the $W^{\pm}$ blocks map to charged currents, while the A and Z blocks map to neutral currents.*

Example 27.13 (Charged-current image)**.** After flavour resolution, the charged-current sector takes the standard form

$$\mathcal{L}_{CC} = \frac{g}{\sqrt{2}}\,\bar{\nu}_L \gamma^\mu U_{\mathrm{PMNS}}^\dagger \ell_L\, W_\mu^+ + \frac{g}{\sqrt{2}}\,\bar{u}_L \gamma^\mu V_{\mathrm{CKM}} d_L\, W_\mu^+ + \mathrm{h.c.}$$

with the corresponding neutral currents carried by the A and Z channels.

27.5 Yukawa Dictionary

Proposition 27.14 (Yukawa dictionary)**.** *The Higgs bridges map to the standard Yukawa sector*

$$\mathcal{L}_Y = -\bar{Q}_L Y_d \Phi d_R - \bar{Q}_L Y_u \widetilde{\Phi} u_R - \bar{L}_L Y_\ell \Phi \ell_R - \bar{L}_L Y_\nu \widetilde{\Phi} \nu_R + h.c.$$

Remark 27.15. After vacuum selection one reads the familiar mass matrices

$$M_d = \tfrac{v}{\sqrt{2}} Y_d, \qquad M_u = \tfrac{v}{\sqrt{2}} Y_u, \qquad M_\ell = \tfrac{v}{\sqrt{2}} Y_\ell, \qquad M_\nu = \tfrac{v}{\sqrt{2}} Y_\nu,$$

or a seesaw-type light neutrino matrix when the heavy right-handed neutrino block is integrated out. Thus the same bridges that enforced hypercharge consistency now become the observer-level origin of mass-generating Yukawa couplings.

27.6 Flavour Dictionary

Proposition 27.16 (Flavour dictionary)**.** *The family module maps to the texture sector of the Standard Model. More precisely, the family operator and the flavour residue law determine the structure of the Yukawa matrices and hence the emergence of CKM and PMNS mixing data.*

Remark 27.17. This is a particularly important methodological point. The effective Standard Model sees the Yukawa matrices as input parameters. The kernel-residue theory does not replace the Yukawa sector with something different; rather, it supplies a structural origin for the texture pattern encoded by those matrices.

27.7 QCD Dictionary

Proposition 27.18 (QCD dictionary)**.** *The color kernel maps to the local QCD sector*

$$\mathcal{L}_{\mathrm{QCD}} = -\tfrac{1}{4} G^a_{\mu\nu} G^{a,\mu\nu} + \sum_q \bar{q} \left(i\gamma^\mu \partial_\mu - g_s \gamma^\mu G^a_\mu T^a \right) q.$$

Remark 27.19. The singlet rule from the previous chapter does not appear as an extra local term in the Lagrangian. It remains an observer-level selection rule on physical asymptotic states. This separation must be kept strict: the local QCD dictionary and the confinement or observability rule are related, but they are not the same statement.

27.8 The Audited Effective Image

This section records the audited effective image reached at the end of the dictionary construction. The term "effective image" is used here in the restricted sense fixed by the observer-level projection and low-energy interpretation of the preceding sections. It does not claim a full continuum derivation, a complete phenomenological fit, or uniqueness beyond the admissible representative and scope conditions explicitly recorded in the final audit chapter.

Proposition 27.20 (Audited low-energy Standard-Model-type dictionary statement)**.** *After observer decategorification and low-energy continuum expansion,*

$$\chi_{\mathrm{obs}}(S_{\mathrm{ker}}) = \int d^4x \left(\mathcal{L}_{\mathrm{gauge}} + \mathcal{L}_H + \mathcal{L}_{\mathrm{ferm}} + \mathcal{L}_Y \right),$$

with an audited Standard-Model-type broken-phase interaction pattern, including charged and neutral weak currents, Higgs-generated mass terms, flavour mixing, and the local QCD sector, up to the open normalization and nonperturbative color points recorded in the final audit chapter.

Remark 27.21 (Compressed correspondence map)**.** The entire dictionary may be summarized schematically as

$$\mathcal{D}_B \longleftrightarrow \mathcal{L}_{\mathrm{gauge}} + \mathcal{L}_{H,\mathrm{quad}}, \qquad \mathcal{D}_F^{\mathrm{diag}} \longleftrightarrow \bar{\psi} i\gamma^\mu D_\mu \psi,$$

$$W^\pm\text{-blocks} \longleftrightarrow \mathcal{L}_{CC}, \qquad A/Z\text{-blocks} \longleftrightarrow \mathcal{L}_{NC},$$

$$\text{Higgs bridges} \longleftrightarrow \mathcal{L}_Y, \qquad \text{family module} \longleftrightarrow \text{Yukawa texture origin},$$

$$\text{color module} \longleftrightarrow \mathcal{L}_{\mathrm{QCD}} + \text{singlet selection on asymptotic states}.$$

Remark 27.22 (What the chapter proves)**.** The content of this chapter is not that the Standard Model Lagrangian is independently assumed to exist. The content is that the

already constructed kernel-residue theory has an observer-level effective image whose audited low-energy continuum form is a Standard-Model-type field theory with the scope limitations made explicit in the final audit.

27.9 Chapter Audit Summary

- Definitions introduced: the observer-decategorification map, the kernel-side action to be translated, and the dictionary blocks for bosonic, fermionic, weak-current, Yukawa, flavour, and QCD sectors.
- Standing assumptions: admissible observer decategorification, low-energy continuum expansion, the previously fixed one-generation anomaly-compatible matter sector, and the already constructed flavour and color modules.
- Internally established results: the structural correspondence between kernel blocks and the observer-level bosonic, fermionic, weak-current, Yukawa, flavour, and local QCD sectors.
- Imported identifications used: ordinary continuum gauge-field notation and the standard local field-theoretic reading of the translated low-energy terms.
- Open targets deferred: exact normalization matching, complete nonperturbative color observability, and full numerical calibration of the translated parameters.
- Downstream dependence: the final audit chapter uses this dictionary only in its explicitly audited low-energy and imported-identification scope.

27.10 Lean Orientation for the Next Chapter

For later formalization, the dictionary chapter should be split into seven files.

1. An observer-map file defining χ_{obs} and the kernel-side action to be translated.
2. A bosonic-dictionary file mapping the bosonic kernel to the gauge and Higgs quadratic sectors.
3. A fermionic-dictionary file mapping diagonal fermionic kernels to covariant-derivative terms.
4. A weak-current file mapping off-diagonal electroweak blocks to charged and neutral currents.
5. A Yukawa-and-flavour file mapping Higgs bridges and family structure to Yukawa terms and texture data.

6. A QCD-dictionary file mapping the color kernel to the local QCD sector while keeping asymptotic singlet selection separate.

7. A final-image file assembling the full effective Standard Model action and recording the exact hypotheses needed for the dictionary theorem.

This split preserves the key methodological boundary: translation into the Standard Model language is a derived image of prior structure, not a fresh postulate.

27.11 Conclusion of the Chapter

This chapter built the observer-level dictionary from kernel-residue theory to an audited Standard-Model-type low-energy Lagrangian image. It translated the bosonic and fermionic kernels, identified the weak-current structure, mapped Higgs bridges to Yukawa terms, interpreted the family module as the origin of texture and mixing data, translated the color kernel into the local QCD sector, and assembled the audited effective image.

The resulting message is precise: the Standard Model Lagrangian is not inserted by hand at the end of the program. Rather, an audited Standard-Model-type effective field description emerges as the low-energy observer-decategorified image of the internal kernel-residue construction, with exact scope to be stated in the final audit.

The next chapter can therefore close the book by assembling the full theorem-chain of the program, auditing dependencies and failure modes, classifying imported versus internal content, and stating exactly which parts are complete proofs and which remain target theorems.

Chapter 28

The Audited Reconstruction Statement, Failure Modes, and Final Appendices

28.1 Why the Manuscript Ends with an Audit Rather than a Declaration of Completion

The previous chapter completed the observer-level dictionary from kernel-residue theory to an audited Standard-Model-type low-energy field description. The manuscript therefore now turns from construction to audit. A program of this size should not end by merely asserting that all desired structures have been obtained. It should record, in the most standard language available, what is derived internally, what is reconstructed only as an effective image, what uses imported continuum interpretation, and what remains an explicit target theorem.

This final chapter is therefore not another construction chapter. Its function is to provide an explicit audit of scope, dependencies, proof status, and failure modes. It records the master assumptions, the dependency structure, the classification of internal versus imported versus open content, the principal failure modes, the observer-level comparison map into ordinary Standard Model language, an algorithmic computational schema, a minimal worked example, the current quantitative status, and the Lean-oriented formalization boundary. No claim of global completion is intended beyond the results explicitly audited below.

28.2 Standing Assumptions and Provenance Labels

The consolidated statement of this chapter uses the following master assumptions.

1. **A1. Structural core.** The structural core of symbolic objects, admissible transformations, quotient levels, and observer-compatible semantic maps is well posed.

2. **A2. Observer projection.** Observer projection and observer-level coarse access are defined in a way compatible with the previously introduced structural invariants.

3. **A3. Transfer-kernel closure.** Transfer kernels, marked observables, dominant branches, and residue extraction are available on the sector class under study.

4. **A4. Electroweak bridge data.** Higgs residue data, the neutral vacuum condition, and the admissible Yukawa bridges are available in the electroweak sector.

5. **A5. Family geometry.** The family core, flavour residue law, and the audited low-energy closure regime are available for the flavour sectors.

6. **A6. One-generation anomaly compatibility.** The internally proved anomaly result used in the present manuscript is the one-generation anomaly-compatibility statement of Chapter 25, unless explicitly stated otherwise.

7. **A7. Observer decategorification.** There exists an observer-decategorification step that preserves the residue labels, sector projectors, and pole data needed for low-energy interpretation while forgetting the higher structural data listed in Chapter 26.

8. **A8. Low-energy continuum admissibility.** The observer-level low-energy continuum expansion exists in the regime in which the dictionary chapter is applied.

The chapter also uses the following provenance labels.

- **[Internal]** means derived inside the manuscript from previously established definitions and results.
- **[Internal → Effective]** means internally constructed and then interpreted at observer level as an effective regime.
- **[Imported continuum identification]** means the internal object is mapped into standard continuum or field-theoretic language using imported classical notation or interpretation.
- **[Asymptotic low-energy]** means the statement is meant only in the audited low-energy or projected observer regime.
- **[Open target]** means the statement remains a target theorem or open completion point rather than a closed proof.

28.3 Formal Audit Specification of Observer Decategorification

For the purposes of the final audit, the observer-decategorification map introduced in Chapter 26 is summarized here in compact formal form.

Definition 28.1 (Audited observer-decategorification specification)**.** Write

$$\mathcal{D}_{\mathrm{obs}} := \chi_{\mathrm{obs}} : \mathcal{C}_{\mathrm{int}}^{\mathrm{adm}} \longrightarrow \mathcal{O}_{\mathrm{eff}}^{\mathrm{adm}}.$$

Here $\mathcal{C}_{\mathrm{int}}^{\mathrm{adm}}$ denotes the class of admissible internal structured sectors satisfying the standing assumptions A1–A8 together with the observer-compatibility and projected-branch conditions used in the Standard Model block, while $\mathcal{O}_{\mathrm{eff}}^{\mathrm{adm}}$ denotes the class of observer-level effective equivalence classes modulo the reparametrizations that preserve the audited charge labels, current structure, projected sector content, and low-energy pole data.

Remark 28.2 (Preserved data, forgotten data, and canonicity)**.** The map $\mathcal{D}_{\mathrm{obs}}$ preserves the residue labels, projected sector identities, and low-energy pole information used in the downstream dictionary. It forgets assembly history, higher categorical multiplicity, and distinctions internal to equivalence classes that do not survive the stated observer regime. It is canonical only when the admissibility and branch-selection conditions isolate a unique observer-level representative; otherwise the manuscript uses a selected admissible representative and states this choice explicitly when relevant.

Proposition 28.3 (Well-posedness under audited assumptions)**.** *Assume A1–A8. Then $\mathcal{D}_{\mathrm{obs}}$ is well defined on the admissible rewrite/projection equivalence classes used in the final dictionary. Moreover, if two admissible internal representatives differ only by an internal re-encoding that preserves residue labels, projected sector content, current structure, and the low-energy pole data required downstream, then their images under $\mathcal{D}_{\mathrm{obs}}$ determine the same audited observer-level effective class.*

Remark 28.4 (Representative dependence and observer-level invariance)**.** When the admissibility and branch-selection conditions isolate a unique observer-level representative, the downstream low-energy dictionary is canonical at the level encoded by $\mathcal{O}_{\mathrm{eff}}^{\mathrm{adm}}$. When uniqueness fails and a selected admissible representative is used instead, the claims made in this chapter are asserted only up to the observer-equivalence relation built into $\mathcal{O}_{\mathrm{eff}}^{\mathrm{adm}}$, unless stronger representative-independence is stated explicitly.

28.4 Main Audited Reconstruction Statement in Standard Language

The consolidated statement below is written as far as possible without relying on the densest internal notation of the earlier chapters.

Remark 28.5 (Status note for the main reconstruction statement)**.** The proposition below is a consolidated audited statement rather than a homogeneous theorem of a single proof type. Its clauses have different formal status: some are exact internal results under A1–A8, some are observer-level effective consequences after audited decategorification, some are imported continuum-language identifications, and some are included only as boundary markers for what is not claimed. Each clause is therefore labeled locally by provenance and scope.

Proposition 28.6 (Main audited reconstruction statement)**.** *Assume A1–A8. Then, within the audited assumptions and imported identifications of this manuscript, the framework yields the following.*

1. ***[Internal → Effective] Scope:*** *asserted at the observer-level low-energy dictionary fixed by the audited decategorification map. It constructs observer-level continuum, quantum, statistical, hydrodynamic, and field-theoretic effective regimes from one common kernel-residue framework.*

2. ***[Internal with open upper-layer closure] Scope:*** *exact for the internal dependency chain, with quantitative upper-layer closure remaining open where explicitly indicated. It constructs many-body, chemical, biological, cognitive, and formal-symbolic regimes as higher-order organization layers of the same structural dynamics rather than as disconnected external additions, with the upper-layer quantitative closure points explicitly audited as open where not yet proved.*

3. ***[Asymptotic low-energy; Internal one-generation anomaly result; Imported continuum identification] Scope:*** *interpreted only in the audited low-energy observer regime and only with the imported continuum-language identifications fixed in Chapters 26–27. Its low-energy microphysical image reconstructs an audited Standard-Model-type effective description with the standard electroweak and color sector organization, chiral matter content at the one-generation anomaly-compatible level proved in Chapter 25, Higgs-broken phase, Yukawa couplings, CKM- and PMNS-type flavour structure, and the color-singlet admissibility rule for physical asymptotic states.*

In particular, the effective low-energy field description is not introduced as an independent postulate. Rather, it is reconstructed as the observer-level effective image of

the internal kernel-residue construction, with exact scope limited by the open targets and imported identifications audited in this chapter.

Remark 28.7 (Status note)**.** The proposition above is not a single homogeneous theorem in the narrow formal sense. It is a consolidated audited reconstruction schema whose clauses have mixed status: internal derivation, internal-to-effective reconstruction, imported continuum interpretation, or explicit open-target status. No clause should be read as asserting completion of the open target theorems listed later in this chapter.

Remark 28.8 (Scope limitation on anomaly results)**.** The anomaly result proved internally in the present manuscript is explicitly the one-generation compatibility statement unless otherwise indicated. Any stronger reading involving a complete multi-generation anomaly classification must be treated as external, conditional, or open.

28.5 What Is Not Claimed in This Manuscript

Not claimed here as closed proofs or completed validations. The manuscript does not claim:

1. full numerical reproduction of all observed particle data;
2. a precision fit of masses, mixing angles, or couplings;
3. a complete nonperturbative color spectrum or a full confinement theorem;
4. uniqueness of all effective dictionary identifications unless explicitly stated under audited assumptions;
5. uniqueness of the anomaly-free electroweak-color branch beyond the explicitly audited one-generation scope;
6. full phenomenological validation across all measured observables; or
7. complete quantitative closure of the higher-organization layers.

28.6 Three Levels of Success and Failure

The manuscript uses three different evaluative levels, and they must not be conflated.

Level	Success criterion	Failure criterion
Internal consistency	well-posed structural objects, admissible transformations, stable kernels, and nonempty audited sector classes	contradiction, instability, or empty admissible class at the structural level
Effective interpretability	a well-defined observer projection and audited dictionary from residue data to low-energy sector language	non-well-defined projection, loss of required invariants, or dictionary mismatch
Phenomenological adequacy	partial qualitative checkpoints only, exactly as listed later in this chapter	failed qualitative checkpoint or, for deferred quantitative claims, no assessment yet because full fit is not claimed here

28.7 Dependency Table of the Manuscript

The manuscript was written in dependency order rather than in order of familiarity. The following table records the major dependency layers.

Layer	Chapters	Depends on
Structural core	0–4	none beyond the initial symbolic and categorical setup
Geometric and dynamical core	5–8	structural core, observer projection, transfer kernels
Quantum, information, and thermodynamic layer	9–11, 15	dynamical core, kernel-residue interpretation, observer records
Continuum and field-theoretic layer	12–14	thermodynamic closure, effective propagation, pole-residue analysis
Higher-organization layer	16–20	field-theoretic and informational layers, effective stability, memory, symbolic persistence
Standard Model specialization	21–26	residue calculus, electroweak charges, flavour module, color module, anomaly checks, observer dictionary
Final audit	27	all previous layers

Remark 28.9 (Methodological invariant)**.** This table is not only editorial. It records the intended dependency invariant of the manuscript: later layers do not introduce new axioms into earlier layers, but only instantiate, interpret, or constrain structures already

built below them. This staged dependency structure is intended to support incremental formalization once package interfaces are made explicit.

28.8 Explicit Coverage Matrix for the Standard-Model-Type Claim

The phrase "Standard-Model-type effective description" is used only with the following explicit coverage and non-coverage conditions.

Remark 28.10 (Reading rule for the coverage matrix)**.** The table below does not list everything loosely reminiscent of the Standard Model. It records only those components for which the manuscript asserts an audited correspondence, together with their formal status: internal, internal-to-effective, imported continuum identification, or not claimed. Absence from the table should be read conservatively as absence of an audited claim.

SM-type component	**Status**	**Provenance**	**Chapter**	**Comment**
one-generation charge organization	established in audited low-energy form	internal residue data → effective charge reading	21–22	observer-level reading only
one-generation anomaly compatibility	established in the audited one-generation scope	internal	25	no stronger multi-generation claim
audited observer-level low-energy dictionary	established up to imported continuum reading	internal → effective plus imported continuum notation	26	asserted only up to the observer-level equivalence and low-energy scope fixed by the audited dictionary
electroweak bridge and weak-current structure	established in audited low-energy form	internal → effective plus imported continuum notation	22, 26	broken-phase reading only
flavour organization	partial, structurally organized, not calibrated	internal closure plus open numerical completion	23, 26	qualitative rather than fitted
color-singlet asymptotic admissibility	partial at the projected observer level	internal local dictionary; full confinement open	24, 26	no closed confinement theorem
exact normalization constants	not claimed	open / calibration layer	27	deferred quantitative layer
precision phenomenology and full fit	not claimed	deferred beyond the present manuscript	27	outside the audited scope of the present text

28.9 Classification of Internal, Imported, and Open Content

Class	Content	Downstream use
Internal derived content	symbolic structures, sequence objects, h-patterns, EFU assembly, transfer kernels, observer projection, structural action, residue calculus, Higgs bridges, flavour residue law, anomaly reduction, and standard one-generation anomaly compatibility	supports the field-theoretic, higher-organization, and Standard Model chapters as internal constructions
Effective dictionary content	the observer-level dictionary from kernels and residue data into the audited low-energy Standard-Model-type image	supports the final comparison map and the low-energy reconstruction claim
Imported classical background	ordinary continuum and field-theoretic language, local gauge-field notation, standard charge-language conventions, and the usual interpretation of the translated low-energy terms	used only at the observer-level interpretation stage
Open target theorems	full confinement theorem, uniqueness of the anomaly-free electroweak-color branch beyond the minimal assumptions, exact normalization derivation, complete nonperturbative color spectrum, and a universal flavour-edge law beyond minimal closure	delimit the exact scope of Chapters 24–27
Open program extensions	quantitative chemistry, full biological selection dynamics, rigorous cognitive semantics, and full-scale computational calibration of the higher-organization layers	delimit the exact scope of Chapters 16–20 and the computational recipe

Remark 28.11. This classification is the final safeguard against overclaim. The manuscript develops a large amount of internal structure, but not every research target has yet been converted into a closed theorem or a reproducible quantitative pipeline.

28.10 Failure Modes

Each failure mode below is stated together with the block of the program that would be damaged by it.

1. **Failure of observer-level closure.** If observer decategorification does not preserve the structural invariants needed to pass from kernels to effective observables, then the bridge from internal dynamics to measurable physics breaks.

2. **Failure of low-energy residue closure.** If the low-energy pole-residue sectors do not stabilize under the required projections, then the Standard Model specialization does not close even at the audited effective-image level.

3. **Failure of the confinement target.** If colored one-particle poles survive in a physically admissible projected observer resolvent or singlet composites fail to survive, then the color chapter does not yield the intended asymptotic physical sector.

4. **Non-uniqueness explosion of admissible branches.** If genuinely distinct chiral branches satisfy the same Higgs and anomaly constraints, then the standard charge pattern is not uniquely selected by the present structural criteria.

5. **Mismatch of dictionary normalization.** If the observer dictionary matches the correct term structure but not the correct normalization of couplings, masses, or currents, then the framework remains structurally successful but quantitatively incomplete.

6. **Failure of universal flavour closure.** If the family residue law does not robustly control the observed texture hierarchy and mixing data across sectors, then the flavour chapter remains only partially explanatory.

7. **Failure of higher-layer stability.** If the many-body, chemical, biological, or cognitive layers do not admit stable effective closures, then the upper half of the manuscript remains a programmatic extension rather than a closed derived hierarchy.

28.11 Internal-to-Observer Comparison Map

The table below records the final comparison between the internal language of the manuscript and the ordinary language of particle physics.

Internal object	**Effective SM-type image**	**Observable shadow**	**Status**	**Chapter**
T^3, Y, Q, C^a	electroweak and color charge observables	charge assignments and current labels	internal → observer dictionary	21–22
Higgs residue and neutral vacuum	Higgs doublet and broken vacuum image	broken-phase mass splitting pattern	internal → effective, imported continuum notation	22
Bosonic kernel blocks	gauge and Higgs quadratic sectors	pole masses and propagator structure	internal → effective	22, 26
Diagonal fermionic kernels	covariant-derivative kinetic terms and neutral currents	charged-sector propagation data	internal → effective	22, 26
Off-diagonal $W^{\pm}$ blocks	charged weak currents	charged-current amplitudes	internal → effective	22, 26
Higgs bridges	Yukawa couplings and mass matrices	mass generation channels	internal construction with imported continuum reading	22, 26
Family module and residue law	Yukawa textures, CKM-type and PMNS-type mixing	mixing matrices and flavour hierarchies	internal closure plus open numerical completion	23, 26
Color kernel and singlet rule	local QCD sector plus asymptotic singlet admissibility	color-singlet asymptotic sectors	local dictionary internal; full confinement open	24, 26
Observer-decategorified action	audited Standard-Model-type effective action	an audited low-energy field-theoretic image	asymptotic low-energy, imported continuum interpretation	26–27

Remark 28.12. The table is a reading aid rather than a replacement for the dictionary theorem. Its purpose is to localize which parts of the final comparison are internal, which are effective, and which rely on imported continuum interpretation.

28.12 Novelty Localization

The manuscript separates genuinely new internal results from new framework organization and from imported background language. The purpose of this table is not to diminish the manuscript's claims, but to prevent novelty inflation by distinguishing new derivations from new organizational synthesis and from imported continuum vocabulary.

Claim block	**New internal result?**	**New organizational synthesis?**	**Imported back-ground?**	**Comment**
structural core and transfer architecture	yes	yes	no	provides the manuscript's internal base language
observer-level dictionary chain	partial	yes	partial	new organization plus partly new bridge claims
one-generation anomaly compatibility	yes	no	no	localized internal consistency result
Standard-Model-type comparison map	no in the continuum vocabulary itself	yes	yes	novelty lies in localizing which parts are internal, observer-level, and imported background language
low-energy continuum interpretation	no	limited organizational use	yes	used only as observer-level reading layer

28.13 Glossary and Notation Map

The audit chapter includes a compact glossary so that the reader can quickly locate the internal terms on which the final claims depend.

Term	First formal location	Status	Observer-level analogue	Warning
structural core	Chapters 0–4	Internal	none directly	not a spacetime or particle ontology by itself
observer projection	Chapters 4–5	Internal → Effective	coarse observable image	not identical with an ordinary measurement postulate
transfer kernel	Chapters 3–8	Internal	effective propagation or interaction generator	internal first, physical second
residue calculus	Chapters 14, 21	Internal	pole-residue extraction language	numerical shadows appear only after observer projection
observer decategorification	Chapters 21, 26	Internal → Effective	low-energy field-theoretic translation	forgets higher structural data and is not a new dynamics
family module	Chapter 23	Internal → Effective	flavour texture and mixing organization	not an independent charge sector
physical observer resolvent	Chapter 24	Internal → Effective	projected asymptotic resolvent	canonical only when selection conditions isolate a unique projector

The main symbols used in the final audit are summarized below.

Symbol	Meaning	Primary location
$\mathcal{D}_B$, $\mathcal{D}_F$	bosonic and fermionic effective kernels	Chapters 14, 22, 26
$\mathcal{R}_{\mathrm{micro}}$, $\mathcal{R}_{\mathrm{phys}}$	microscopic and projected observer resolvents	Chapter 24
T^3, Y, Q, C^a	residue observables for electroweak and color data	Chapters 21–22
χ_{obs}	observer-decategorification map into low-energy field language	Chapter 26
$Z_i^{(f)}$, $\lambda_i^{(f)}$	residue factors and dominant branches	Chapters 14, 21, 23
V_{CKM}, U_{PMNS}	quark and lepton mixing matrices as relative diagonalizations	Chapters 23, 26

28.14 Algorithmic Computational Schema

The manuscript supports an algorithmic schema even before full quantitative calibration is complete. Each step below records input, operation, output, current status, and main failure mode.

1. **Structural initialization.** Input: a structural object class, observer rules, admissible transformations, and transfer kernels. Operation: fix the admissible symbolic dynamics and coarse access rules. Output: a structural-physical base model. Status: **[Internal]**. Failure mode: incompatible observer rules or missing kernel closure.

2. **Pole-residue extraction.** Input: admissible kernels and observer-compatible sectors. Operation: compute dominant branches and identify stable pole-residue sectors. Output: candidate low-energy sectors with residue data. Status: **[Internal]** with **[Asymptotic low-energy]** interpretation. Failure mode: no stable projected branches.

3. **Internal observable extraction.** Input: residue-labelled sectors. Operation: extract charge observables, family labels, and sector projectors. Output: observer-relevant residue data. Status: **[Internal]**. Failure mode: transfer instability of residue observables.

4. **Electroweak and anomaly restriction.** Input: candidate residue-labelled matter sectors. Operation: impose Higgs-bridge and anomaly-consistency constraints. Output: admissible low-energy matter candidates at the one-generation audited level. Status: **[Internal]** for one-generation compatibility, **[Open target]** for stronger uniqueness. Failure mode: empty admissible set or non-unique branch proliferation.

5. **Flavour and color construction.** Input: admissible electroweak-color sectors. Operation: construct flavour blocks, color blocks, and the relevant projected observer resolvents. Output: candidate mixing data and admissible asymptotic sector structure. Status: **[Internal → Effective]**. Failure mode: unresolved projector ambiguity or unstable flavour closure.

6. **Observable extraction.** Input: pole-residue data and projected branch structure. Operation: construct the quantities from which masses, mixing matrices, and admissible asymptotic sectors are to be extracted after the stated calibration and closure conditions are met. Output: symbolic observable data. Status: **[Asymptotic low-energy]** with quantitative calibration still open. Failure mode: normalization mismatch or insufficient low-energy closure.

7. **Observer-level dictionary.** Input: kernel-side action and observer-relevant residue data. Operation: apply observer decategorification and low-energy continuum expansion. Output: an audited Standard-Model-type effective local field description. Status: **[Internal → Effective]** plus **[Imported continuum identification]**. Failure mode: loss of necessary invariants under the translation step.

The main computational ambiguities currently known are summarized below.

Source of ambiguity	**Where it enters**	**Physical or gauge-like?**	**Current fixing rule**
projector choice	observer-resolvent construction	partly physical, partly representational	use the admissible projected observer sector selected by the current branch conditions
normalization conventions	low-energy dictionary and continuum reading	physical once calibrated	left explicit in the final audit as an open normalization point
branch multiplicity	residue extraction and flavour construction	physical if stable under projection	keep only branches surviving the audited projection and pole criteria
closure regime choice	flavour relations and higher-layer effective descriptions	asymptotic effective choice	restrict claims to the stated low-energy closure regime

28.15 Toy Worked Micro-Example

The manuscript does not yet contain a full numerical Standard Model extraction. It does, however, support the following minimal worked micro-example.

Example 28.13 (Toy one-generation electroweak-color extraction)**.** Start with a candidate residue sector carrying electroweak data (T^3, Y), family-neutral data, and color multiplicity. Impose the Higgs-bridge relations of Chapter 22 and the one-generation anomaly constraints of Chapter 25. This reduces the admissible hypercharge assignments to an audited admissible one-generation branch after neutral-neutrino selection. Then project a surviving projected observer sector through the observer-resolvent architecture of Chapter 24 and translate the surviving electroweak and color blocks by the observer-decategorification map under the current admissibility assumptions of Chapter 26. The result is not a full numerical Standard Model fit, but it is a complete symbolic demonstration of the chain

$$\begin{aligned} \text{residue data} &\longrightarrow \text{audited one-generation branch} \\ &\longrightarrow \text{a surviving projected observer sector} \\ &\longrightarrow \text{an audited low-energy field-theoretic image.} \end{aligned}$$

Remark 28.14. This example is intentionally modest. It is included to make the computational chain less schematic, not to claim a completed numerical extraction of masses or couplings.

28.16 Quantitative Status and Immediate Checkpoints

The current manuscript distinguishes symbolic extraction from numerical calibration.

Quantity class	**Current status**	**Pass criterion**	**Failure signal**
charge organization and one-generation anomaly compatibility	internally established	reconstruction of an audited admissible one-generation branch under the stated Higgs and anomaly assumptions	no stable anomaly-compatible branch under the audited assumptions
flavour organization	structurally organized, not fully numerically calibrated	at least three stable observer-relevant flavour classes, surviving off-diagonal coupling data, constrained but nontrivial mixing, and nonuniform projected residue weights in explicit toy models	collapse to an undifferentiated flavour sector or loss of mixing data under projection
effective low-energy dictionary	structurally established up to imported continuum identification	translated low-energy terms match the intended observer-level sector structure under the stated admissibility assumptions	loss of sector labels, current assignments, or pole data in the translation step
color organization and admissibility	structurally organized at observer-projective level	stable tripartite color labeling before singlet projection, exclusion of isolated observer-level color states, and admissible composite singlet channels	surviving isolated observer-level color states or loss of singlet admissibility
full mass and coupling normalization	open	an explicit calibration scheme tying symbolic parameters to low-energy observables	no stable normalization rule linking symbolic and observer-level parameters
nonperturbative color spectrum and confinement	open target	projected resolvent behavior excludes isolated color poles while retaining admissible singlet	colored asymptotic poles survive or singlet sectors do not persist

Remark 28.15 (Scope of quantitative status)**.** Quantitative status in this manuscript means only the present state of structurally extractable quantities and observer-level effective parameters. Where numerical calibration is not yet supplied, the text refers only to formally available observables or parameter-bearing effective quantities, not to a completed phenomenological fit.

Remark 28.16. The immediate falsifiability of the present manuscript is therefore not "full numerical reproduction of all observed data". It is the failure or success of the stated structural checkpoints: anomaly compatibility, residue-to-observer consistency, projected-sector admissibility, and qualitative flavour or color behavior in explicit model instances.

28.17 Lean Formalization Appendix

The manuscript was designed to admit staged formalization. The following split is the natural Lean-oriented decomposition.

1. A structural-core package covering symbolic structures, sequence objects, h-patterns, EFU assembly, and transfer kernels.
2. An observer-and-dynamics package covering causal rewriting, projection, geometric channels, structural action, and effective equations.
3. A quantum-statistical package covering amplitudes, density structures, reduced dynamics, and thermodynamic objects.
4. A continuum-and-field package covering transport, relativistic limits, propagators, poles, scattering sectors, and many-body closures.
5. A higher-organization package covering chemistry, biological organization, cognition, language, and formal symbolic systems as later effective nodes.
6. A Standard Model package covering residue calculus, electroweak sectors, flavour, color, anomaly checks, and the effective dictionary.
7. An audit package covering dependency graphs, statement classification, failure modes, computational schemas, and target-theorem wrappers.

Each package should expose primitive types, structure morphisms, admissibility predicates, theorem namespaces, and an explicit imported-axiom boundary. The formalization policy of the manuscript is the following.

- internal closed results should be tagged as `theorem`;

- imported background assumptions should be tagged as `axiom` or placed in a clearly separated imported layer;
- definitions and canonical maps should be tagged as `def`;
- purely notational compression should be tagged as `abbrev`;
- open targets should be tagged as `conjecture` or explicit target-theorem wrappers.

Remark 28.17. The Lean appendix should preserve the same discipline as the text itself: each theorem must expose its hypotheses explicitly, imported classical lemmas must be isolated, and target theorems must never be disguised as completed proofs. The formalization success criterion for the present manuscript is therefore staged: first certify the internal dependency chain, then isolate imported continuum identifications, and only after that formalize the open target wrappers as conjectural interfaces.

28.18 Conclusion of the Chapter

This final chapter closed the manuscript by stating the main audited reconstruction statement in conditional form, recording the dependency structure, classifying internal versus imported versus open content, listing the principal failure modes, comparing the internal framework with ordinary Standard Model formalism, and fixing a computational and Lean-formalization boundary.

The resulting conclusion is deliberately restrained. The manuscript develops a structural-physical framework intended to yield, within the audited assumptions and scope restrictions, a unified chain of effective descriptions. Its low-energy microphysical image reconstructs an audited Standard-Model-type effective description, while its higher layers organize thermodynamic, many-body, chemical, biological, cognitive, and formal-symbolic regimes within a proposed dependency chain. The manuscript also records the precise places where mathematical closure is still incomplete. That audit is part of the scientific content of the manuscript, because it identifies both established derivations and precise remaining closure gaps.